



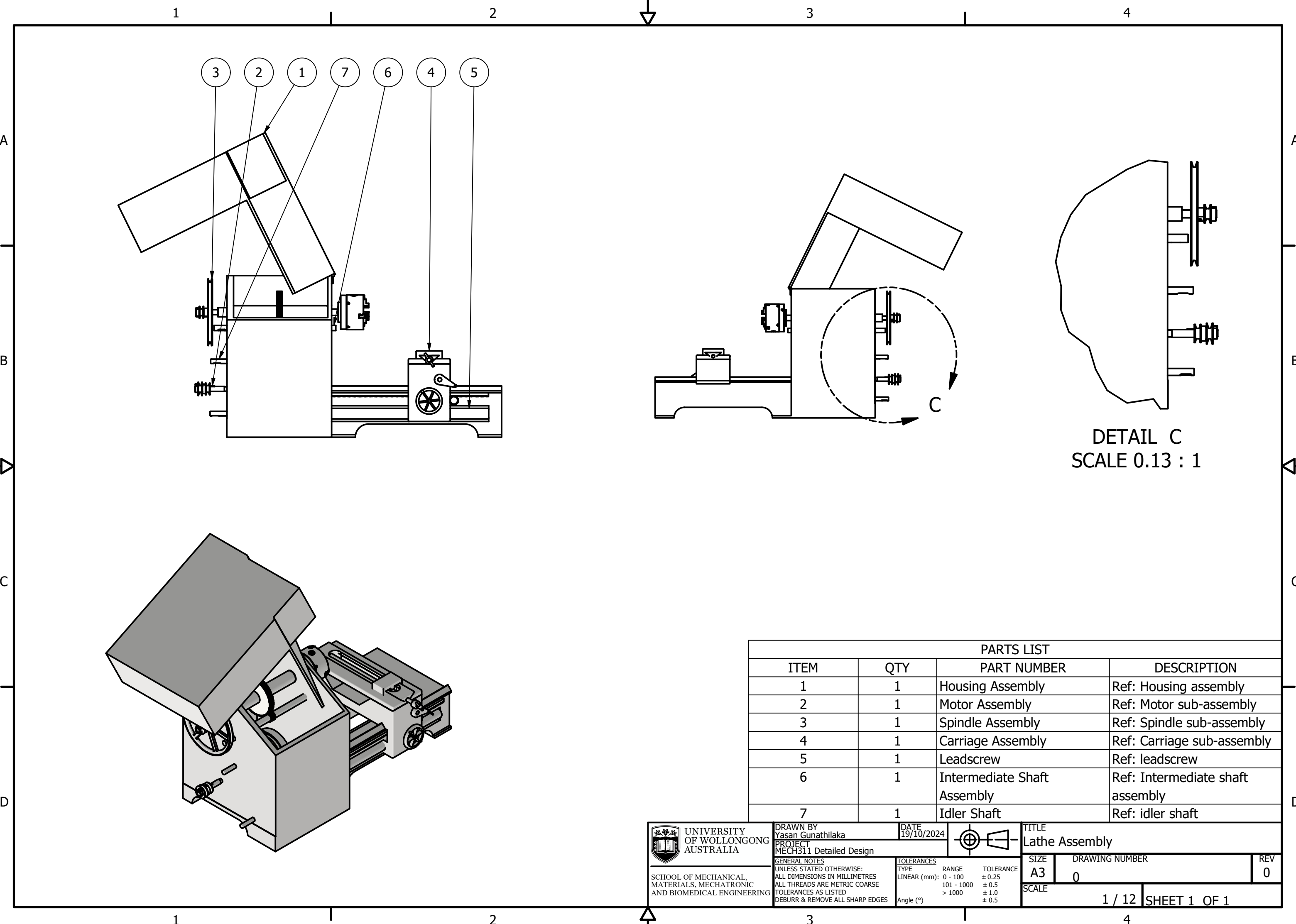
UNIVERSITY
OF WOLLONGONG
AUSTRALIA

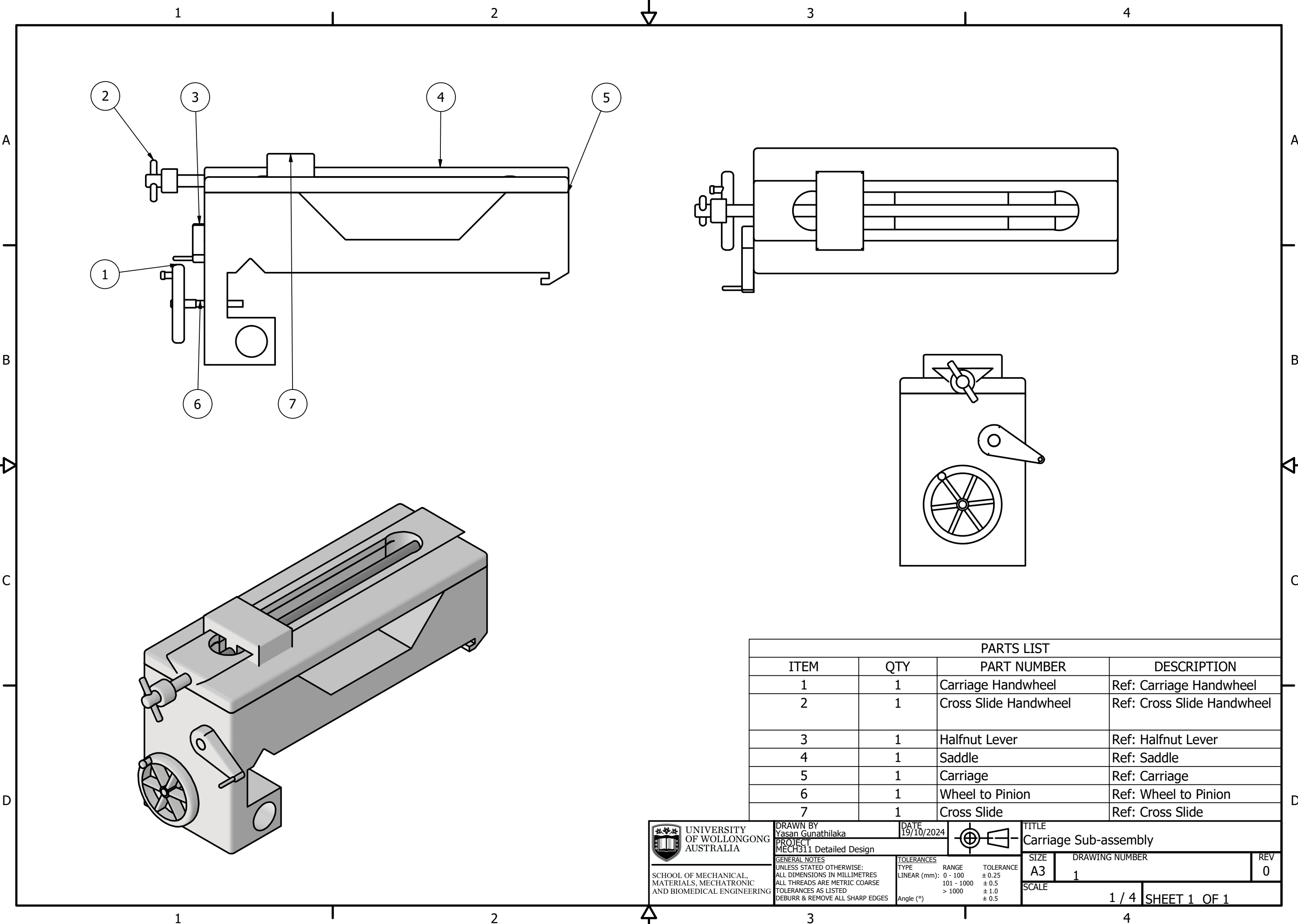
School of Mechanical, Materials, Mechatronic and Biomedical Engineering

Final Design Report Design Workbook

STUDENT NAME	STUDENT NUMBER
YASAN GUNATHILAKA	7549246

MECH311 – Mechanical Engineering Design
Spring 2024





A

B

C

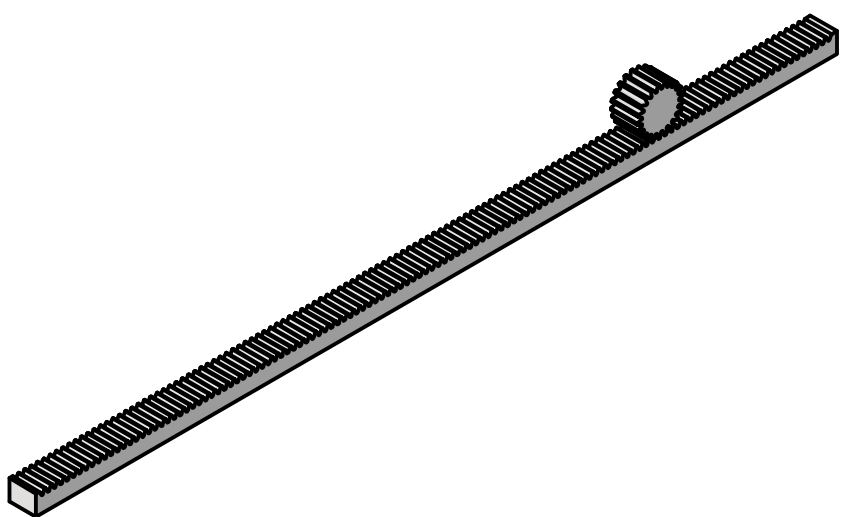
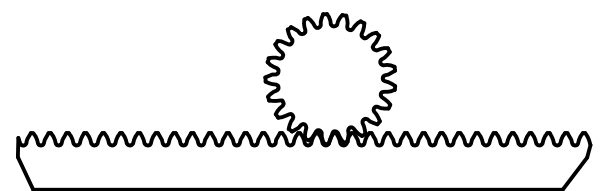
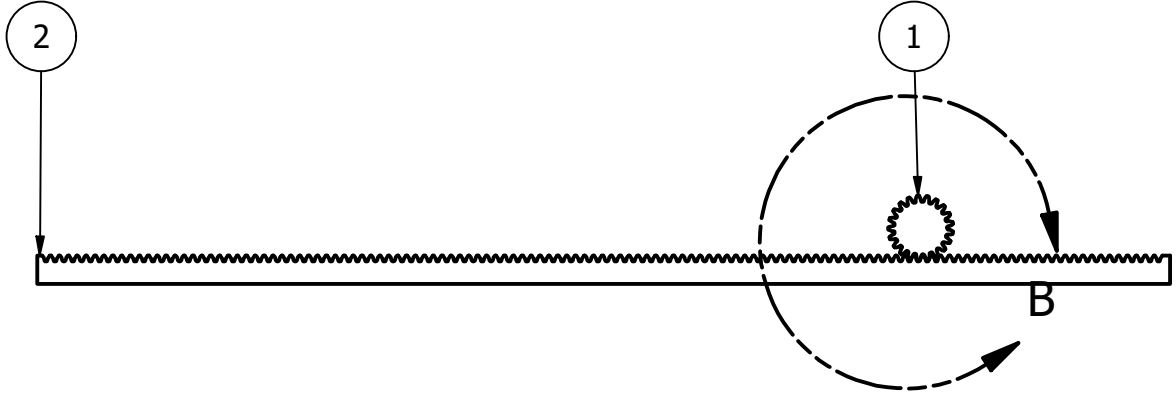
D

A

B

C

D



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Pinion_New	Ref: Pinion_New
2	1	Rack_New	Ref: Rack_New



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SCHOOL OF MECHANICAL,
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AND BIOMEDICAL ENGINEERING

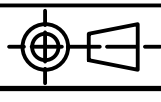
DRAWN BY
Yasan Gunathilaka

DATE
19/10/2024

PROJECT
MECH311 Detailed Design

GENERAL NOTES

UNLESS STATED OTHERWISE:
ALL DIMENSIONS IN MILLIMETRES
ALL THREADS ARE METRIC COARSE
TOLERANCES AS LISTED
DEBURR & REMOVE ALL SHARP EDGES



TOLERANCES

TYPE

RANGE

TOLERANCE

LINEAR (mm):

0 - 100

± 0.25

101 - 1000

± 0.5

> 1000

± 1.0

Angle (°)

± 0.5

TITLE

Rack and Pinion Sub-assembly

SIZE

A3

DRAWING NUMBER

4

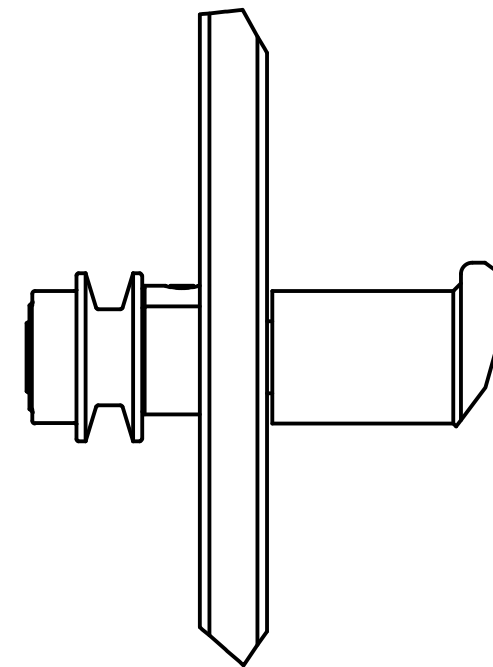
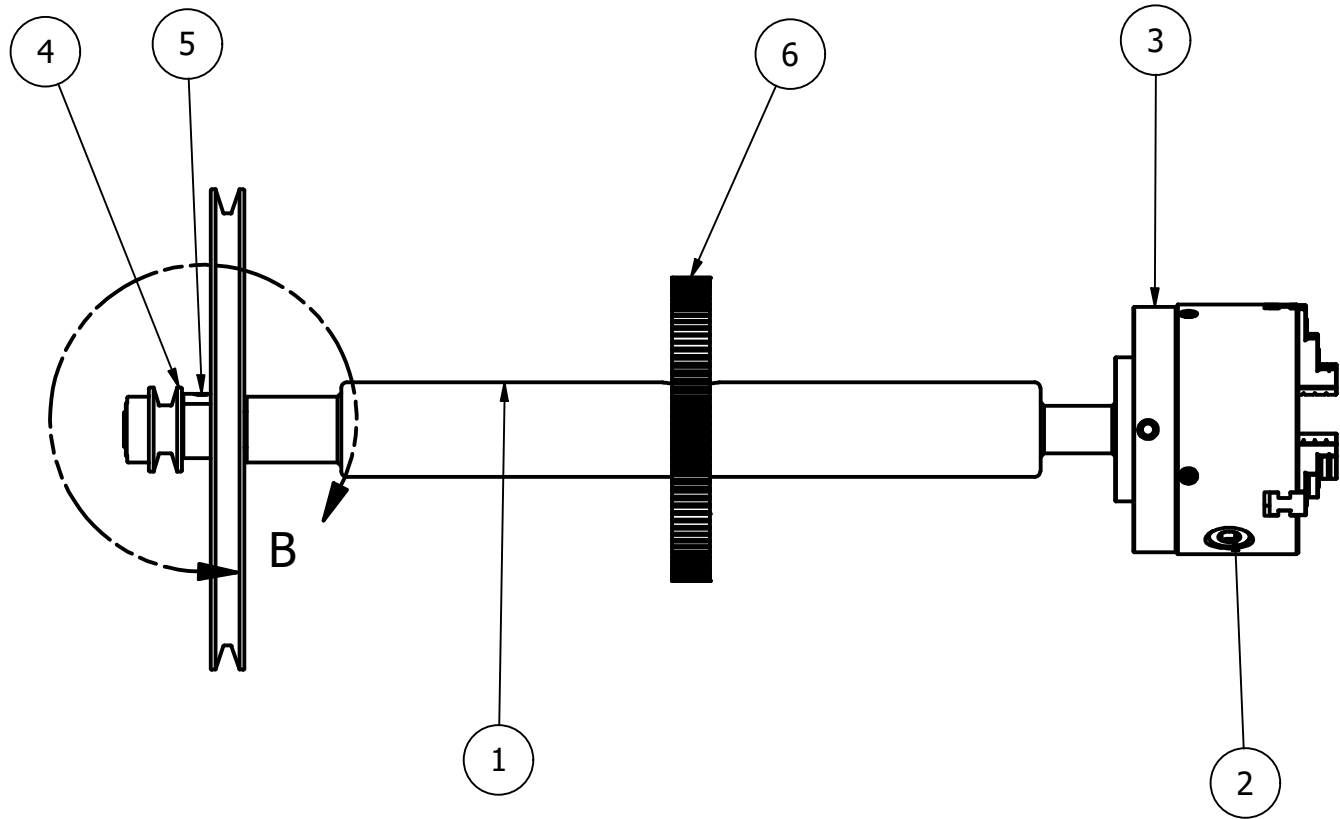
REV

0

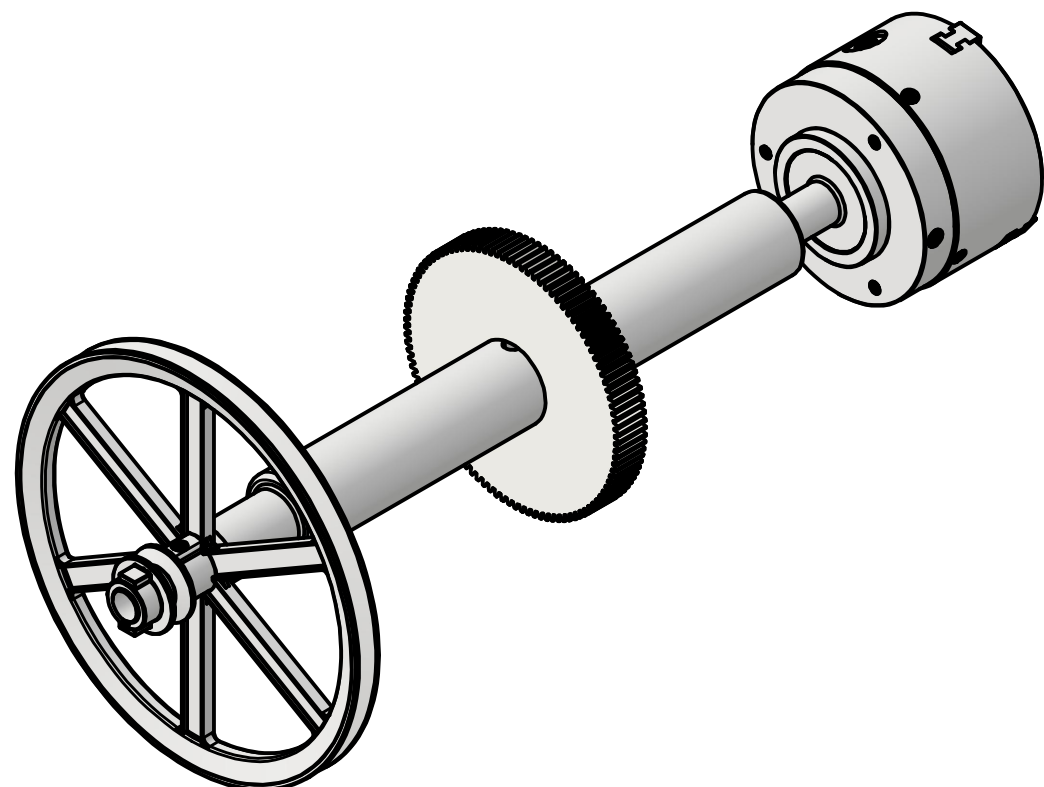
SCALE

1 / 4

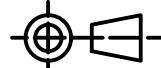
SHEET 1 OF 1



DETAIL B
SCALE 1 / 2



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Spindle	Ref: Spindle
2	1	1506N51_3-Jaw Steel Lathe Chuck	Ref: 1506N51_3-Jaw Steel Lathe Model
3	1	1506N58_4 Degree Hardinge Mounting Plate for 5 Diameter 3-Jaw Steel Lathe Chuck	Ref: 1506N58_4 Degree Hardinge Model
4	1	pulley51mm_motor	Ref: pulley51mm_motor
5	1	pulley254mm_spindle	Ref: pulley254mm_spindle
6	1	gear105	Ref: gear105

 UNIVERSITY OF WOLLONGONG AUSTRALIA	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024		TITLE Spindle Sub-assembly			
	PROJECT MECH311 Detailed Drawings				SIZE A3	DRAWING NUMBER 5a	REV 0	
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES				SCALE 1 / 4 SHEET 1 OF 1			
			TOLERANCES					
		TYPE	RANGE	TOLERANCE				
		LINEAR (mm):	0 - 100	± 0.25				
			101 - 1000	± 0.5				
			> 1000	± 1.0				
		Angle (°)		± 0.5				

A

B

C

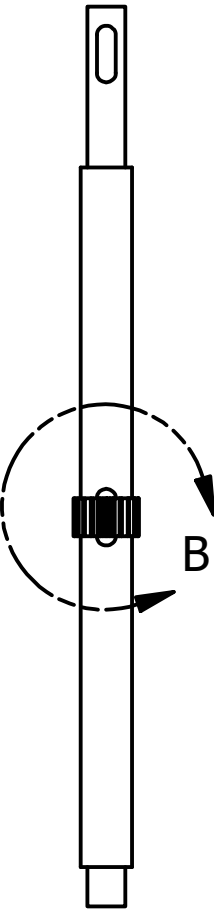
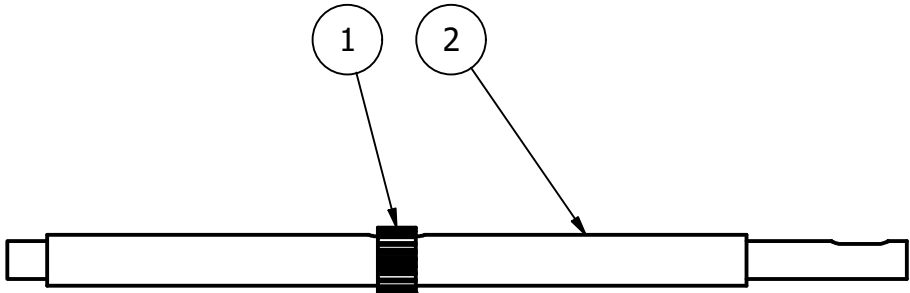
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A

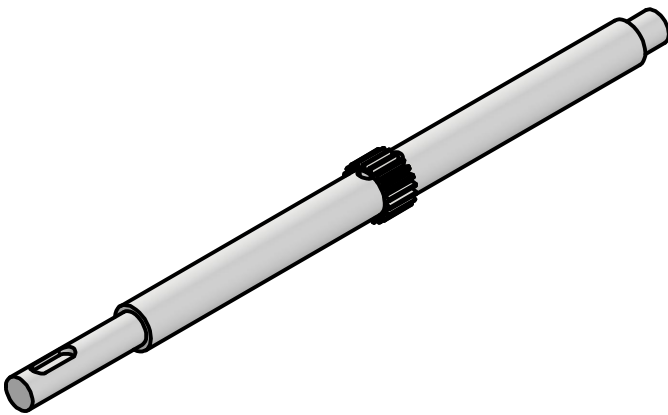
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C

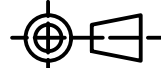
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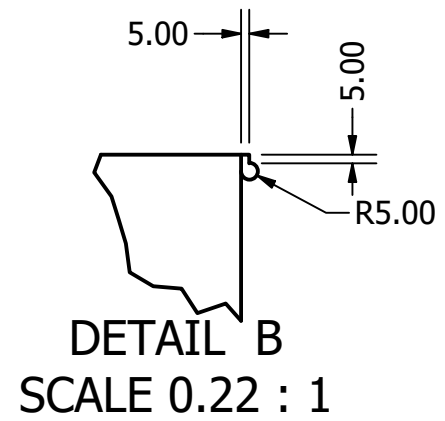
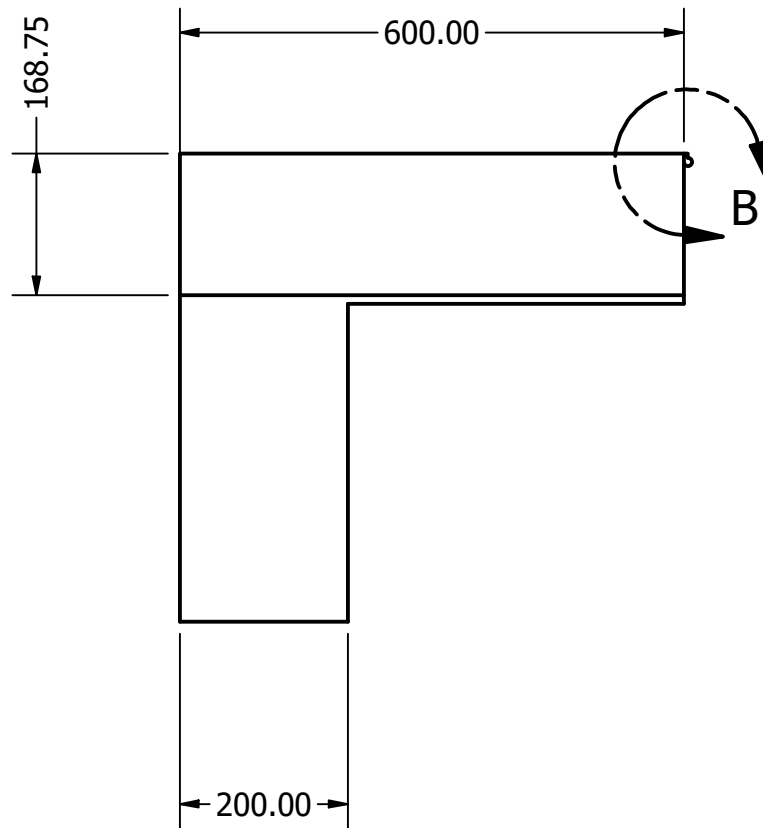
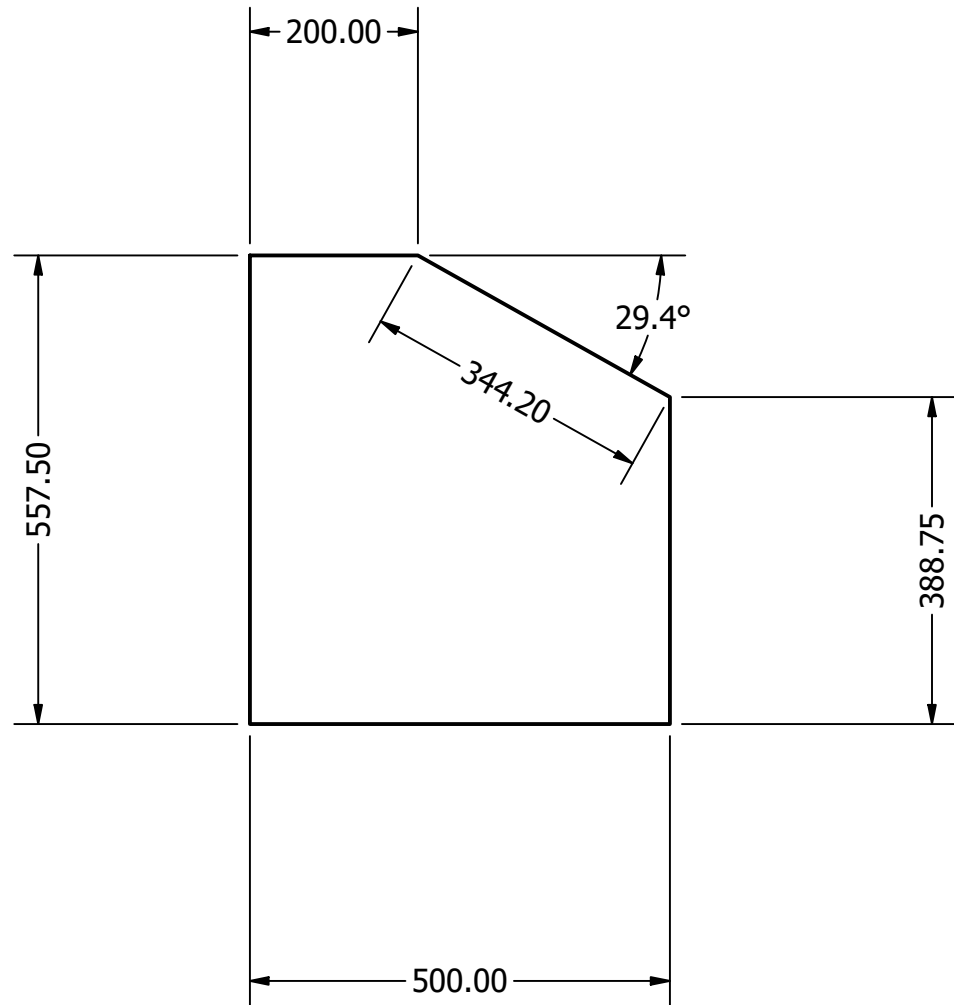


DETAIL B
SCALE 1 / 2

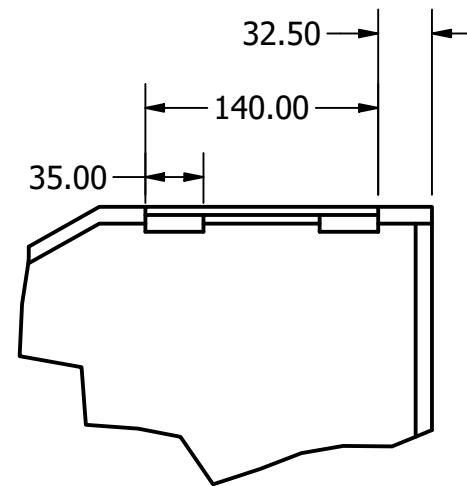


PARTS LIST						
ITEM		QTY	PART NUMBER		DESCRIPTION	
1		1	gear21		Ref: gear21 model	
2		1	Intermediate Shaft		Ref: Intermediate Shaft	

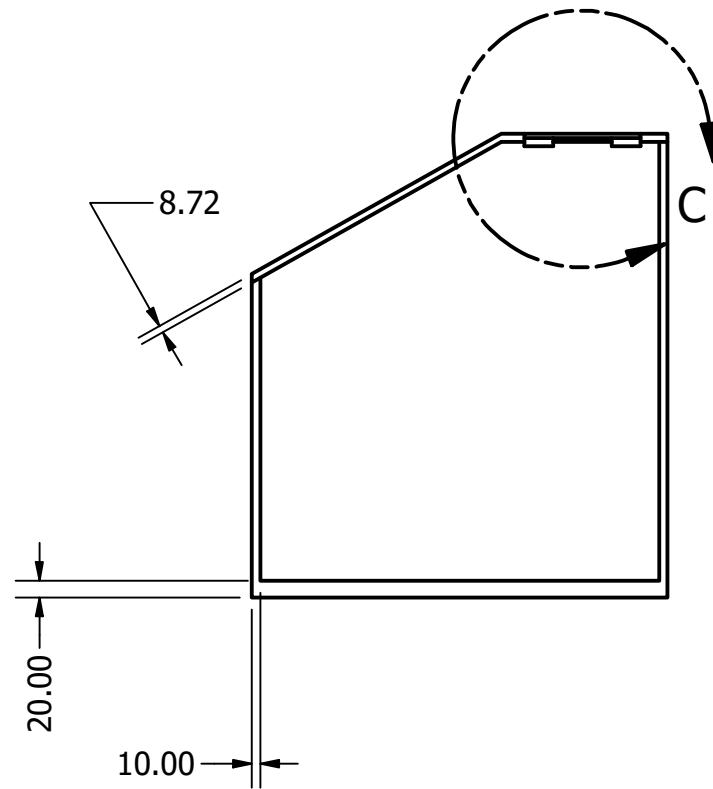
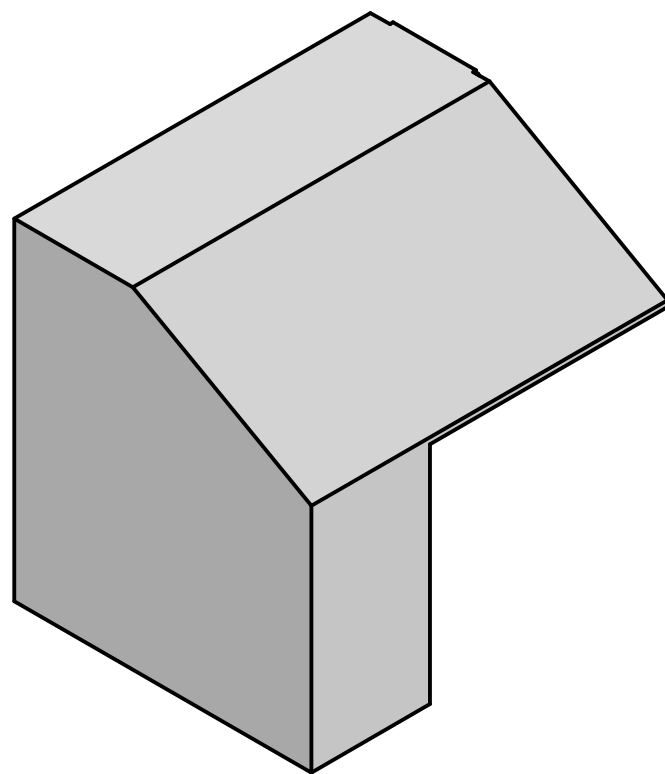
UNIVERSITY OF WOLLONGONG AUSTRALIA SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024		TITLE Intermediate Shaft Sub-Assembly		
	PROJECT MECH311 Detailed Design				SIZE A3	DRAWING NUMBER 5b	REV 0
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES TYPE RANGE TOLERANCE LINEAR (mm): 0 - 100 ± 0.25 101 - 1000 ± 0.5 > 1000 ± 1.0 Angle (°) ± 0.5		SCALE 1 / 4		
		SHEET 1 OF 1					



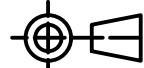
DETAIL B
SCALE 0.22 : 1



DETAIL C
SCALE 0.22 : 1



Material: Stainless Steel
Qty: 1

 UNIVERSITY OF WOLLONGONG AUSTRALIA	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024				TITLE Gearbox Cover		
	PROJECT MECH311 Detailed Drawing								
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES				SIZE A3	DRAWING NUMBER 1b	REV 0
			TYPE		RANGE	TOLERANCE	SCALE		
LINEAR (mm):			0 - 100	± 0.25	1 / 9 SHEET 1 OF 1				
			101 - 1000	± 0.5					
			> 1000	± 1.0					
			Angle (°)	± 0.5					

A

B

C

D

A

B

C

D

1

2

3

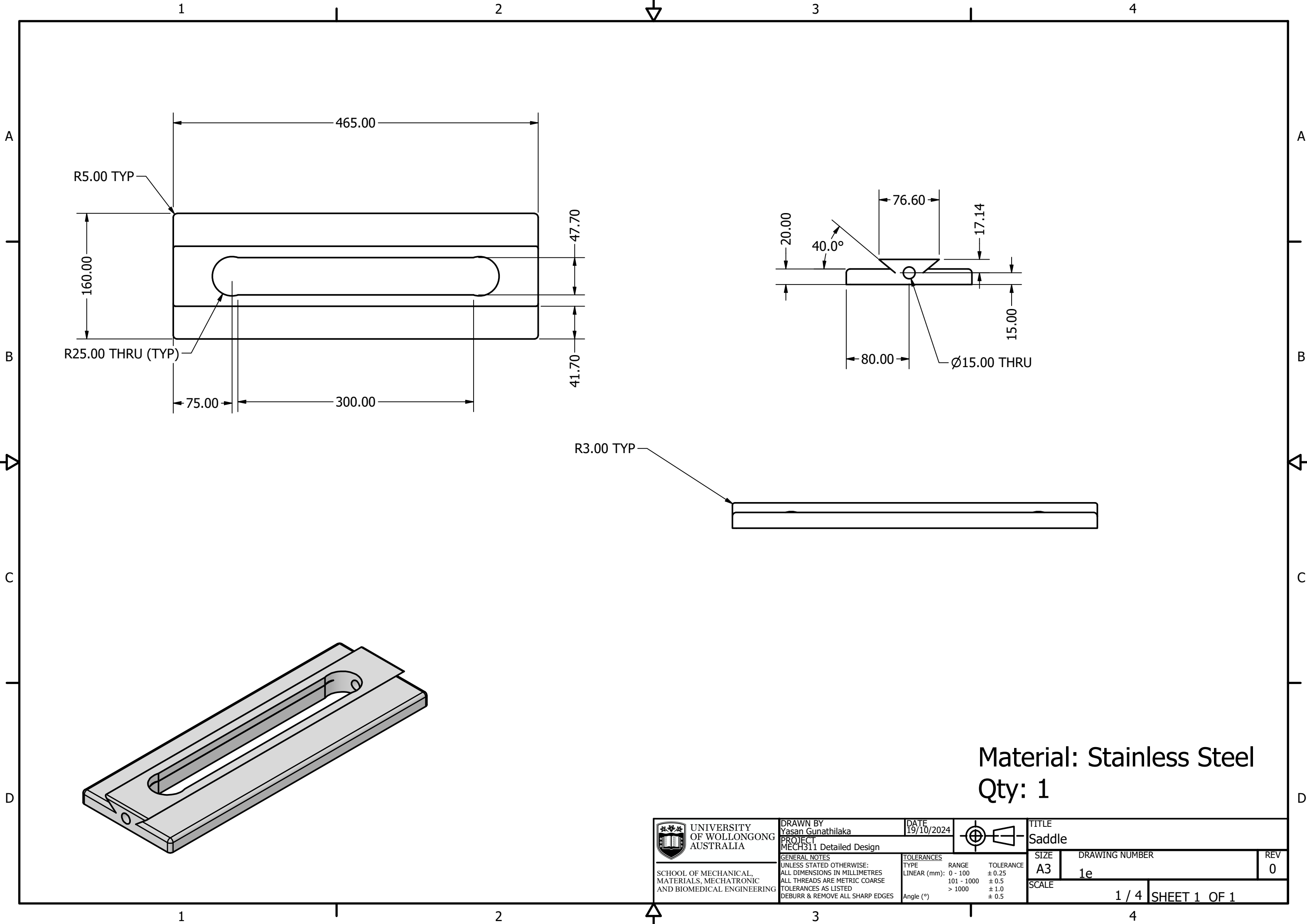
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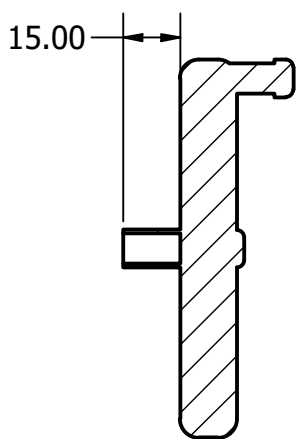
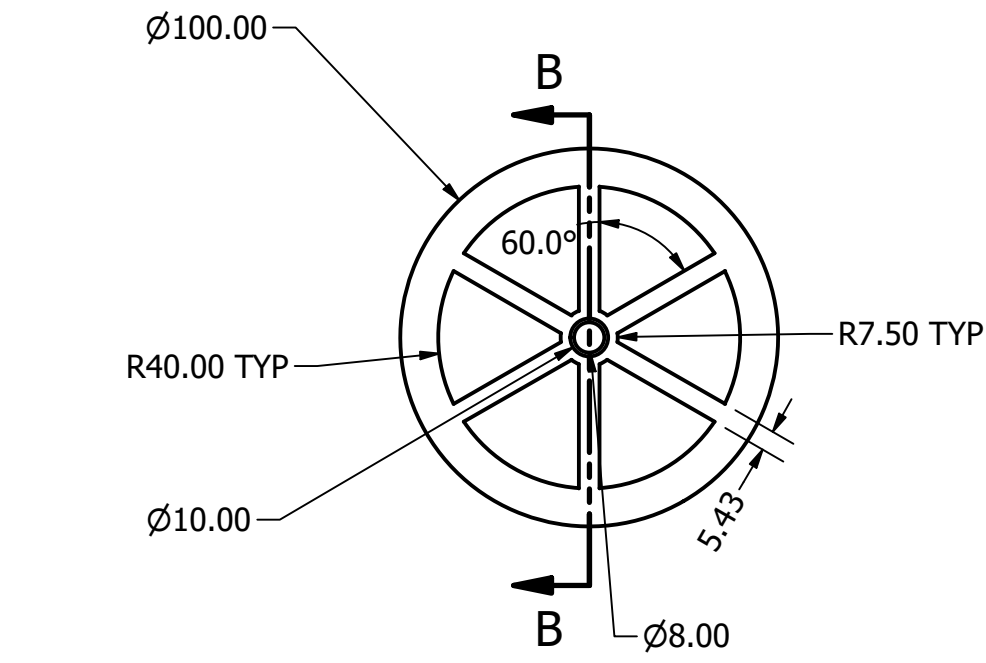
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75.00

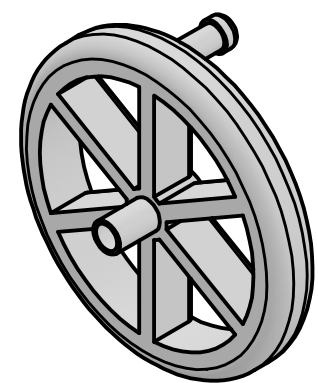
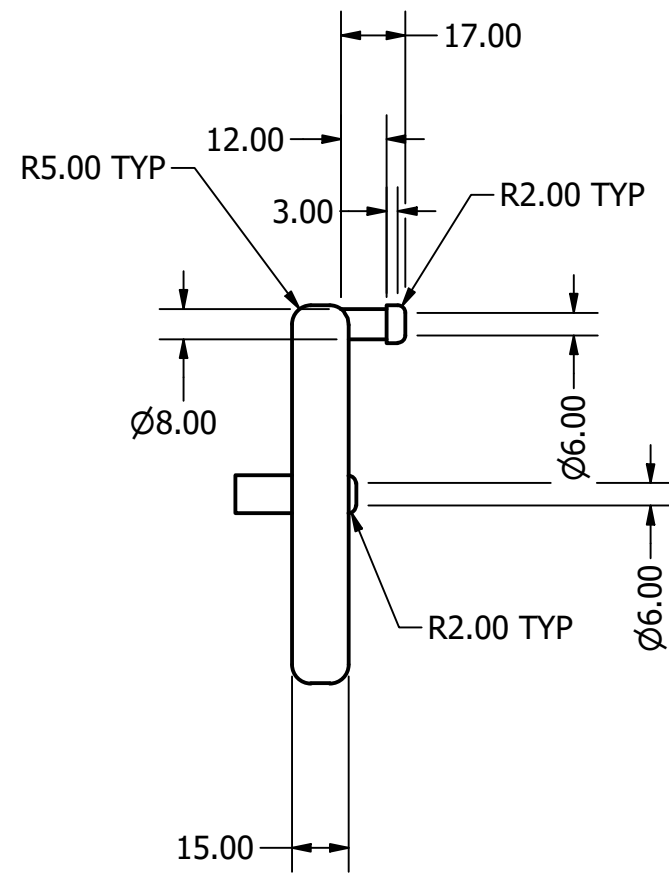
Material: Stainless Steel
Qty: 1

	UNIVERSITY OF WOLLONGONG AUSTRALIA		DRAWN BY Yasan Gunathilaka		DATE 19/10/2024				TITLE Wheel to Pinion		
	SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING		PROJECT MECH311 Detailed Design						SIZE A3	DRAWING NUMBER 1d	REV 0
			GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES TYPE LINEAR (mm):		RANGE 0 - 100 101 - 1000 > 1000	TOLERANCE ± 0.25 ± 0.5 ± 1.0 ± 0.5	SCALE 2 : 1		SHEET 1 OF 1

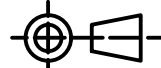


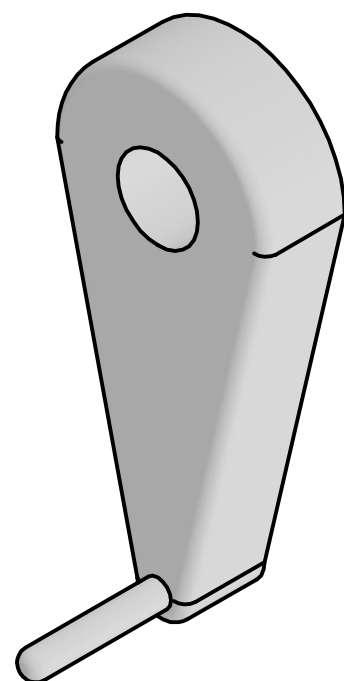
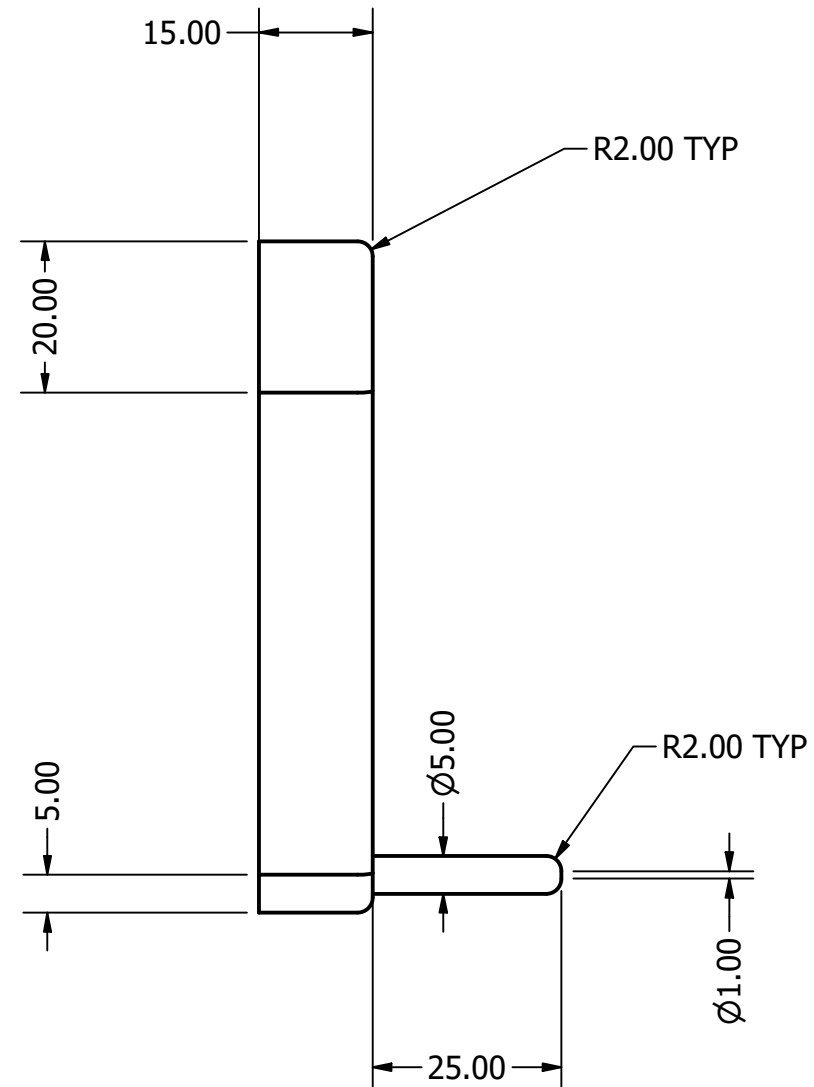
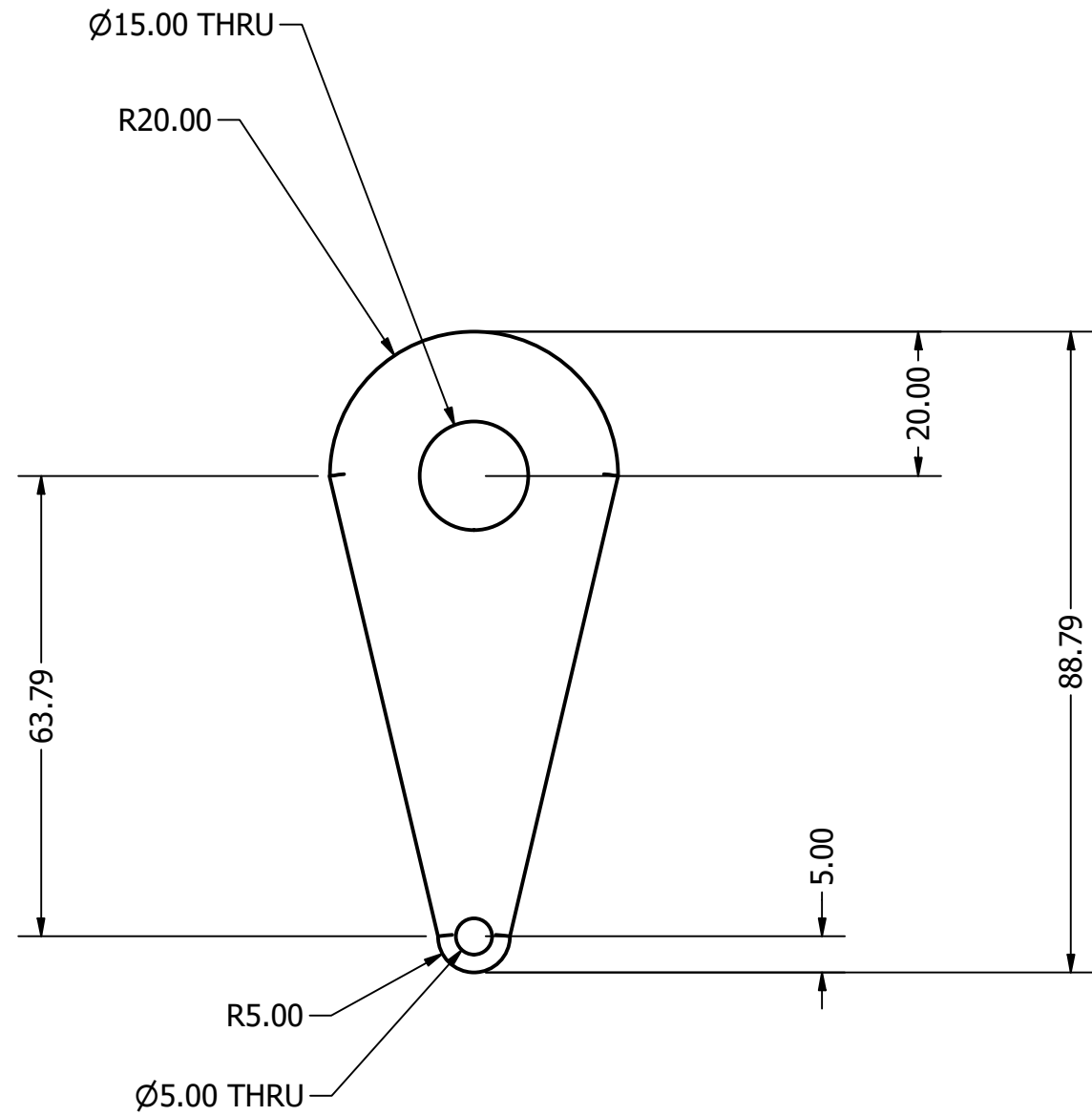


SECTION B-B
SCALE 1 / 2

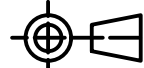


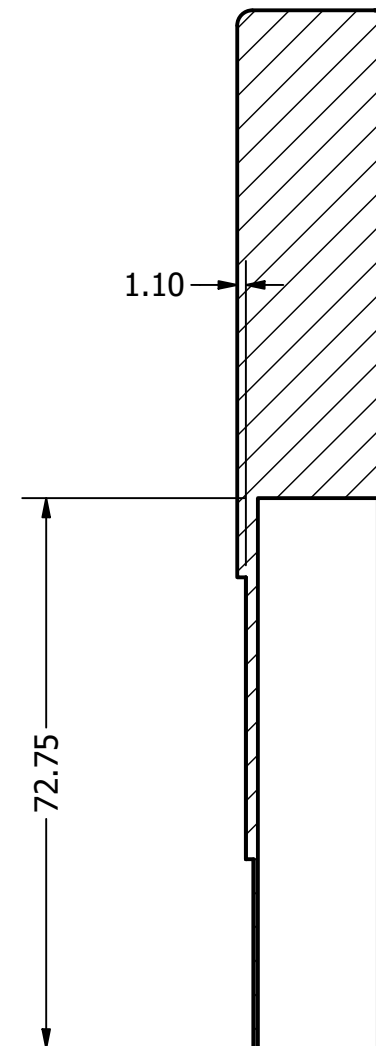
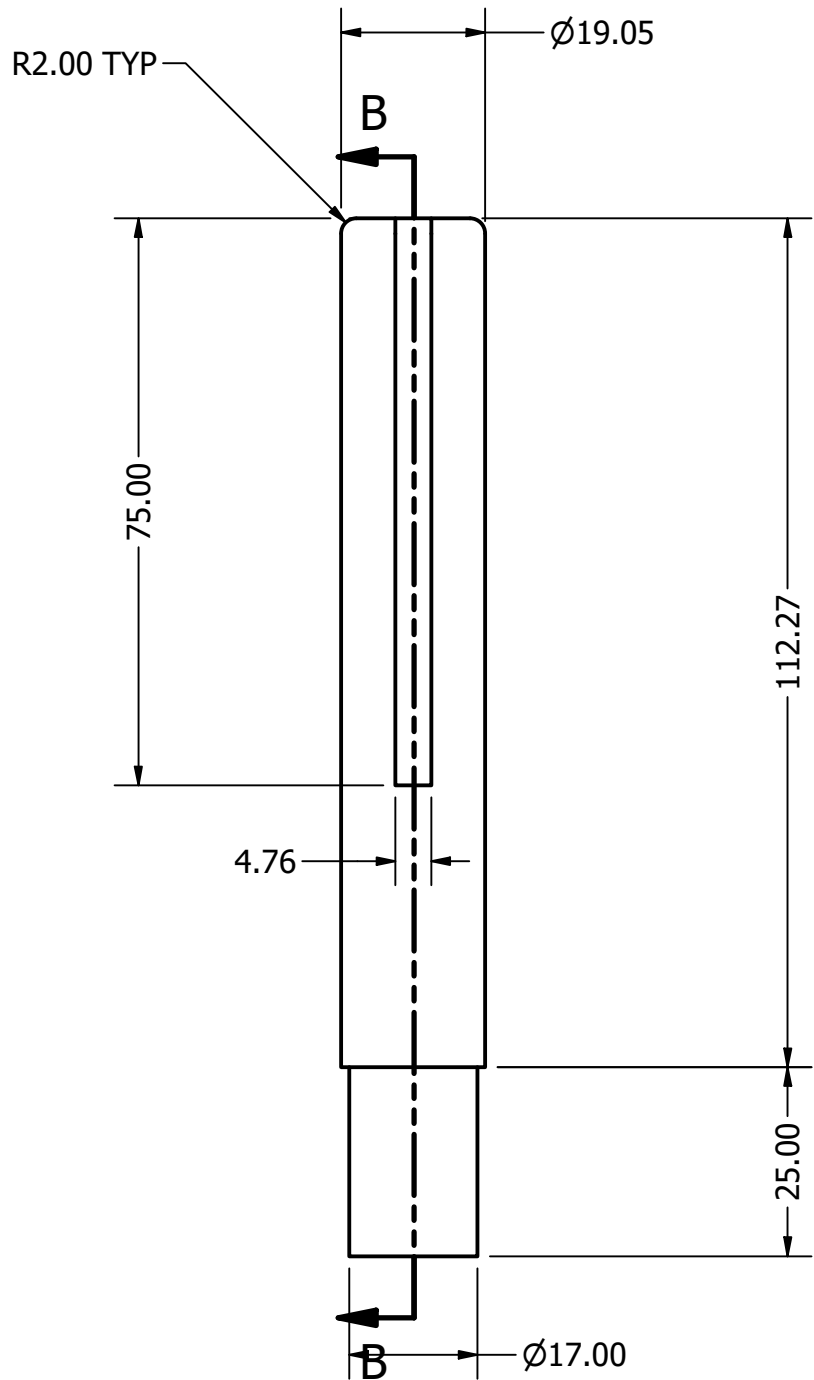
Material: Stainless Steel
Qty: 1

 UNIVERSITY OF WOLLONGONG AUSTRALIA SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	DRAWN BY Yasan Gunathilaka		DATE 19/11/2024		TITLE Carriage Handwheel		
	PROJECT MECH311 Detailed Design				SIZE A3	DRAWING NUMBER 1f	REV 0
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES TYPE RANGE TOLERANCE LINEAR (mm): 0 - 100 ± 0.25 101 - 1000 ± 0.5 > 1000 ± 1.0 Angle (°) ± 0.5		SCALE 1 / 2		
			SHEET 1 OF 1				

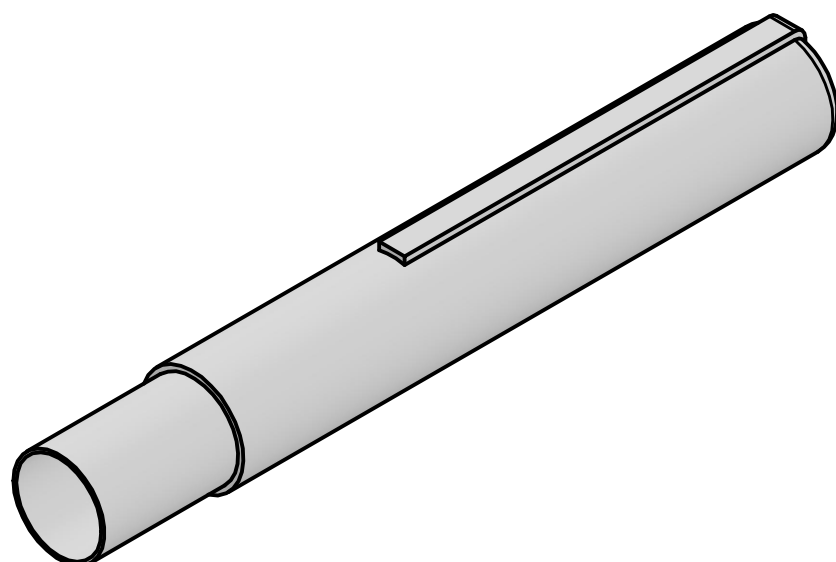


Material: Stainless Steel
Qty: 1

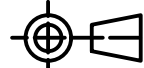
 UNIVERSITY OF WOLLONGONG AUSTRALIA SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	DRAWN BY Yasan Gunathilika		DATE 19/10/2024		TITLE Halfnut Lever			
	PROJECT MECH311 Detailed Design				SIZE A3	DRAWING NUMBER 1f	REV 0	
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES				SCALE 1 : 1 SHEET 1 OF 1			
TOLERANCES								
TYPE				RANGE	TOLERANCE			
LINEAR (mm):				0 - 100	± 0.25			
				101 - 1000	± 0.5			
				> 1000	± 1.0			
Angle (°)					± 0.5			



SECTION B-B
SCALE 1 : 1



Material: Stainless Steel
Qty: 1

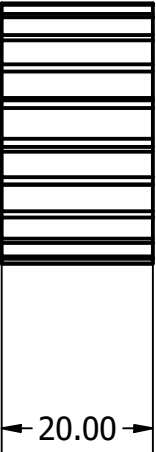
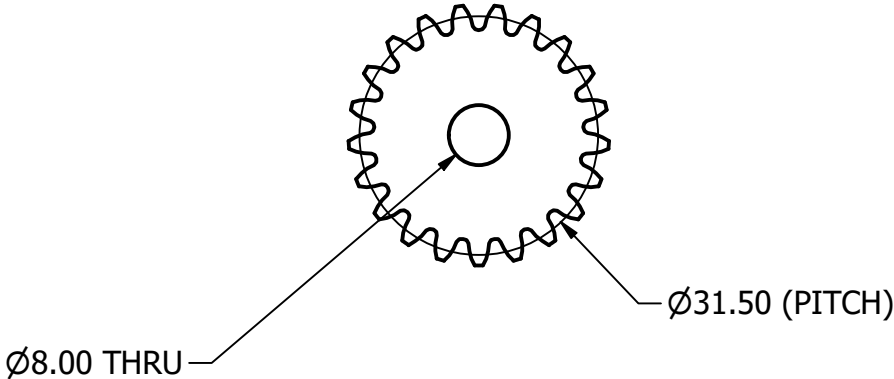
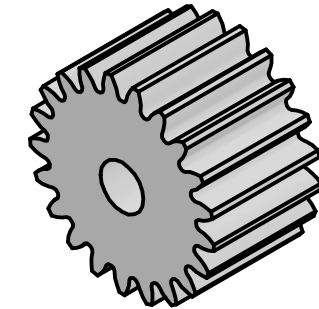
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	PROJECT MECH311 Detailed Design						SIZE A3	DRAWING NUMBER 2a	REV 0
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES				SCALE		
			TYPE LINEAR (mm):	RANGE 0 - 100 101 - 1000 > 1000	TOLERANCE ± 0.25 ± 0.5 ± 1.0 ± 0.5			1 : 1	
		Angle (°)				SHEET 1 OF 1			

A

B

C

D



Material: Stainless Steel
Qty: 1



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AUSTRALIA

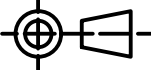
SCHOOL OF MECHANICAL,
MATERIALS, MECHATRONIC
AND BIOMEDICAL ENGINEERING

DRAWN BY
Yasan Gunathilika

PROJECT
MECH311 Detailed Design

GENERAL NOTES
UNLESS STATED OTHERWISE:
ALL DIMENSIONS IN MILLIMETRES
ALL THREADS ARE METRIC COARSE
TOLERANCES AS LISTED
DEBURR & REMOVE ALL SHARP EDGES

DATE
19/10/2024



TOLERANCES		
TYPE	RANGE	TOLERANCE
LINEAR (mm):	0 - 100	± 0.25
	101 - 1000	± 0.5
	> 1000	± 1.0
Angle (°)		± 0.5

TITLE
Pinion

SIZE	DRAWING NUMBER	REV
A3	1d	0

SCALE	
1 : 1	SHEET 1 OF 1

A

B

C

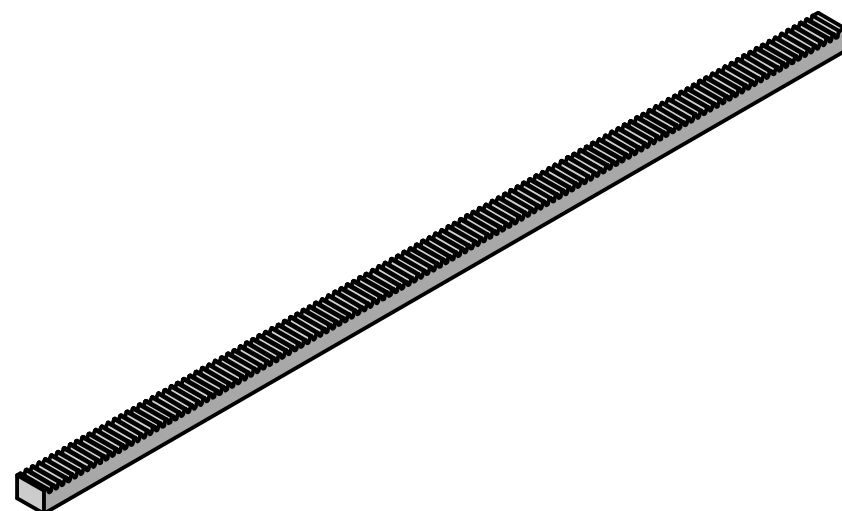
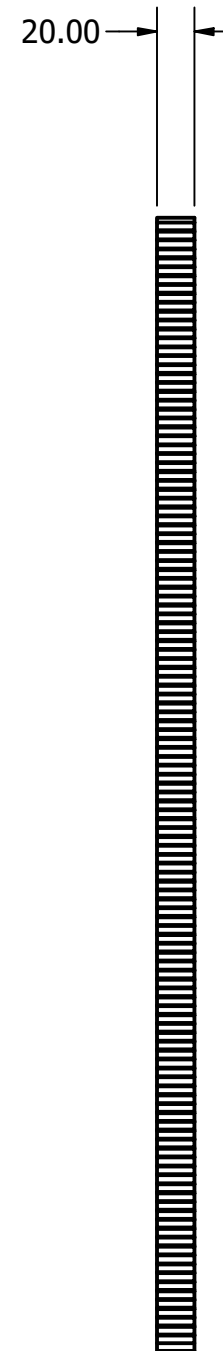
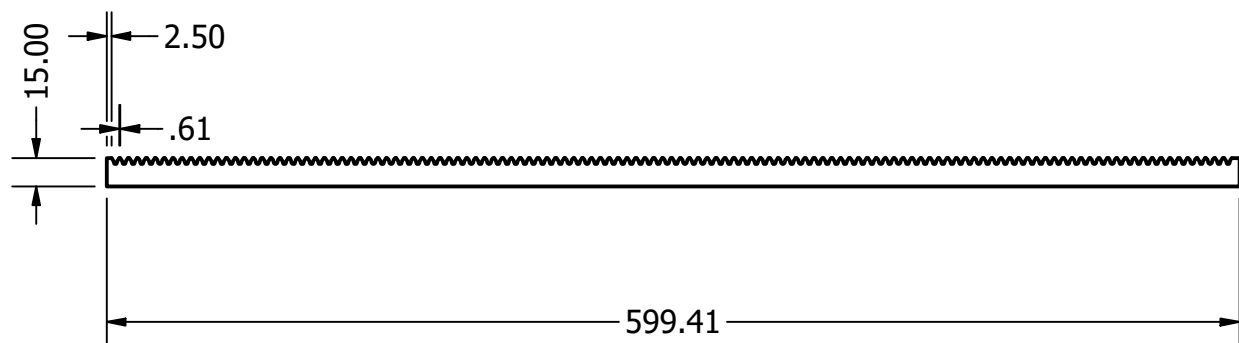
D

A

B

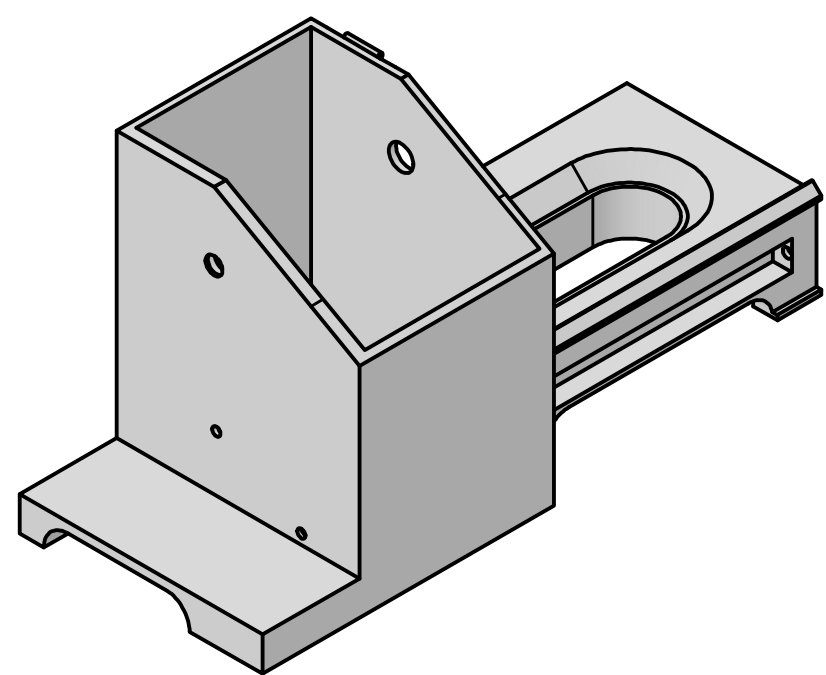
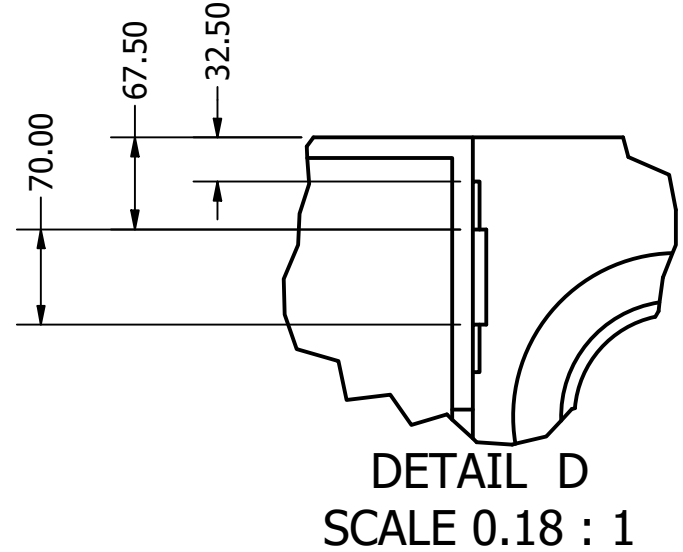
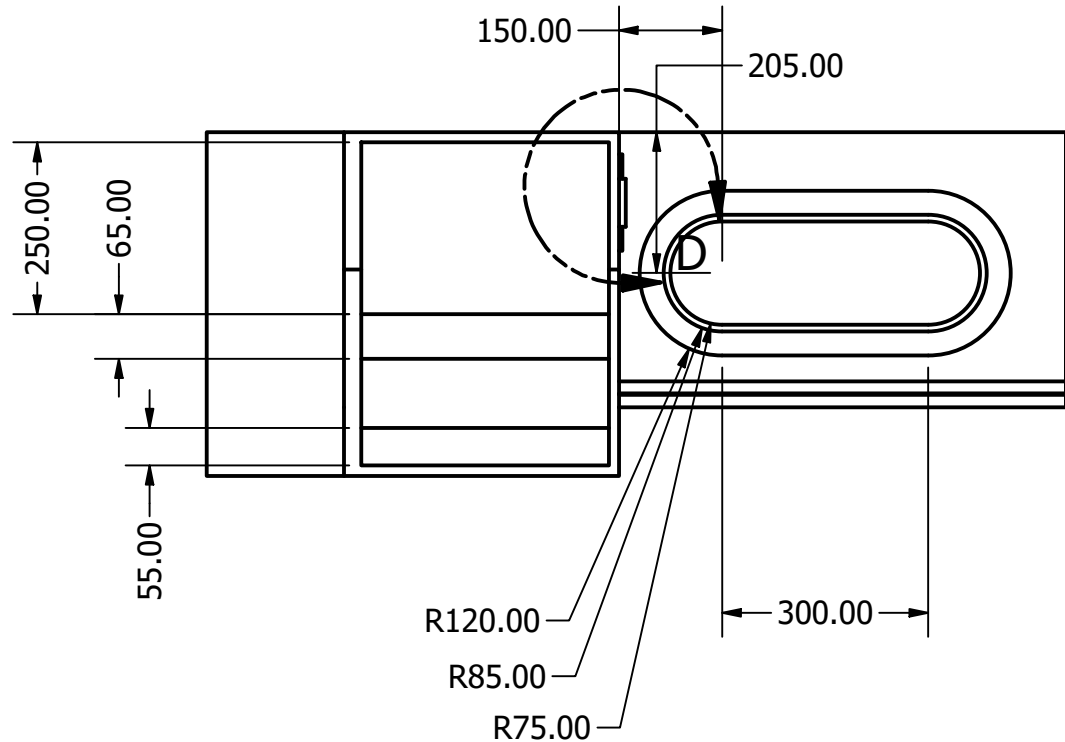
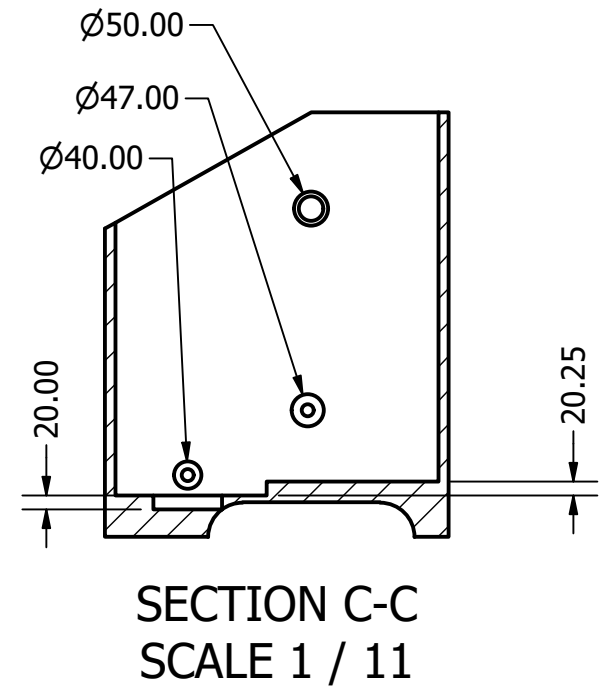
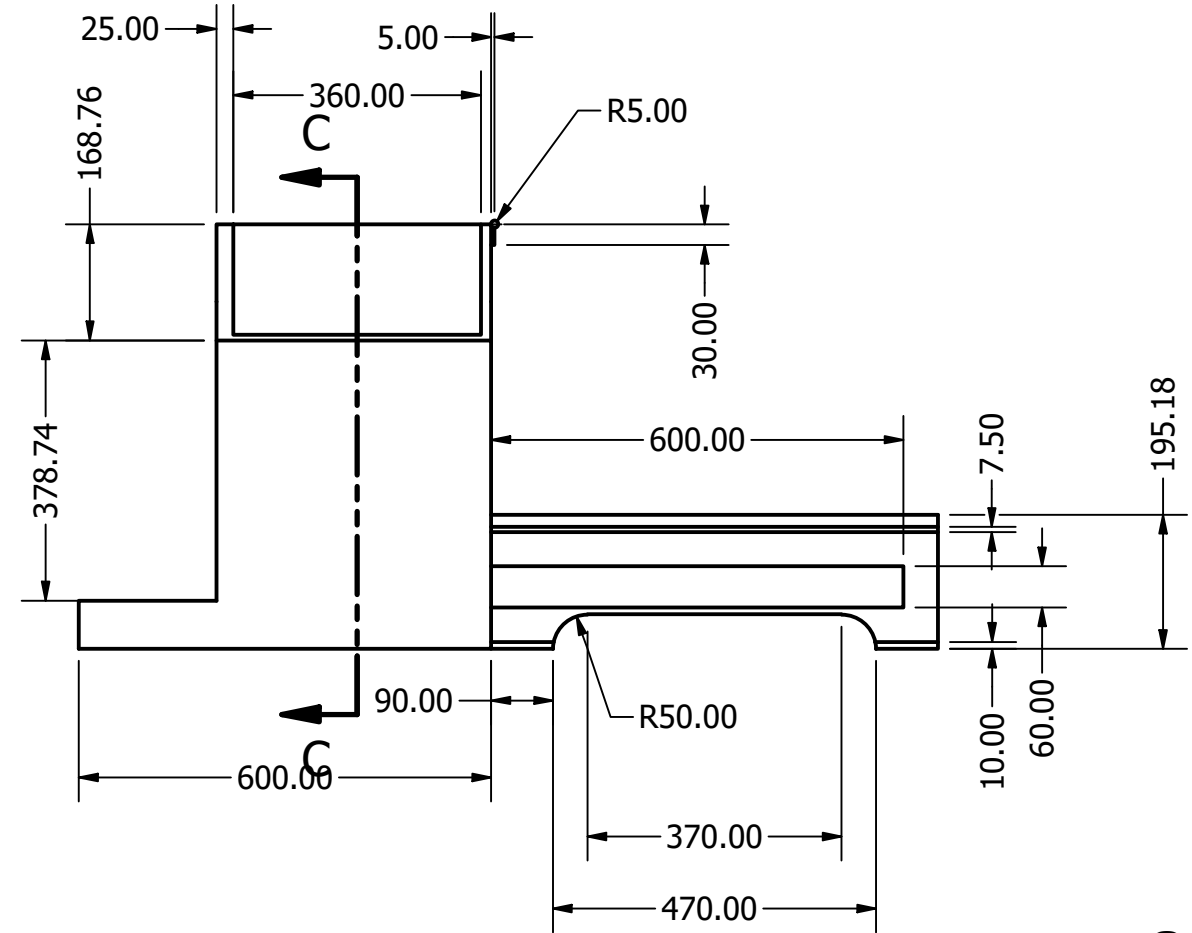
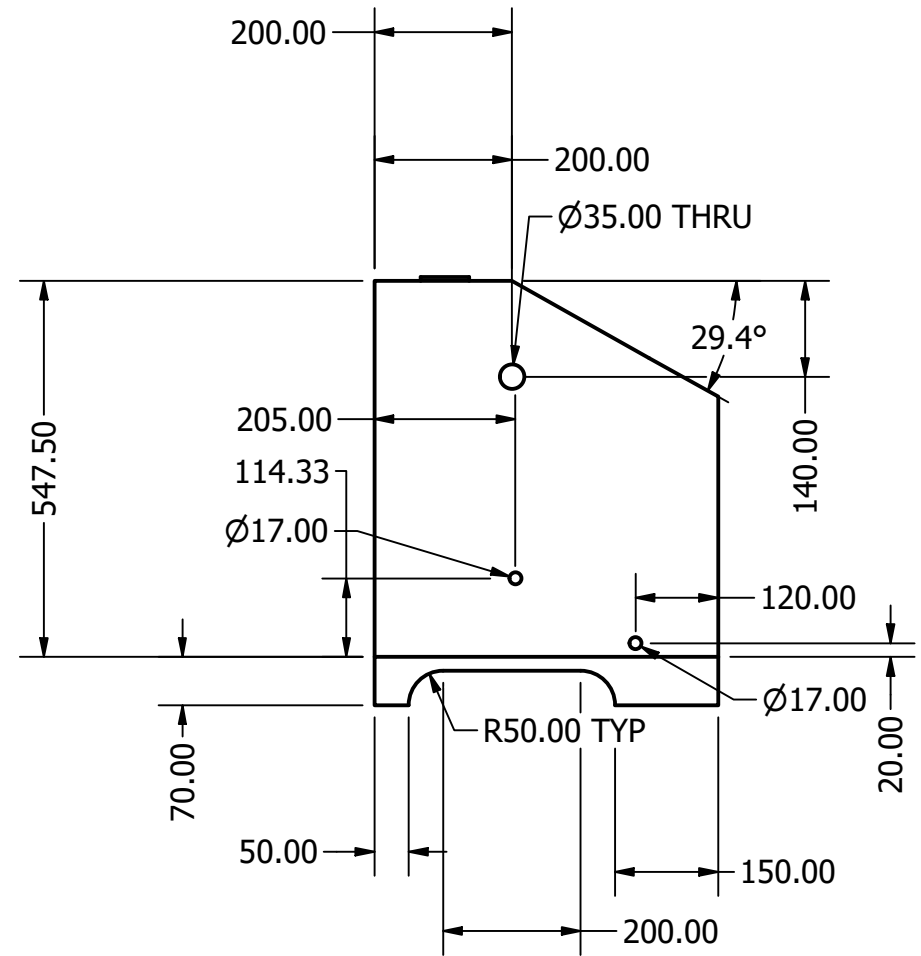
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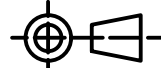
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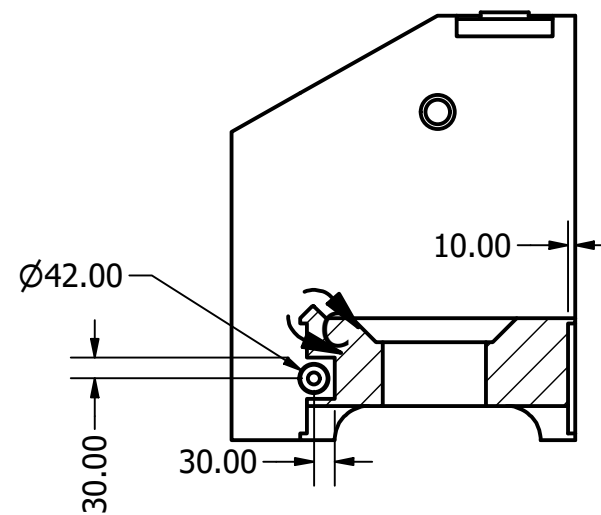
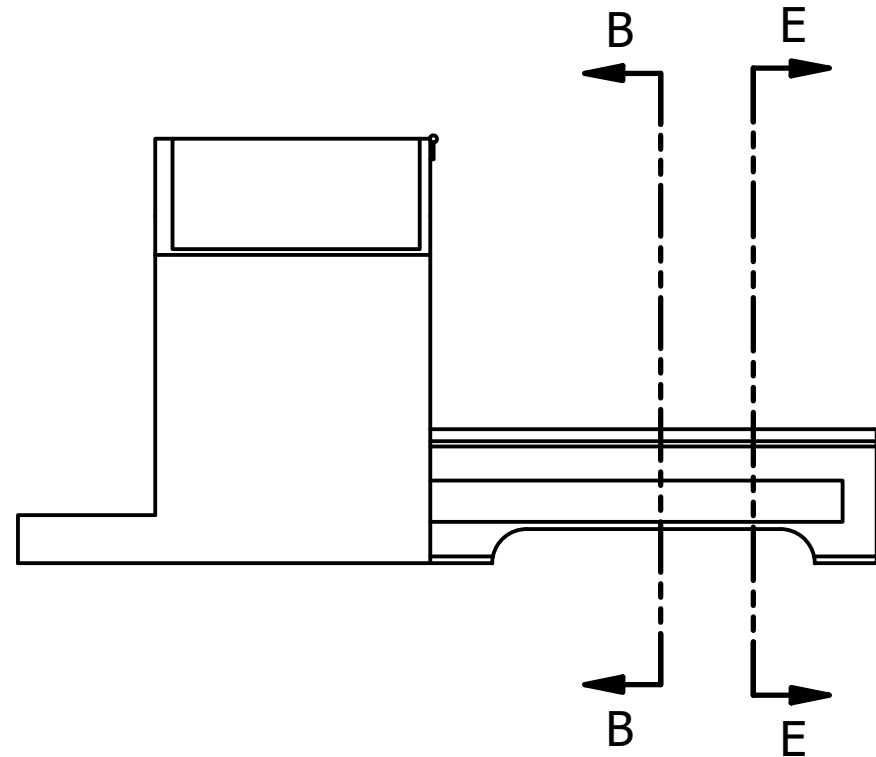


Material: Stainless Steel
Qty: 1

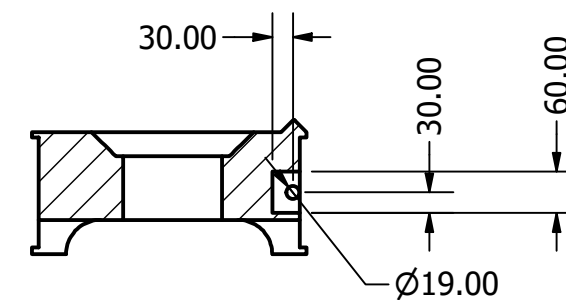
 UNIVERSITY OF WOLLONGONG AUSTRALIA	DRAWN BY Yasan Gunathilika		DATE 19/10/2024			TITLE Rack	
	PROJECT MECH311 Detailed Design					SIZE A3	DRAWING NUMBER 1c
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES			TOLERANCES		SCALE	
TYPE LINEAR (mm):				RANGE 0 - 100 101 - 1000 > 1000	TOLERANCE ± 0.25 ± 0.5 ± 1.0 ± 0.5	1 / 4	SHEET 1 OF 1



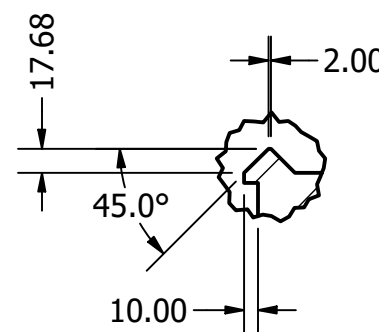
 UNIVERSITY OF WOLLONGONG AUSTRALIA SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024				TITLE Body and Ways				
	PROJECT MECH311 Detailed Design										
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES								
			TYPE LINEAR (mm):		RANGE		TOLERANCE				
		0 - 100		± 0.25		SIZE A3		DRAWING NUMBER 3a.1		REV 0	
		101 - 1000		± 0.5		SCALE					
		> 1000		± 1.0		1 / 11		SHEET 1 OF 1			
		Angle (°)		± 0.5							



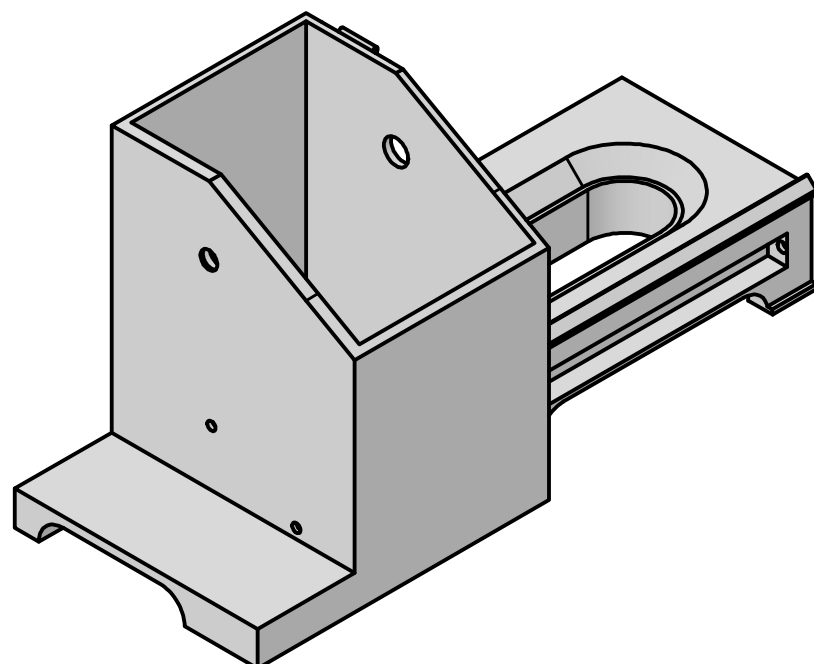
SECTION B-B
SCALE 1 / 11

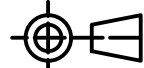


SECTION E-E
SCALE 1 / 11



DETAIL C
SCALE 0.18 : 1



 UNIVERSITY OF WOLLONGONG AUSTRALIA	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024			TITLE Body and ways				
	PROJECT MECH311 Detailed Drawing									
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES			SIZE A3		DRAWING NUMBER 3a.2		REV 0
			TYPE RANGE TOLERANCE							
LINEAR (mm):			0 - 100 ± 0.25			SCALE 1 / 11		SHEET 1 OF 1		
			101 - 1000 ± 0.5							
			> 1000 ± 1.0							
			Angle (°)	± 0.5						

A

B

C

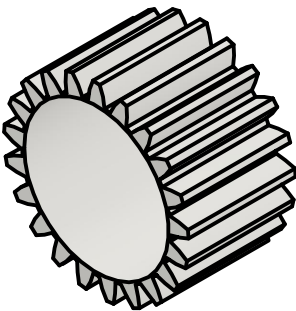
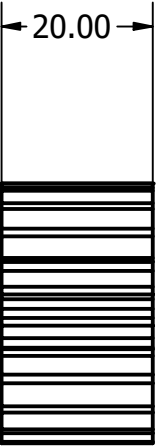
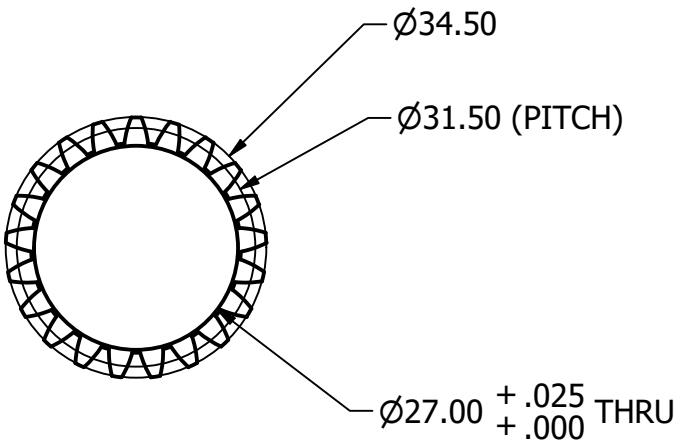
D

A

B

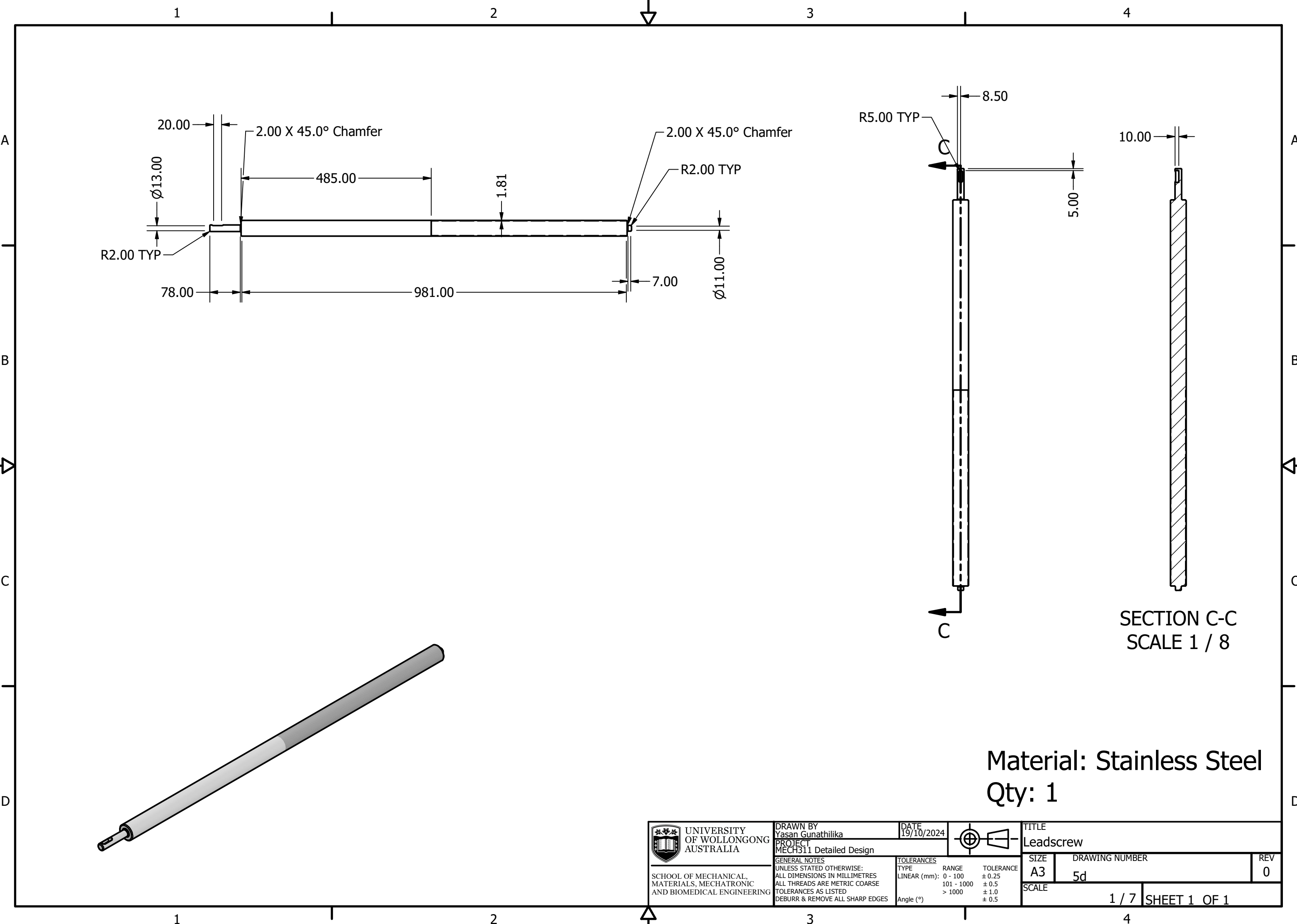
C

D



Material: Stainless Steel
Qty: 1

 UNIVERSITY OF WOLLONGONG AUSTRALIA	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024		TITLE Interm Gear		
	PROJECT MECH311 Detailed Design				SIZE A3	DRAWING NUMBER 5b.1	REV 0
SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES TYPE RANGE TOLERANCE LINEAR (mm): 0 - 100 ± 0.25 101 - 1000 ± 0.5 > 1000 ± 1.0 Angle (°) ± 0.5		SCALE 1 : 1		SHEET 1 OF 1



A

B

C

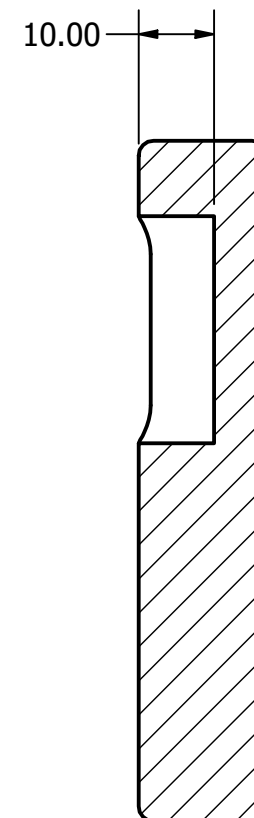
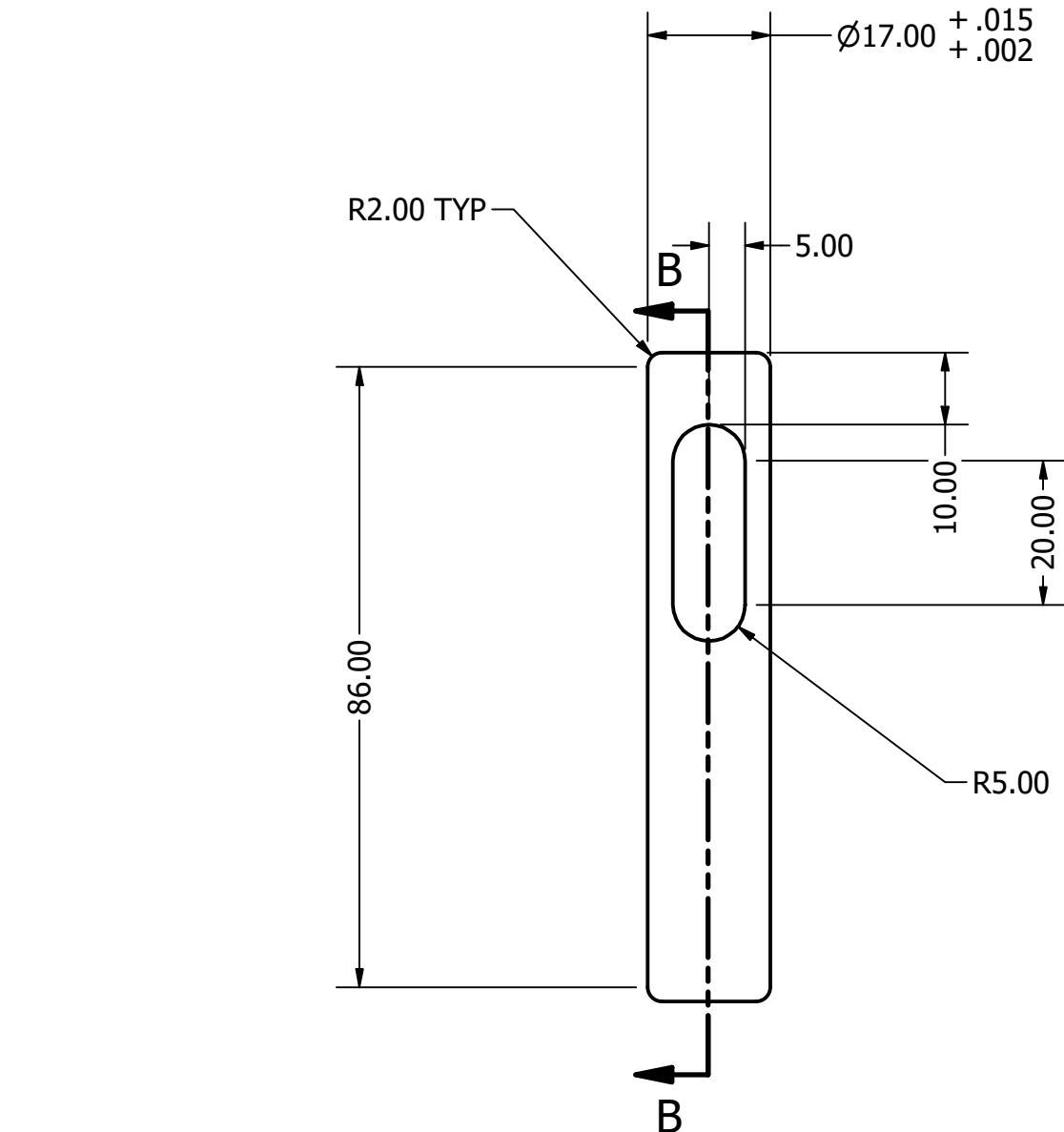
D

A

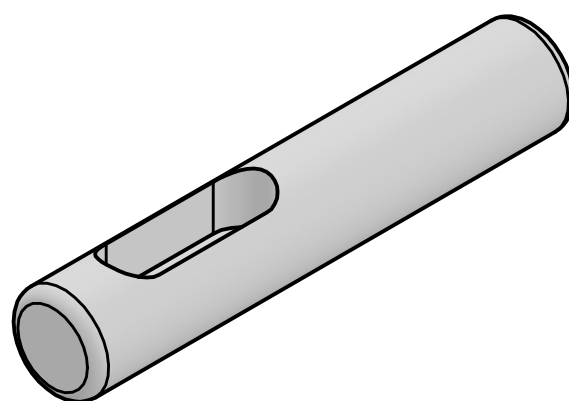
B

C

D

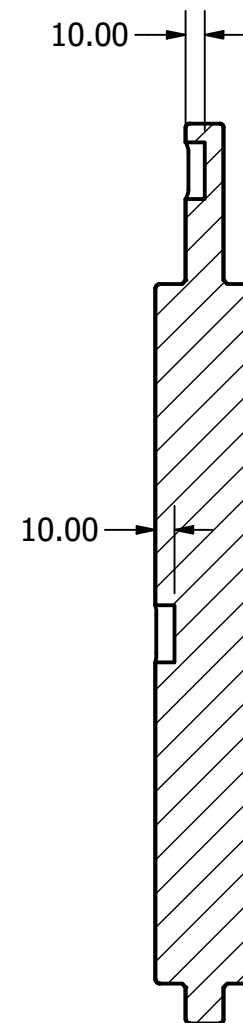
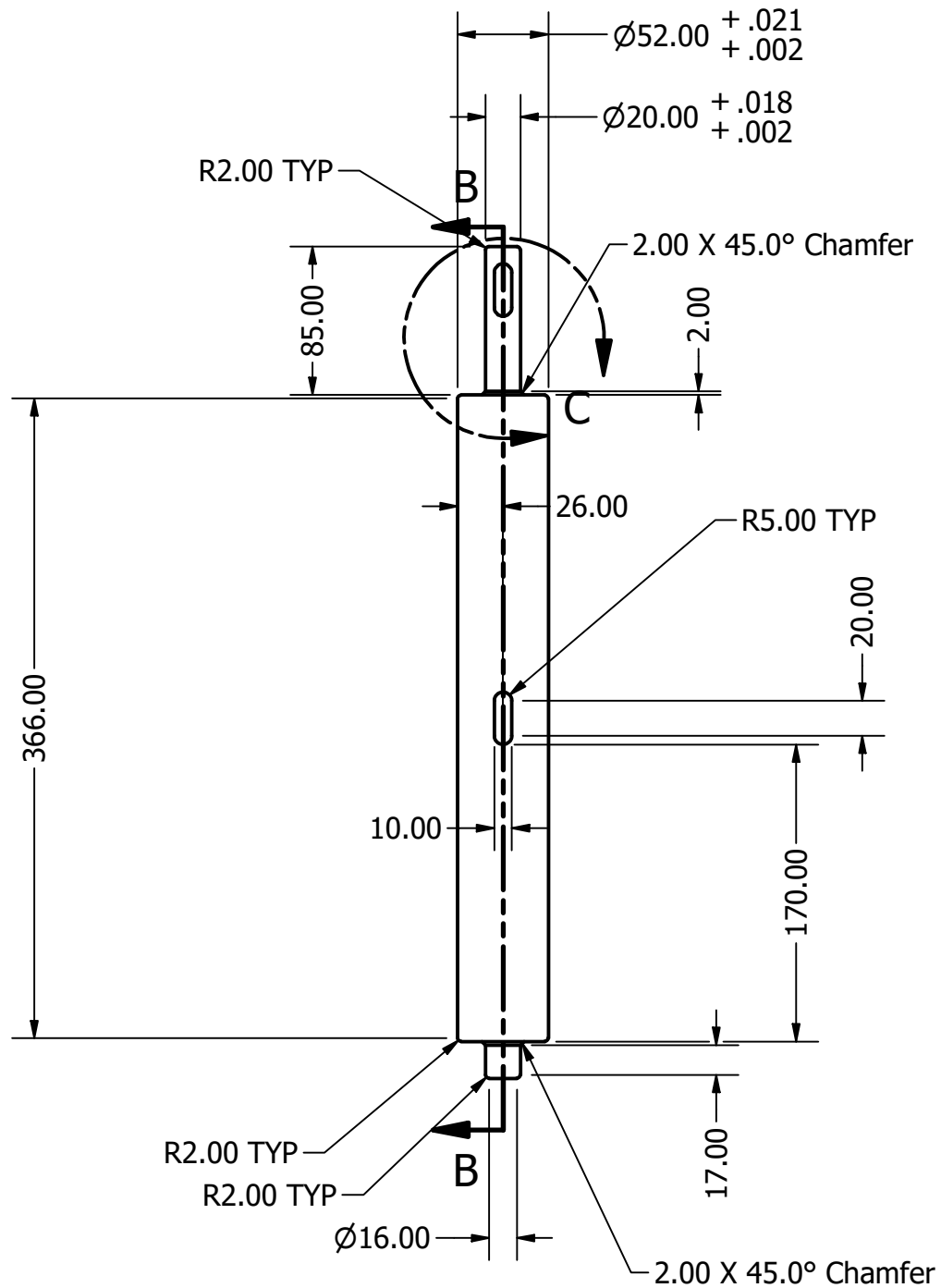


SECTION B-B
SCALE 1 : 1

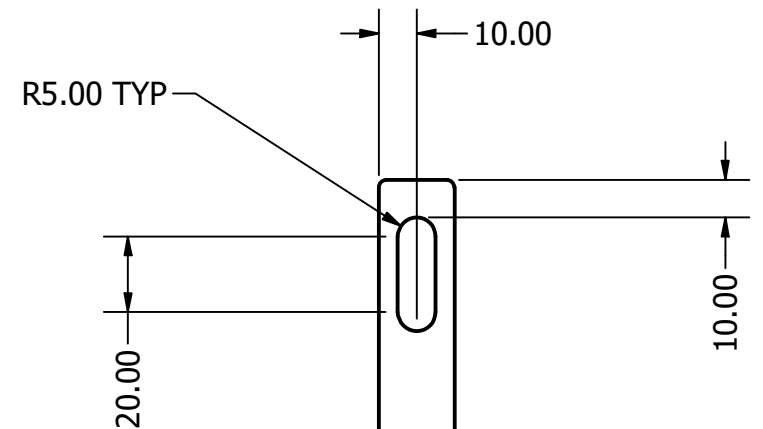


Material: Stainless Steel
Qty: 1

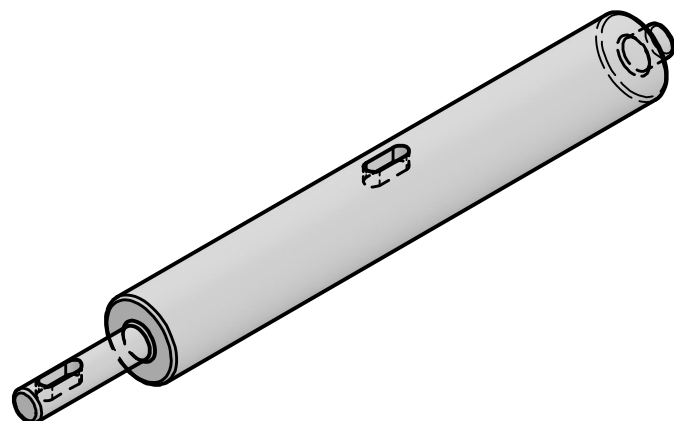
 UNIVERSITY OF WOLLONGONG AUSTRALIA SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	DRAWN BY Yasan Gunathilaka		DATE 19/10/2024		TITLE Idler Shaft		
	PROJECT MECH311 Detailed Design				SIZE A3	DRAWING NUMBER 6A	REV 0
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES TYPE RANGE TOLERANCE LINEAR (mm): 0 - 100 ± 0.25 101 - 1000 ± 0.5 > 1000 ± 1.0 Angle (°) ± 0.5		SCALE 1 : 1 SHEET 1 OF 1		



SECTION B-B
SCALE 1 / 4



DETAIL C
SCALE 1 / 2



Material: Stainless Steel
Qty: 1

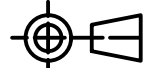
 UNIVERSITY OF WOLLONGONG AUSTRALIA SCHOOL OF MECHANICAL, MATERIALS, MECHATRONIC AND BIOMEDICAL ENGINEERING	DRAWN BY Yasan Gunathilaka		DATE 30/3/2018				TITLE Intermediate Shaft				
	PROJECT MECH311 Detailed Design						SIZE A3	DRAWING NUMBER 6B	REV 0		
	GENERAL NOTES UNLESS STATED OTHERWISE: ALL DIMENSIONS IN MILLIMETRES ALL THREADS ARE METRIC COARSE TOLERANCES AS LISTED DEBURR & REMOVE ALL SHARP EDGES		TOLERANCES				SCALE 1 / 4 SHEET 1 OF 1				
			TYPE LINEAR (mm):	RANGE 0 - 100 101 - 1000 > 1000	TOLERANCE ± 0.25 ± 0.5 ± 1.0 ± 0.5						

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1 DESIGN REQUIREMENTS	2
1.1 Final Design Specifications	3
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1 DESIGN REQUIREMENTS

My primary objective is to develop a conceptual design for a hobbyist. I will design a metal lathe tailored to the Australian market. This lathe will be geared towards hobbyists, with a focus on balancing cost-effectiveness and functionality. The goal is to create a budget-friendly design that remains capable of handling some steel machining tasks. By prioritizing simplicity and minimalism in the design, I will aim to keep production costs low while ensuring practicality for hobbyist use.

The following table highlights the design requirements allocated to me by my stakeholder:

	Detailed	Concept
Design Specification	2	N/A
Drilling Capacity (mm)	13	13
Spindle Bore Diameter (mm)	18	90
Centre Distance (mm)	500	700
Minimum Speed	240	100
Maximum Speed	2000	2000

The table below outlines the final design specifications, accompanied by drawings of the CAD model I created in Inventor. The design sheets at the end of the report include the calculations and analyses carried out to justify and develop the hobbyist lathe.

1.1 Final Design Specifications

Design Specifications:

Variable	Value	Notes/Reference
Design specification version	2	A3 - requirement under allocated design specification.
Drill size capacity	13 mm	A3 - requirement under allocated design specification. F4 - Deriving desired drill size A6 - Drill analysis and results
Spindle bore diameter	18 mm	A3 - requirement under allocated design specification. Designed in CAD model and highlighted on detailed drawings .
Centre distance	500 mm	A3 - requirement under allocated design specification. Designed in CAD model and highlighted on sub-assembly drawings.
Minimum speed	240 RPM	A3 - requirement under allocated design specification.
Maximum speed	2000 RPM	A3 - requirement under allocated design specification.
MOTOR SPECIFICATIONS		
Motor max speed	1725 RPM	A5 - Analysis on picking a motor for design requirements, motor specifications.
Motor power	750 W	
Variable Frequency Drive Compatibility	< 1725 RPM	Motor will be used in combination with a VFD to get variable speeds. A5 - Justification for using VFD and guide for the hobbyist to pick a VFD within their budget.
Other features	-	230 V AC, Single phase, Base mount, Keyed Shaft, Overload protection, \$830.45, McMaster-Carr.
Motor Dimensions	L - 330 mm W - 254 mm H - 235 mm	A5 - Supplier and motor specifications
CUTTING LIMITS		
Feed rate (mm/rev)	0.35 @240 RPM 0.05 @2000 RPM	F3 - Reiteration and derivation of calculations A6 - Results analysis
Required power (W)	739 @240 RPM 695 @2000 RPM	
DRILLING LIMITS		
Feed rate (mm/rev)	2.2 @240 RPM 0.3 @2000 RPM	F4 - Reiteration and derivation of calculations A6 - Results analysis
Required power (W)	715 @240 RPM 699 @2000 RPM	
PULLEY & BELT		
Pulley ratios	5.71 @302 RPM 0.875 @1971 RPM	A7 - Results and selection of pulleys from supplier F5 - Calculations and pulley specification analysis
Number of Pulley	4	

Outer pulley diameters (mm)	Spindle Pulley: 240 RPM - 254 2000 RPM - 44.45 Motor Pulleys: 240 RPM - 44.45 2000 RPM - 50.8	2 distinct pulley ratios are chosen for the two extreme options, highest torque (240 RPM) and highest speed (2000 RPM). Conditions in between can then be derived from the VFD used with the motor
Number of pulley drive ratios	2 + VFD	
Number of belts	2	A7 - Belt calculations and justification for type selection. Justification for picking 2 separate belts. F5 - Calculations and belt analysis
Belt type	V-belt, A section, Polyamide	
Belt lengths	L ₁ = 711.2 mm L ₂ = 1066.8 mm	
GEARS		
Gear module	1.5 mm	A8 - Gear force analysis and calculations
Number of changeable gears	9	From N ranging 21 to 158 for different threading and turning set ups
Number of fixed gears	2	Spindle gear: N = 105, D = 157.5 mm Intermediate gear: N = 21, D = 31.5 mm
Number of possible gear ratios	15	F5 - Finding suitable gear ratios and their respective combinations
Number of feed ratios	4	2 for cutting, 2 for threading
BEARINGS		
Number of bearings	8	-
OVERALL PARAMETERS		
Overall Dimensions:	L - 1250 mm W - 500 mm H - 627.5 mm	Slightly bigger height as the housing was designed with generous room for the gear box and other components Other housing dimensions are present in the drawing
Overall weight	70 kg	Approximation only Base and frame made of steel which would be the bulk of the weight. Motor approximately 5-10 kg
Morse taper size	MT2/MT3	Approximation based on my spindle bore diameter of 18 mm.
Threading	1, 1.25, 1.5 mm	A2

1.2 Drawings of Final Design

2 DESIGN SHEETS

Design sheet register

MECH311

The following design sheet register which was created by **Lachlan Tuck** would be used among the team for easy reference between the design sheets.

Designation has 2 characters; a letter denoting the topic of the design sheet, followed by a number denoting the index of that sheet. For example, "A3" refers to topic A, sheet number 3.

Also note, where a design sheet was used across several weeks, only the last iteration is listed here. I.e. if a sheet was started week 3 and completed in week 5, only the week 5 version will be provided.

DESIGNATION LEGEND	
A	Major design alterations/decisions/documentation
B	Cutting/drilling power calculations
C	CAD drawings
D	General classwork
E	Design Requirements, market research & feature discussion
F	Excel sheet calculator
G	Detailed design guide

DESIGN SHEET REGISTER	
Designation	Topic [# of pages]
CONCEPT DESIGN	
D1	Initial lathe design questions [3]
C1	Initial CAD drafts and conceptualisation [2]
A1	Drivetrain & motor selection conceptualisation [4]
E1	Market research & feature discussion [2]
C2	Detailed CAD design draft [1]
E2	Identifying design requirements [1]

F1	Drilling power estimation
F2	Cutting power estimation
F3	Revised cutting power estimation
A2	Threading decisions [2]
B1	Revised cutting/drilling power calculations [1]
B2	Cutting/drilling power calculations [8]
<i>DETAILED DESIGN</i>	
A3	Identifying design requirements [1]
A4	Carriage design [1]
A5	Picking a new motor [3]
F3	Revised cutting power estimation (v2)
F4	Drilling calculations
A6	Refining machining calculations [5]
F5	Pulley, belt & gearing
A7	Pulley and belt [10]
A8	Gear ratios [1+8]
A9	Shaft layout [3]
A10	Revising lathe layout [4]
F6	Shaft sizing
A11	Shaft design [12]
F7	Bearing selection
A12	Bearing selection [8]
A13	Lubrication [3]
A14	Shaft Tolerancing [4]

2.1 Design Sheets - Concept Report

This section is a compilation of the design sheets that were used by the team for the Concept report of the Hobbyist Metal Lathe. The sheets are in order of the design process undertaken by the group over the period of 29/07/24 to 23/08/24.

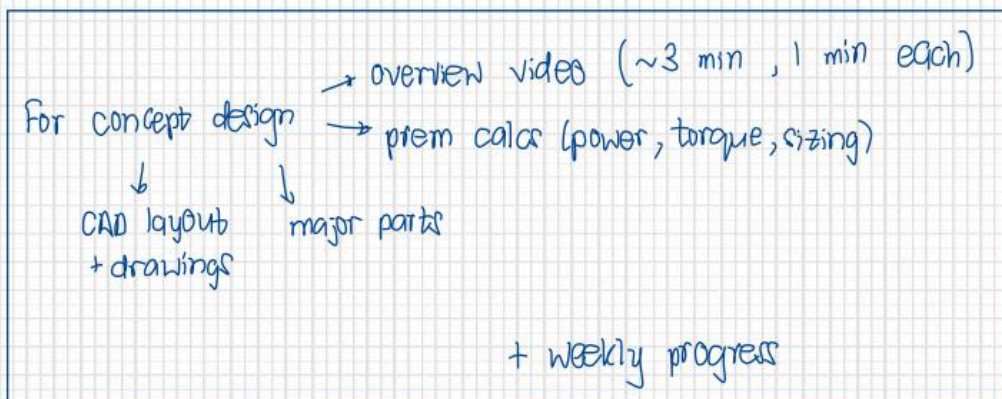
Problem:

- Aus & international standards

*** Assumptions**

Designed for Aus market

Operated by hobbyist (not too complicated)



*** Team Discussion;**

- Implementing a clutch that transfers power between 2 drive trains for 2 different speeds/torques?
- Belt driven vs Gears?
- Slot/Drawer under lathe to remove waste
 ↳ out of scope?
- Design process steps:

Find required Torque → Get motor → Layout ...
 for Drilling/turning

*** MAKE GANTT CHART ***

Problem: Design process planning

Lathe - Precisely removes material from cylindrical workpiece, cut thread, create smooth surface finish.

↳ hobbyist lathe - more simple, cheap. Generally used in a home garage. Expected to cut wood, polymers, some soft metals.

Major Parts of a lathe:

- Electric Motor
- Drive trains
- Leadscrew
- Ways / bed
- Body / chassis
- Headstock

Materials machined:

- Wood
- Polymers
- Metals
- Composites

NOTE: look into difference between drill press vs lathe

Problem: 240V single phase vs 415V 3-phase

* Electric AC motor :

240V Single Phase

- Very accessible as it can be directly connected to the power supply.
- Not as powerful
- For smaller loads & appliances
- Less efficient $\rightarrow$ lower power factor

415V 3-phase

- Very powerful
- Requires additional infrastructure, electrician



huge drawback for a hobbyist

- additional expenses $\rightarrow$ installation
- takes up more space
- consistent, efficient power supply, high torque

* 415V 3-phase would be an OVERKILL! *

Although 415V 3-phase would be better in performance, its drawbacks of needing additional infrastructure, cost... outweigh the benefits.

$\therefore$ 240V Single Phase is chosen

$\rightarrow$ sufficient for a hobbyist.

Problem: Which drive system should we use?

Belt Driven

VS

Gears

Cost effective

↳ suitable for hobbyist

Smooth (less vibration)

High serviceability

Less noise

Can cause slippage

Belt wear & tear

Less power transmission



Should be used wherever possible

More precise

Durable

Higher cost

Shock sensitive

Requires servicing



Is needed for it's precision in situations like threading

∴ we will be using a hybrid system,

Motor → Drive shaft : Belt drive system

Drive shaft → Spindle : Gears

Drive shaft → Power feed : Gears

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG  School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	Task: Analyse target market and propose requirements tied to this target market	Designer: Lachlan	Ref: E1
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Functional Analysis

Target market: Hobbyists who want to cut metal.

This will likely include a large number of people operating out of their own garage, and on a relatively small budget compared to industrial and commercial lathes. This also consequently means that users' cutting requirements will be lower; compared to commercial and industrial lathes our lathe does not need to be as large, powerful, or accurate, and these factors will largely contribute to lowering the manufacturing and design cost, and therefore cost for the customer. It also means our lathe can be physically smaller.

We have identified some key requirements that we think will suit our customers:

- **Small footprint**; this lathe should be reasonably and realistically small enough and light enough for one or two people of reasonable physicality to lift and set it up the first time
- The lathe should have **easy mounting points** to be secured to a bench or worktop
- A **single phase motor** should be selected rather than 3-phase 415V, since many hobbyists will be operating from a home garage and 415V 3 phase availability is unlikely (and if they need to install it, it will cost a significant amount of money).
- Belt drive from the motor to spindle *may* be preferred; Gears may be more prone to dirt/debris intrusion
- Lead screw ideally should be protected from debris intrusion
- Enough cutting power to cut steel comfortably, without having to do many passes vs other steels
- We have considered a reversible power feed with a reversible spindle, but we think that this is too complicated for a hobbyist lathe. We will reconsider later if necessary or recommended by stakeholders

Comparison with the market:

- Ability to cut steel (specific grades), brass, aluminium, plastic cutting abilities and wood
- Large range of speeds ie 300-3000 RPM available. Lower speeds ideal for cutting thread.
- Most seem to use multi stage belt drive systems, allowing easy maintenance and changing of drive ratios
- Ability to cut multiple thread types
- Lowest wattage I have seen so far was 150 W of cutting power, but material cutting types are not specified. A higher price one at \$2700 had a 750 W motor and metal thread cutting abilities.

Summary

Based on looking at a few competing lathes, some valued features for a well rounded hobbyist lathe would be:

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- strong enough motor to cut most metals, at least several hundred watts
- several possible belt drive ratios, to assist with value for money cutting power without a massive motor
- if we have more than one possible thread pitch, a lever to select the correct gear ratio would be more convenient than physically swapping gears on shafts

Potential motor candidates:

[0.55kW 3/4 HP Single Phase Cap Start/Cap Run Electric Motor \(mnpelectrical.com.au\)](http://mnpelectrical.com.au)

[Air Movement Motor 600W 4P 240V PSC FR48 3 speed vented, resilient rubber single shaft | Motor Gearbox Products \(electricmotor.com.au\)](http://electricmotor.com.au)

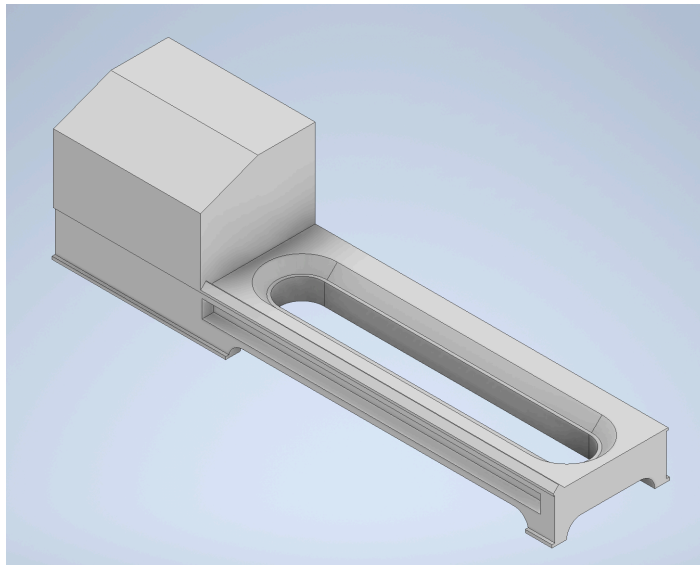
Or for a bit more power: [1.1kw 1.5HP 1400rpm shaft size 24mm Electrical motor single phase 240v \(cononmotor.com.au\)](http://cononmotor.com.au)

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	Task: Early CAD drafts	Designer: All	Ref: C1
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Problem: We have sketched several drafts of our lathe design, see below.

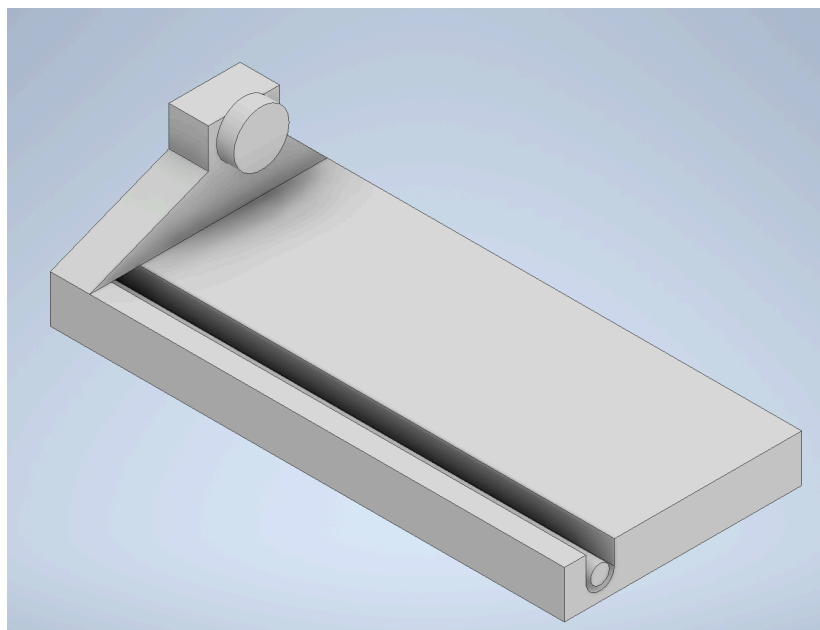
We are thinking having the motor, electronics, and gearing/belting mounted on the far left is suitable along with having the rails run only on one side for cost cutting measures. A motor power from 500W-1kW has also been determined as a reasonable range to design with for now.

Early CAD draft from Wil



Implements a slot for the waste to fall into. The team discussed the idea of adding a removable tray below for easy waste removal after operation.

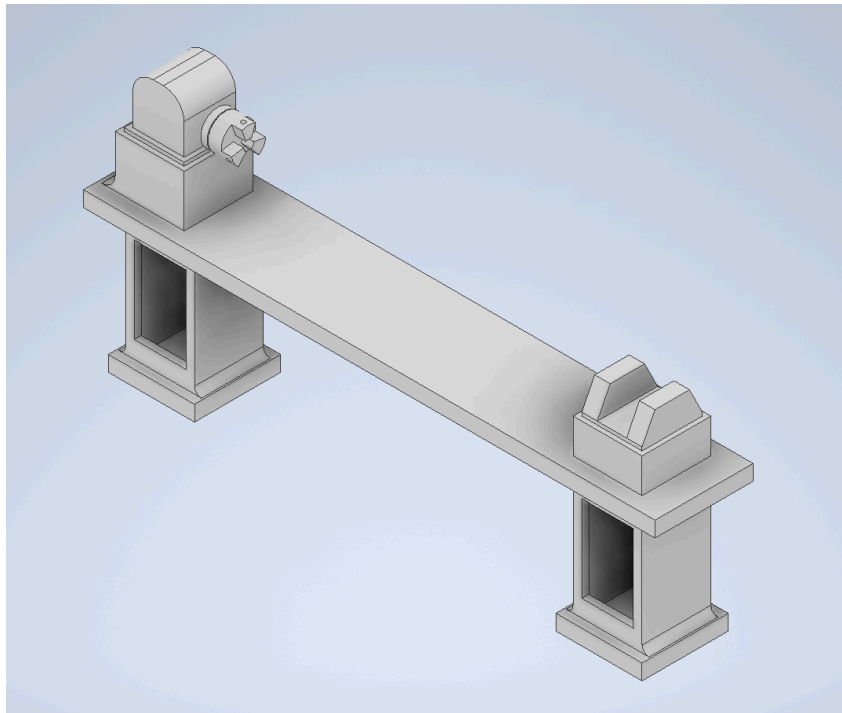
Early CAD draft from Lachie



FACULTY OF ENGINEERING & INFORMATION SCIENCES School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	Task: Early CAD drafts	Designer: All	Ref: C1
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Highlight the use of one rail on the left side as this was seen as a feasible option. It also reduces cost and operation complexity.

Early CAD draft from Yasan



Compares the other CAD drafts to a more industrial lathe kind of design with stands. However this could be removed to save up costs and the hobbyist would be able to place the lathe on a table due to its smaller size.

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	Task: Picking the Motor	Designer: Yasan	Ref: A1
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Problem: Finding a motor

As per the specifications derived by Lachlan from his drilling and turning powers, we are essentially looking for a motor that would ideally be over 1670 W and lower than 1800 W.

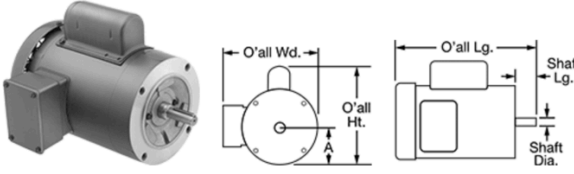
EDIT - this was before the reiteration

We will look into many suppliers, but the we found that most of the selections are provided by the 2 following suppliers, these two suppliers are renowned and frequently used by projects so we will focus on them:

- McMaster-Carr
- TECO Australia

On McMaster we filtered Single Phase electric motors with the power range: 2 hp to 4.9hp as we want the power to be lower than 1,800 W (2.41 hp).

Face-Mount AC Motors



Also known as C-face motors, these bolt directly to your equipment at any angle. They're rated for continuous duty and have a keyed shaft that comes with key stock. You can wire for clockwise or counterclockwise rotation; instructions are included.

To use a motor with a variable frequency drive (AC motor speed control), choose a three-phase motor that's inverter rated.

 For technical drawings and 3-D models, click on a part number.

Power, hp	Max. Speed, rpm	Full Load Current, A	Inverter Rated	Motor Enclosure Type	Overall			Shaft		Insulation		Each	
					Lg.	Wd.	Ht.	Dia.	Lg.	Center to Base	Class		Max. Temp., ° F
115/208-230 Volts AC, Single Phase													
NEMA 56C													
2	3,450	23/12-11.5	No	Totally Enclosed Fan Cooled (TEFC)	14 1/4"	9 1/2"	9 1/4"	5/8"	1 7/8"	3 5/8"	F	311°	6135K209 \$1,047.31

We found this motor to be the closest to our desired power, however as this produces only 1,500 W - this option would be eliminated due to our minimum requirement of 1670 W.

Motors were offered as Face-Mount vs Base-Mount. Upon discussion our preference was to go with a face mount as this could easily be mounted to the side wall of our gearbox at any height without having to add additional panels - to save costs and complexity.

However we do need to take into consideration the vibrations and how sturdy the motor would be when mounted to the side panel, we would need to design thicker walls if this becomes our choice.

Where do we place the motor?

Since we plan on using different belts to change the spindle speed, we have decided to mount the motor within the metal enclosure but have the key stick out to a separate enclosure that would hold the belts. This way, we'd be able to easily change the belts by allowing the hobbyist to have access to this separate enclosure.

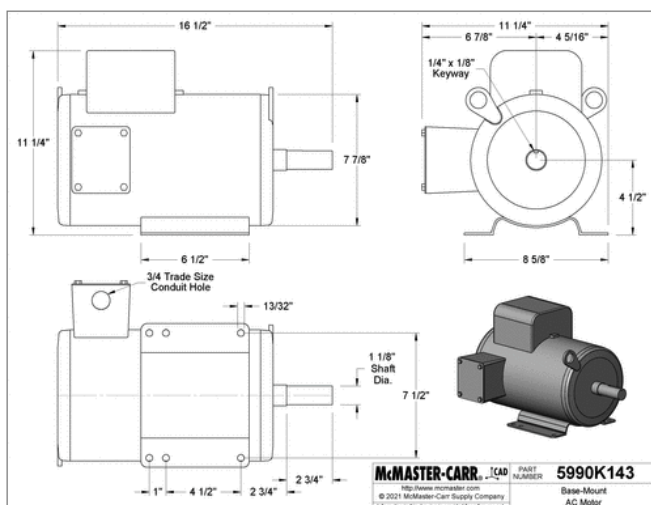
FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	Task: Picking the Motor	Designer: Yasan	Ref: A1
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After looking into several suppliers, we were not able to find a motor that was close to but above the value of 1,670W. It looks like it's a common trend for the values to step from 1,500 W to 2,200 W (this was seen by so many suppliers). Therefore we would have to settle for a 2,200 W and change our initial calculations.

A higher power would also increase the drilling times that we compromised for earlier in our calculations.

While there were a couple options available to us within the 2,200 W range. We preferred McMaster-Carr as they provide us with the CAD model and the pricing unlike TECO Australia. However there were no Face-mount options, this is not a huge disadvantage so we had to compromise.

The following motor is our final selection:



\$1,415.08 Each
Ships by next week
5990K143

☐ Each

ADD TO ORDER



3-D STEP

Download

Learn more about our [Solidworks Add-In](#)

Power Source	Electric
Mounting Style	Base
Voltage	208V AC-230V AC
Electrical Phase	Single
Motor Frame Size	NEMA 184T
Power	3 hp
Maximum Speed	1,750 rpm
Full Load Current	12.5-11.2 A
Frequency	60 Hz
Electrical Connection Type	Hardwire
Wire Connection Type	Wire Leads
Single-Phase Start Type	Start/Run Capacitor
Inverter Rated	No
Duty Cycle	Continuous
Service Factor	1.15
Motor Enclosure Type	Open Dripproof (ODP)
Enclosure Material	Steel
Overall	
Length	16 1/2"
Width	11 1/4"
Height	11 1/4"
Mounting Orientation	Any Angle, Horizontal, Vertical
Bearing Type	Ball
Shaft	
Diameter	1 1/8"
Length	2 3/4"
Center to Base	4 1/2"
Direction of Operation	Clockwise or Counterclockwise

It should be noted that the cost is \$1,415. This is slightly on the higher end but we were happy to designate some amount for the motor to get one of high quality. This is because the hobbyist would

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG  School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	Task: Picking the Motor	Designer: Yasan	Ref: A1
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have less trouble repairing/servicing the motor if it breaks down. It's also hidden at the bottom of our gearbox which would be a huge hassle to replace or inspect.

This cost is also justifiable as it is a 4-pole motor, while 4 pole motors are more expensive - it is better suited for a hobbyist as it would run smoothly and quietly. We don't necessarily require very high RPM's so as long as we can get a constant good torque at around 1750 RPM, it will get the job done.

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG  School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	TASK: Determine motor power specifications based on some drilling and machining operation requirements.	Designer: Lachlan	Ref: B2
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Problem: In order to select an appropriate motor in terms of torque and RPM, we need to do some calculations on machining requirements for particular machining processes on a particular grade of metal.

It is likely this will take several iterations for us to settle on a balanced solution where our lathe is capable of performing reasonable cutting operations on reasonable materials, without causing excessive cost or power requirements.

We will be referring to the Machinery's Handbook for relevant calculations and theory.

As a starting point which may prove to be excessively optimistic, we will test the viability of being able to cut metals with a power factor **K_p of 2.4** which includes alloyed steels and monel metals. This means that we expect to require 2.4kW of cutting power to remove 1 cm³/s.

In order to estimate required motor power, we will use the following equations:

$$P_c = K_p C Q W \quad (1)$$

$$P_m = \frac{P_c}{E} = \frac{K_p C Q W}{E} \quad (2)$$

where P_c = power at the cutting tool; hp, or kW

This requires an efficiency estimate of the transmission system between the motor and spindle. Values are given in table 4:

Table 4. Machine Tool Efficiency Factors, E

Type of Drive	E	Type of Drive	E
Direct Belt Drive	0.90	Geared Head Drive	0.70-0.80
Back Gear Drive	0.75	Oil-Hydraulic Drive	0.60-0.90

We are likely going to use a hybrid drive system wherein both gear and belt drive is utilised, so as a conservative number we will use **$E=0.9*0.75=0.675$** as our E efficiency number.

To estimate motor cutting power, we are still missing some parameters; C , Q , and W .

C is a feed factor for power constant. This value is selected based upon a particular feed rate (mm/rev for turning). I am not sure which number to start with, so I have picked a lower value of 0.07 mm/rev based on the assumption that we will need a lower feed rate to cut the given grades of steel, assuming our lathe drive motor is being pushed to its upper limit of capacity. So, for a feed rate of 0.07 mm/rev, our C value is 1.3. **$C=1.3$** .

Q is material removal rate, given by the formula below and has units cm³/s.

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For now, I will pick dummy values to get a value of Q to work with, and then later we can drop the power requirements to a particular point and then see what the Q value is after that.

$$\frac{V}{60}fd$$

Where, V is cutting speed (linear speed of the outer edge of the rotating material to be cut), f is feed rate, and d is depth of cut. Our f was previously chosen to be **0.07 mm/rev**, so we need to pick a depth of cut value and a cutting speed value. Note that V/60 is when V has units meters per minute, but if V has units meters per second we do not need to divide by 60.

Lets try a 2mm cut depth, and assuming that our spindle (not necessarily our motor, due to the unknown drive ratios) runs at 2000 RPM (a common motor speed for motors in our power range). Let the spindle rpm be N. **Let us also assume we are cutting at most a shaft of radius 10 cm.** This yields:

$$Q = Vfd = \frac{2\pi rN}{60}fd = \frac{2\pi(10)(2000)}{60}(0.007)(0.2) = 2.932 \text{ cm}^3/\text{s}$$

Note all units must be in cm for the above formula.

Now we need a tool wear factor W. I have selected **1.6** as it should allow for a semi conservative design, but it is not the worst case scenario design.

Now we can estimate the motor cutting power:

$$P_m = \frac{K_p CQW}{E} = \frac{(2.4)(1.3)(2.932)(1.6)}{0.675} = 21.684 \text{ kW}$$

This number is way too high. A standard 240V single phase wall outlet can only deliver 2400W, and we would like to have a power safety factor in place along with an allowance for other devices to be powered from the same outlet. The design parameters must be adjusted significantly.

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I have created a Google Sheet calculator where parameters can be changed and the required cutting power will be calculated. I am hoping to use this to iterate through cutting operation parameter sets until I find a viable set of cutting parameters within the motor power limits.

Motor power limits

A standard wall outlet offers **10A at 240V AC**, so 2400W peak. With 400W allocated for additional loads on the power outlet, I have then applied a 10% safety buffer which leaves us with 1800 W available for the lathe. I will iterate the cutting parameters until this limit is reached.

Key changes

I think we may have chosen an overly optimistic K_p value initially. Therefore, I will try modelling for a lower grade of metal; softer grades of stainless steel have K_p of 1.64 so I will try that.

I have also drastically reduced the maximum cutting radius to 4 cm, in order to drop the RMR (material removal rate).

I feel that a cut depth of 1 mm is already a little bit low so I will avoid dropping that for now.

The above parameters still yield a required motor power of 2.05kW, which is above our desired limit of 1800 W. One parameter not yet changed was W , the tool wear factor; initially I had set it to 1.6 for extra heavy rough turning, but I will take it down a notch to 1.3 for normal rough and semi finish turns. I think this is reasonable given that cutting stainless steel will be the absolute upper limit of what our hobbyist lathe should be cutting.

Results

The following input parameters have yielded a required motor power of 1.67 kW.

	Total power from wall (W)	Power allocated for other loads (W)	Safety buffer (%)	
	2400	400	10	
Results	Allowable power for lathe (W)			
	1800			
	Spindle RPM N	Cutting radius (cm)	Cut depth (cm)	Feed (cm/rev)

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	1800	4	0.1	0.005
Results	Cutting speed V (cm/s)	Rmr (Q) (cm ³ /s)		
	753.9822369	0.3769911184		
	Feed factor for power C	Drivetrain efficiency E	Power constant Kp	Tool wear factor W
	1.4	0.675	1.64	1.3
Results	Required motor power Pm (kW)	Pass/Fail		
	1.67	Pass		

Next, we should have a look at the required power to cut threads, and also the required power to drill (where the drilling operation would have the drill bit stationary and fed into the end of the rotating workpiece).

Drilling power estimation

To estimate the power required to drill into a piece of low hardness stainless steel, we are again following Machinery's Handbooks process.

Determining the viability of a drill process involves calculating several parameters:

- T, thrust of the drill relative to the workpiece (N)
- Pc, power of cutting (W). This is the amount of motor power that actually makes it to cutting.
- M, torque required (Nm)

For SI metric units only:

$$T = 0.05 K_d F_f F_T B W + 0.007 K_d D^2 J W \quad (4)$$

$$M = \frac{K_d F_f F_M A W}{40,000} = 0.000025 K_d F_f F_M A W \quad (5)$$

$$P_c = M N / 9550 \quad (6)$$

Use with either inch or metric units:

$$P_m = \frac{P_c}{E} \quad (7)$$

I'll start with calculating available cutting power.

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$$P_c = P_m E = (1.67)(0.675) = 1.127 \text{ kW}$$

(assuming we have a hybrid drive system involving both gears and belts, which means we use the efficiencies of both and hence 0.675).

Next, we will try to estimate the required torque for a drilling operation. Since we have already used 2000 RPM as our maximum cutting speed, and our motor power is at most 1.67kW, our maximum torque is therefore limited to $1.67\text{kW}/2000\text{ RPM} = 1.67 \cdot 1000 / (2000 \cdot 2 \cdot \pi / 60) = \mathbf{7.973 \text{ Nm}}$.

So, we cannot use any more torque than 7.97Nm at 2000 RPM but the option still exists for us to use gear reduction to drop the speed and increase the torque. I'll try 2000 RPM and 7.97 Nm first and see how it goes, however I expect that there will not be enough torque.

Torque sufficiency testing

Using the torque equation (equation 5 as listed above) we can see what drilling parameters are possible with 7.97 Nm of torque. So, **M=7.97Nm**.

What other parameters do we have? **Kd is 24,000** for stainless steels of a 200 Brinell Hardness Number (BHN), so we will use that since the stainless steel we modelled off for turning had a BHN of 150-175 (from table 1a in the turning section for power estimation). Unfortunately this Kd is for a steel of higher BHN but there isn't much I can do except interpolate later if we need to lower the value.

Feed factor **Ff directly varies with the drill feed rate**. I think we can either use the same feed rate we used for turning, or leave it unsolved to see what feed rates are possible with the torque we have available.

Torque factor **Fm** depends on drill diameter; I have selected a 22 mm diameter which gives **Fm=260.8**.

"A" is a chisel edge factor for torque; for a split point drill (more suitable than standard drill for hard materials) we are to use $c/d = 0.03$, for which **A=1**.

"W" is a tool wear factor; I have chosen **W=1.5** initially as a "worst case scenario" value which means we can also drop it to the lower 1.3 value if our torque requirement is too low.

Using these input parameters gives **Ff=0.03**, which equates to a feed rate a little over **0.01 mm/rev** (we can interpolate to find an exact value, but this has been skipped for now).

At 2000 RPM, 0.01mm/rev as a feed rate **means a millimetre of distance drilled will take about 3 seconds**. We think this is acceptable within the context that this is a hobbyist budget friendly lathe, drilling the largest possible hole in the hardest material. However, since we do plan to have several selectable gear ratios, I will reiterate through several more times to try and find a difference in speed/torque ratio that gives a higher feed rate.

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In summary, the first configuration works fine but is a little slow. I will try a lower speed and higher torque to see whether we can improve the drilling feed rate.

Attempt 2

For attempt 2, we are seeking to optimise the results from test 1 without completely overhauling the input parameters. For the same drilling conditions I think we can optimize the spindle speed/torque to get a feed rate that allows for faster drilling per millimetre, essentially trialing what changing the drive ratio does.

The initial attempt used 2000 RPM, 7.97 Nm of torque. I think a good place to start for attempt 2 is to try 2x gear reduction; 1000 RPM and 2*7.97 Nm of torque.

Total power from wall (W)	Power allocated for other loads (W)	Safety buffer (%)	
2400	400	10	
Allowable power for lathe (W)			
1800			
Spindle power (W)	Spindle RPM N	Drill diameter (cm)	
1670	1000	2.2	
Spindle torque (Nm)			
15.9473253			
Torque factor for drill diameter Fm	Work material factor Kd	Chisel edge factor A	Tool wear factor W
260.8	24000	1	1.3
Permissible feed rate factor Ff	Applicable feed rate		
0.08	around 0.045		

Note that values have not been interpolated yet from the tables within the Machinists Handbook; I'm just working with approximate values for now since we are only at conceptual design phase.

The higher torque and lower RPM has given us an improvement in drilling time. It will take **around 0.75 s to drill 1 mm** compared to 0.33 mm per second - which is significant. Based on this finding we can justify having a drive ratio of 2, along with the main drive ratio of 1.

I think we should look for one more suitable drilling ratio. That would leave us with 3 possible drive ratios for drilling/turning, after that we can look at threading ratios.

Attempt 3

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For attempt 3 we will try the same input parameters as attempt 2, but for a different drive ratio. Ie again with 1.67 kW of cutting power but this time geared down even further. If we can find some kind of an optimal balance between speed and torque this would be perfect to maximise material cutting.

For this run I have tried a 700 RPM output with about 22 Nm of torque. This results in a feed rate of around 0.064 mm/rev, which is quicker than attempt 2 however the lower spindle RPM essentially negates any advantage. Attempt 3 has not justified another drive ratio; I will try a fourth attempt with a higher RPM spindle and see what the result is there.

We do not need to know the exact ratio just yet for a conceptual design. I will confirm whether or not there is a need for a third gearing ratio and leave it at that for now.

Essentially I'll try another set of input parameters and see if it seems there is further optimisation to be had. If so, then a 3rd drive ratio is justified but otherwise 2 drive ratios is fine.

Attempt 4

Attempt 1 used a 2000 RPM spindle speed, attempt 2 used a 1000 RPM speed which was a significant improvement, and attempt 3 tried a 700 RPM speed with negligible advantage. Attempt 4 will try a 1500 RPM speed.

The resulting Feed factor for a 1500 RPM spindle is 0.05, which gives roughly 0.025 mm/rev in feed rate. This approximates to 0.625 mm/s drilled, or 1.6 s to drill 1 mm.

At this stage, a third drive ratio is NOT justified. 2 drive ratios are justified, one with an output of 2000 RPM and the other with a 1000 RPM output.

EDIT: after completing 4 drilling torque viability tests I have realised I forgot to apply the efficiency factor of our drivetrain, which was originally 0.675 as a result of both gears and belt systems. The need for both belt and gear systems was an assumption based on needing multiple gear ratios, however so far we have only got 2 justified ratios. This means we will likely only need gears or belts but not both for the main spindle drive system.

I will select $E = 0.9$ for a direct belt drive system, and redo the calculations for my two gear ratios. See results below.

For drive ratios 1000 RPM and 2000 RPM, the updated applicable feed rates are given below now that a 90% efficiency has been factored in.

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See excel sheet for approximate numbers, but the bottom line is that a 2000 RPM spindle speed can drill 1 mm deep in about 3 s, while the 1000 RPM spindle speed can drill 1 mm in about 2s. **During the detailed design phase we should also re-check turning capabilities with a 90% efficiency rate instead of 67.5%.**

Power feed is not intended to be used for drilling, only for threading and at far lower speeds, so drilling thrust does not need to be calculated here. Drilling thrust would also need fine control so as to not drill too deep, hence I think it is better to only use manual hand winding to control the drill thrust.

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Problem: Since we have chosen a more powerful motor (2,200 W) after being unable to find a 1670 W one, we would like to revisit turning operations and see whether we can increase the maximum spindle bore capacity from 4 cm. We will be reusing the excel sheet calculator for turning operations used initially to estimate cutting capabilities.

We will be referring to the Machinery's Handbook for relevant calculations and theory.

Note, much of this sheet has been recycled from the initial calculations as the exact same process is being followed, just with different values. Initially, we start with 2000 RPM, a 4 cm cutting radius, 0.1 cm cut depth, a feed rate of 0.005 cm/rev, feed factor for power C of 1.4, $K_p = 1.64$ and tool wear factor 1.3. In order to verify required motor power, we will use the following equations:

$$P_c = K_p C Q W \quad (1)$$

$$P_m = \frac{P_c}{E} = \frac{K_p C Q W}{E} \quad (2)$$

where P_c = power at the cutting tool; hp, or kW

This requires an efficiency estimate of the transmission system between the motor and spindle. Values are given in table 4:

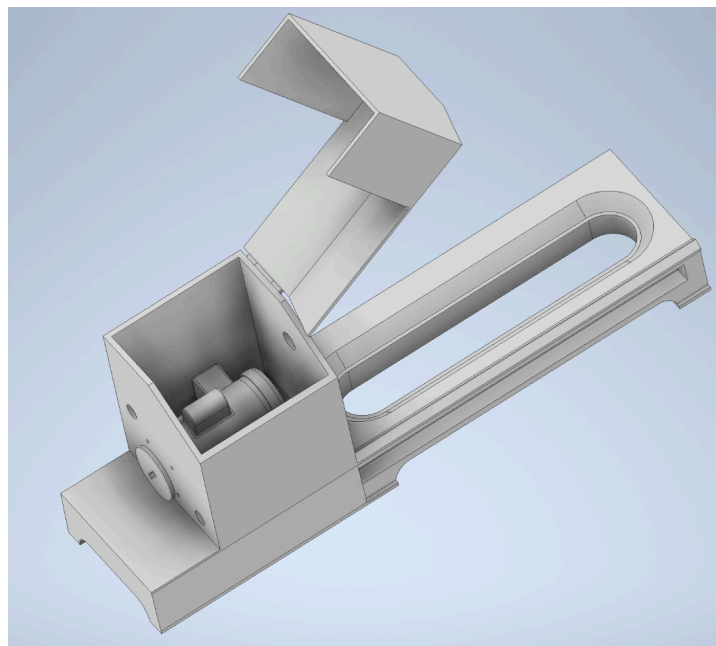
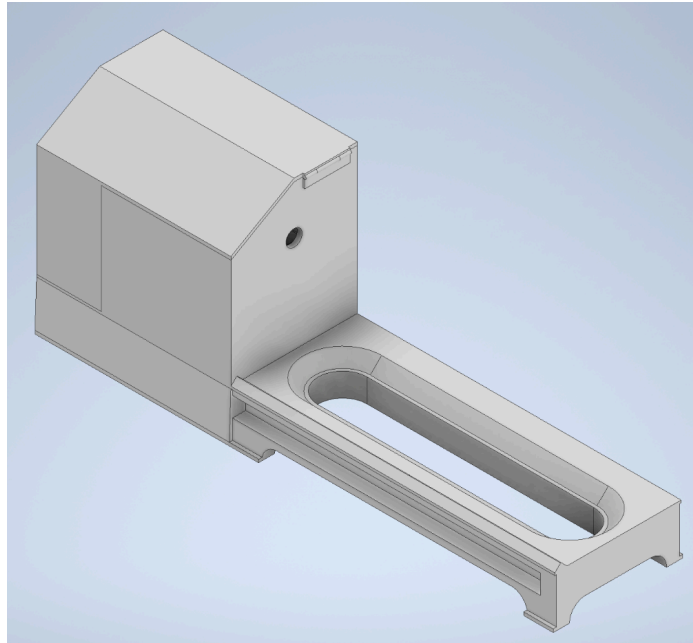
Table 4. Machine Tool Efficiency Factors, E

Type of Drive	E	Type of Drive	E
Direct Belt Drive	0.90	Geared Head Drive	0.70-0.80
Back Gear Drive	0.75	Oil-Hydraulic Drive	0.60-0.90

I will skip the explanation of the rest of the estimation process, as it is the same as provided in the design sheet Ref:9. All this document is doing is recalculating the possible bore radius with a 2,200 W motor and the newly justified 1000 RPM speed. Using the excel calculator created for Ref:9, I have recalculated the possible spindle bore diameter as 9.5 cm, utilising the 1000 RPM speed for higher torque and also utilising 100% of our available 2,200 W. We will move forward with 9 cm being quoted as the maximum spindle bore diameter, leaving the motor around 120 W short of its maximum power output.

This calculation has improved our spindle bore diameter to 18 cm from 8 cm.

Problem: CAD Design Progress:



The gearbox cover was designed with the hinge opening vertically to reveal the gears and belts. This was due to the target market we were targeting, which is the average Australian who purchases a lathe like this to be put on a bench in their shed. Due to this, we suspected that other traditional ways of opening the gearbox cover would not work as there would not be any room behind or potentially in front of the lathe, but guaranteed to be room above the chuck and ways.

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Concept Design System: Hobbyist Lathe	TASK: Discuss and determine threading specifications and the effect this has on overall design.	Designer: Lachlan	Ref: A2
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Problem: Determine suitable thread cutting specifications, keeping in mind the need to minimise cost to the end user while providing reasonable machining capabilities. Discuss how this affects overall design.

The number of drive ratios between the spindle and the power feed depends on the number of thread standards we want to cut. For thread cutting, the feed rate and the spindle must be perfectly synchronised, so a gear system is recommended between the two as a belt system may have some variation in drive ratio due to stretching and contracting of the belt, which would result in a ruined thread.

We require a low RPM at the spindle so that the feed rate of the thread cutting blade is slow with respect to time, allowing users sufficient time and precision when cutting threads. We will also require a reverse gear between the spindle and the thread selection gearbox in order to cut both left hand and right hand threads. **It would be beneficial to minimise the amount of possible threads to cut in order to reduce gearbox complexity and cost.**

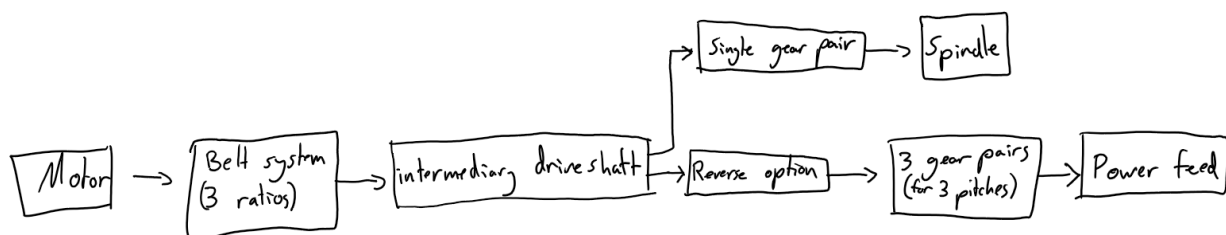
Given the commonality of single threaded threads, we will not be designing for capability to create multi threaded threads. Single threads only. Similarly we will **only be considering metric standards** as this lathe was specified to be for the Australian market.

Key points:

- For single threaded threads, the pitch is equal to the lead.
- M number is irrelevant provided that major diameter does not exceed maximum diameter of turn-able material
- 1 mm seems common
- 1.25 mm seems common
- 1.5 mm seems common
- Upon consulting our stakeholder (MECH311 teacher), it was recommended that 1, 1.25, and 1.5 mm were a good choice for having as our 3 selectable pitches to cut. It was also approved that we include a reverse gear to cut both left handed and right handed gears.

Drivetrain layout - qualitative summary

As per design decisions, we have finalised our drivetrain layout. See below.



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Determination of approximate spindle speed for threading

A third spindle drive ratio is justified for threading, as users need to be able to observe the threading progress and be able to stop threading with a reasonable amount of precision. Our largest pitch is 1.5 mm, which means 1 rotation of the spindle will move the powerfeed 1 mm. Knowing the RPM of the spindle, we can solve backwards from an initial estimate of powerfeed speed and then determine what our third drive ratio should approximately be. As this is at the conceptual design phase, this is likely not the exact number but just an estimate for conceptual purposes.

Let's say that 2.5 mm travelled in 1s is the fastest rate of movement we will allow for the powerfeed.

$$\text{FeedRate} * \text{RPM} / 60 = 2.5 \text{ mm/s}$$

Where feed rate is 1.5 mm/rev, and we then solve for RPM.

$$\text{RPM} = 2.5 * 60 / \text{Feed rate} = 100 \text{ RPM}$$

This seems to be a reasonable value, with the low spindle speed offering high torque and the resulting time taken to thread 2.5mm being long enough for users to carve thread with sufficient precision for a hobbyist.

The conclusion of this is that our third drive ratio is not to output a faster RPM than 100 RPM, and we do have the option to slow it further if torque is an issue for threading.

2.2 Design Sheets - Detailed Design Report

This section is a compilation of the design sheets that were used by Yasan for the Detailed Design report of the Hobbyist Metal Lathe. The sheets are in order of the design process undertaken over the period of 26/08/24 to 18/10/24.

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Identifying Design Requirements	Designer: Yasan Gunathilaka	Ref: A3
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Problem: Ensuring the product meets everything within the scope

Allocated Design Specifications:

	Detailed	Concept
Design Specification	2	N/A
Drilling Capacity (mm)	13	13
Spindle Bore Diameter (mm)	18	90
Centre Distance (mm)	500	700
Minimum Speed	240	100
Maximum Speed	2000	2000

Feedback received (items to rectify) from Jon in Concept Report:

- Credible design meeting project requirements: 13/20
- Analysis requirements and calculations: 25/30
- Design process and progress: 15/25
- 3D models and drawings: 17/25
- Overall: 70/100

Points to be highlighted:

- Video should be showing 3D CAD model
- Purely mechanical should have 5+ speeds (2 min is for high/low torque + electrically adjustable speed).
- Drive spindle directly from pulley? Gearbox too complicated
- Motor power a little too excessive
- Add design sheets for power feed drive system, carriage, base etc
- Just using change-gears to keep it cheap and simple?
- Drawings are not correct (more detail and section views)

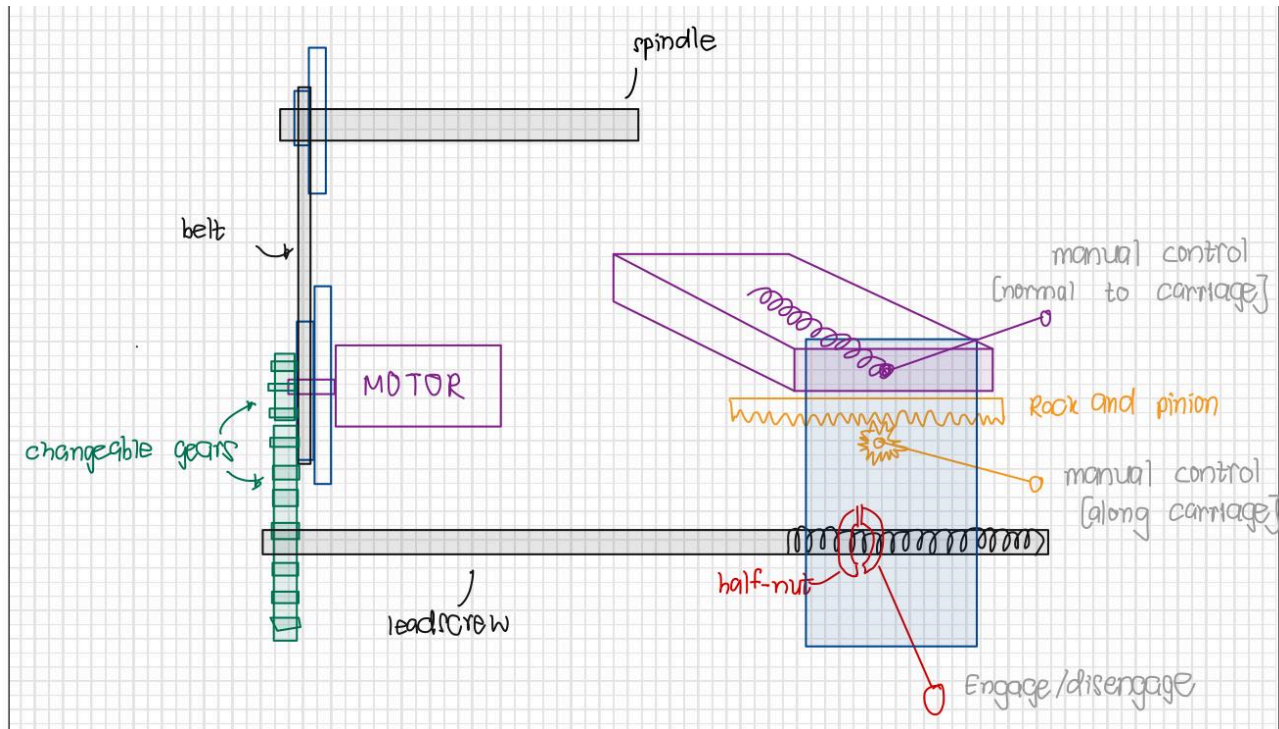
Changes

- Spindle can be connected directly to the pulley, the intermediate shaft is indeed redundant.
- 2 mechanically adjustable speeds (using pulleys) + adjustable speed motor will be used
- Can remove the option to reverse lead screw to keep it simple -> i.e. not needing to add another shaft with gears to change direction, won't need to design a way to disengage leadscrew from the motor either
- Refining machining calculations to settle for a smaller motor power

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Carriage Design	Designer: Yasan Gunathilaka	Ref: A4
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Problem: How would the carriage be moved along and normal to the leadscrew?

New layout:



With the new design we have one less shaft to worry about.

As the lathe is of a smaller scale, I have decided to leave the leadscrew always running, it does consume more energy to constantly keep it running but the pro of not having to add a clutch to engage/disengage leadscrew outweighs the extra energy consumed.

Adding the option to engage/disengage leadscrew complicates the design, to keep the lathe as simple as possible - I have removed the option of having a reverse feed.

I have placed the changeable gears and pulley belt in such a way that it can be easily accessed, this is so the hobbyist can easily replace the gears and adjust the belt to get their required feed rate and spindle speed.

A half nut is now used to engage and disengage the carriage from the leadscrew. This allows us to manually move the carriage when needed using a simple rack and pinion. To move the carriage normal to the lathe, I will use another smaller lead screw which can be manually rotated.

Option to add a gear in between which connects normal-to-carriage movement to the leadscrew was considered, however this would over complicate the design as I would have to add further mechanisms to have the ability to engage and disengage that separately as well. My whole idea with the lathe is to keep it as simple and cheap as I can.

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Picking a new motor	Designer: Yasan Gunathilaka	Ref: A5
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Problem: Motor power is too high (2,200 W)

As per the feedback received, the motor power we had chosen is a little too excessive.

Picking a motor with a lower power would mean we'd have to settle for more softer materials to be machined or it would take longer to machine parts - however considering our centre distance has shortened, making our overall lathe much smaller than the concept design - it would be an overkill to machine very hard materials.

I will look for a motor within the range of 500 W - 1,500 W. Similar to our concept design, I will first look into popular distributors like McMaster-Carr and TECO.

Both base mount and face mount motors can be fitted into my design, upon research I have found that the major advantages of a base mount is its stability and ease of installation whereas face mount is good for situations where the space is limited.

While it's ideal to make the design as compact as I can, considering the safety and ease of installation - I will look for a base mount option.

Specifications I will look for:

- 240V AC Single Phase ✓
- 500 W - 1500 W (1 hp to 1.9 hp) ✓
- Should have a good RPM range to make sure spindle RPM is between 240 - 2000 ✓
- Base mount ✓
- Horizontal mounting position ✓

Preferable specifications:

- Cheap as possible
- Keyed shaft for easy connection to the pulley & changeable gears ✓
- Smaller size ✓
- Some sort of overload protection ✓

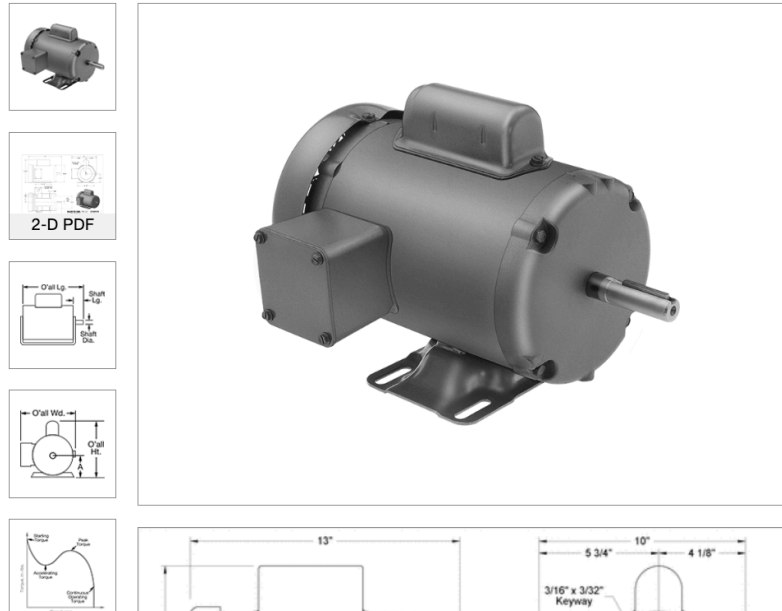
FACULTY OF ENGINEERING & INFORMATION SCIENCES School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Picking a new motor	Designer: Yasan Gunathilaka	Ref: A5
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Problem: Motor power is too high (2,200 W)

NEW MOTOR

Base-Mount AC Motor

115/230V AC, NEMA 56H, 1 hp, Totally Enclosed Fan-Cooled, Auto Overload



I have selected the following motor which ticks out most of the specifications I was looking for. While there were cheaper options for around \$600 - \$800, I have decided to pick this \$830 motor as it features automatic overload protection and totally enclosed fan cooled design (TEFC) which would make it more durable (i.e. prolonged use by the hobbyist).

Motor	115/230V AC, NEMA 56H, Single Phase
Supplier	McMaster-Carr
Mounting style	Base Mount
Power	750 W
Max speed	1,725 RPM
Enclosure	Totally Enclosed Fan Cooled, Steel
Dimensions	Length - 330 mm Width - 254 mm Height - 235 mm
Price	\$830.45
Other Features	Keyed Shaft, Automatic Overload Protection

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Problem: Motor power is too high (2,200 W)

In addition to the motor, as I am planning on using variable speed - a variable frequency drive would be needed.

The consumer would have some flexibility with picking a VFD that best suits their needs and budget, however I will add some guidelines for their reference.

Things to consider:

- Motor compatibility

VFD would need to be compatible with a single phase induction motor

- Power rating

Power rating shouldn't exceed the motor's power rating (1725 W) for the safety of the motor. Similarly the voltage (and current) rating should also be limited to below 230 V AC

- Frequency range

The output frequency of the range should be sufficient to adjust the RPMs of the design, the VFD should be able to comfortably achieve the max and min spindle RPMs: 240 - 2000 RPM.

- Cost

Should fit within the budget of the hobbyist, entry level VFDs typically go for around \$80 - \$230 AUD.

- Sizing

As it would be used by a hobbyist, the more compact the better. This would likely be kept in a garage next to the lathe.

- Other

It would be ideal to have overload protection and safety features. It might also be useful to contact the motor manufacturer to get information/recommendation on which VFDs would be most compatible.

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Problem: New motor would need new machining calculations

NOTES:

- I'll be using two separate pulley ratios + variable speed motor to get different spindle RPMs
- I'll be using 3 separate power feed ratios

I have modified my previous excel files [**F3 - Revised Cutting Power Estimation**, **F4 - Drilling Calculations**] to calculate the new cutting and drilling calculations.

Note - the references made to tables on the excel sheet are for the tables on the Machinery's handbook, the tables are attached on the excel sheet for reference.

Excel Sheet [F3 - Revised Cutting Power Estimation]: At 240 RPM

DETAILED DESIGN		
Power from Outlet (W)	Safety Buffer	Motor Power (W)
2400	5%	750
Desired RPM	240	Max RPM
Drivetrain Efficiency - E	0.9	Table 4 (Direct belt drive)
Tool Wear Factor - W	1.3	Table 3
Cut depth (cm)	0.1	
Cutting radius (cm)	4	
Power Constant - Kp	1.5	Table 1
Feed (cm/rev)	0.035	Table 2, convert to cm
Feed Factor - C	0.97	Table 2
Material Removal Rate (cm ³ /s)	100.531	
Cutting Speed (cm/s) - Q	0.352	
Required Power - Pm (W)	739	Pass

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Problem: New motor would need new machining calculations

At 2000 RPM

DETAILED DESIGN		
Power from Outlet (W)	Safety Buffer	Motor Power (W)
2400	5%	750
Desired RPM	2000	Max RPM
Drivetrain Efficiency - E	0.9	Table 4 (Direct belt drive)
Tool Wear Factor - W	1.3	Table 3
Cut depth (cm)	0.1	
Cutting radius (cm)	4	
Power Constant - Kp	0.82	Table 1
Feed (cm/rev)	0.005	Table 2, convert to cm
Feed Factor - C	1.4	Table 2
Material Removal Rate (cm ³ /s)	837.758	
Cutting Speed (cm/s) - Q	0.419	
Required Power - Pm (W)	695	Pass

Summary of Results

	240 RPM	2000RPM	Description
Feed rate (mm/rev)	0.35	0.05	At lower RPM we have a higher feed, this isn't too high because I also wanted to make sure we can cut harder material (higher Kp) - to give the hobbyist a wider range of materials they can cut from instead of stressing too much about how fast he can get those cut
Kp	1.5	0.82	For lower RPM: Kp of 1.5, this would cover materials like: - most gray cast iron, softer cast iron alloys, soft-medium brass/bronze, aluminium, magnesium alloys and almost all free machining steels. At the highest RPM: Kp of 0.82, we are limited to materials like: - Cast aluminium, magnesium alloys, and the softer cast iron alloys. This isn't much but this is only at the

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Problem: New motor would need new machining calculations

			highest speed setting. The hobbyist wouldn't be able to target the harder stainless steel materials - but as this is just a hobbyist I don't think that would be necessary.
Required Power (W)	739	695	Has to be < 750W

Excel Sheet [F4 - Drilling Calculations]: At 240 RPM

DETAILED DESIGN		
Power from Outlet (W)	Safety Buffer	Motor Power (W)
2300	5%	750
Desired RPM	240	
Drill diameter (mm)	13	
Thrust Factor for Drill Diameter - Ft	7.8	Table 9, interpolation
Torque Factor for Drill Diameter - Fm	101.5	Table 9, interpolation
Drivetrain Efficiency - E	0.9	Table 4, Direct Belt
Tool Wear Factor - W	1.3	Table 3
Cut depth (cm)	0.1	
Chisel Edge Factor A	1.085	Table 7, c/d = 0.18
Chisel Edge Factor B	1.355	Table 7
Chisel Edge Factor J	0.03	Table 7
Work Material Factor - Kd	24000	Table 6
Feed (cm/rev)	0.22	
Feed Factor - Ff	0.298	Table 8
Spindle Torque (Nm)	25.594	
Spindle Thrust (N)	6006.432	
Required Power - Pm (W)	714.674	Pass

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Problem: New motor would need new machining calculations

Drilling Calculations: At 2000 RPM

DETAILED DESIGN		
Power from Outlet (W)	Safety Buffer	Motor Power (W)
2300	5%	750
Desired RPM	2000	
Drill diameter (mm)	13	
Thrust Factor for Drill Diameter - Ft	7.8	Table 9, Interpolation
Torque Factor for Drill Diameter - Fm	101.5	Table 9, Interpolation
Drivetrain Efficiency - E	0.9	Table 4, Direct Belt
Tool Wear Factor - W	1.3	Table 3
Cut depth (cm)	0.1	
Chisel Edge Factor A	1.085	Table 7, c/d = 0.18
Chisel Edge Factor B	1.355	Table 7
Chisel Edge Factor J	0.03	Table 7
Work Material Factor - Kd	14000	Table 6
Feed (cm/rev)	0.03	Table 8
Feed Factor - Ff	0.06	Table 8
Spindle Torque (Nm)	3.006	
Spindle Thrust (N)	1221.321	
Required Power - Pm (W)	699.485	Pass

Interpolation used for drilling calculations - For finding Ft and Fm

Interpolation		
$y = y_1 + (x - x_1) \frac{(y_2 - y_1)}{(x_2 - x_1)}$		
x	Drill diameter	13
y	Ft	7.7775
z	Fm	101.485
x1	12.5	
y1	7.54	
z1	94.28	
x2	14.5	
y2	8.49	
z2	123.1	

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Problem: New motor would need new machining calculations

These results were derived by tweaking factors like the Work Material Factor (Kd), Feed and Feed Factor to get the best feed and work material balance while maintaining the required power (Pm) below the 750W we have from the motor.

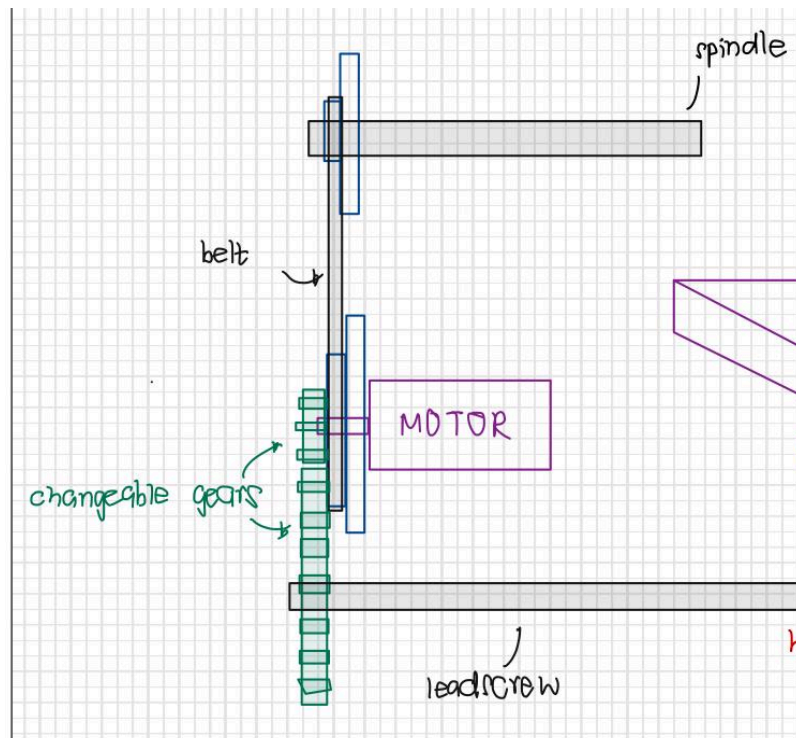
Summary of Results

	240 RPM	2000RPM	Description
Feed rate (mm/rev)	2.2	0.3	At lower RPM we have a higher feed, this isn't too high because I also wanted to make sure we can drill harder material (higher Kd) - to give the hobbyist a wider range of materials they can drill from instead of stressing too much about how fast he can get those drilled
Kd	24,000	14,000	For lower RPM: this would cover most materials like: - Free machining mild steel, Steel with BHN 200, cast iron, most brass/aluminium and magnesium alloys and leaded brass. At the highest RPM: we are limited to materials like: - Cast iron, most brass/aluminium and magnesium alloys and leaded brass
Required Power (W)	715	699	Has to be < 750W

For threading I will be sticking to our concept design analysis:

The spindle linked powerfeed will provide capability to cut 3 thread pitch standards onto bolts and shafts. As the lathe is intended for the Australian market, I have only considered metric threads. **Thread pitches of 1, 1.25 and 1.5 mm** seemed the most common; upon consultation with our stakeholder (MECH311 tutor/teacher) he confirmed that these 3 are a viable choice if we are selecting 3 pitch standards.

Problem: Finding the pulley ratios and belt to run the spindle from the motor



Motor Speed	< 1725 RPM (variable)
Min Speed	240 RPM
Desired Max Speed	2000 RPM

PULLEY DESIGN

Before I begin belting design, I will need to calculate my pulley diameters:

Pulley diameter and speeds can be calculated from: D = driving pulley diameter, d = driven pulley diameter, S = driving pulley speed, s = driven pulley speed

$$D = \frac{d \times s}{S}$$

Minimum pulley diameters are determined by belt manufactures based on min radius that a belt can wrap around pulley.

Velocity Ratio (V) = Pitch diameter driving (Dpi)/Pitch diameter driven (Dpo)

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Problem: Finding the pulley ratios and belt to run the spindle from the motor

Excel Sheet [F5 - Pulley & Belt]:

$$\text{Pulley Ratio} = \frac{\text{Motor Pulley Diameter (Dm)}}{\text{Spindle Pulley Diameter (Ds)}} = \frac{\text{Motor Speed}}{\text{Spindle Speed}}$$

Having attempted different diameters to obtain the ratio, I have settled with the following diameters:

Motor RPM			
1725			
		Diameter (Dm)	Diameter (Ds)
Desired RPM	Pulley ratio	Motor pulley (input)	Spindle pulley (output)
2000	0.8625	50.8	43.815
240	7.1875	44.45	319.484375
Selected Diameters			
RPM Obtained	New Pulley ratio	Dm (mm)	Ds (mm)
1971.428571	0.875	50.8	44.45
301.875	5.714285714	44.45	254

The newly obtained RPM isn't exactly 240 and 2000 but upon confirmation with my stakeholder (MECH311 tutor) I was informed that it doesn't have to be the exact value. And as I will be using a VFD, this can be fine tuned if needed.

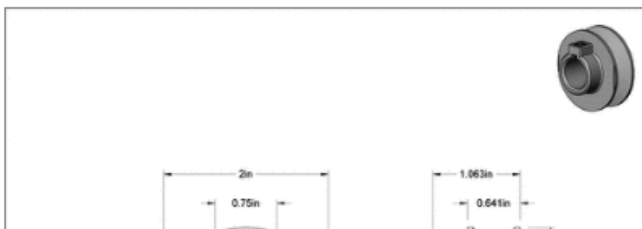
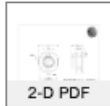
The diameter were tweaked so I can find a standard pulley that's already available, I have searched McMaster-Carr and found the following pulleys:

2000 RPM Motor Pulley

Problem: Finding the pulley ratios and belt to run the spindle from the motor

Pulley

for 4L, A V-Belts, 2" OD, for 3/4" Shaft Diameter



\$6.25 Each
In stock
6245K17

Each

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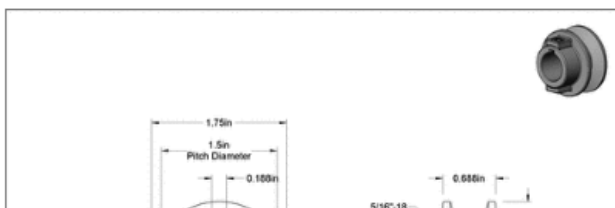
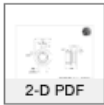
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Pulley Type	Standard
Bore Type	Finished
For Belt Type	V
For V-Belt Type	Standard
For Belt Trade Size	4L, A
OD	2"
For Shaft Diameter	3/4"
ID	3/4"
Pitch Diameter	1.75"
Width	41/64"
Overall Width	1 1/16"
Groove Depth	0.5"
Shaft Mount Type	Set Screw
For Shaft Type	Keyed
Keyway	
Width	3/16"
Depth	3/32"

2000 RPM Spindle Pulley & 240 RPM Motor Pulley

Pulley

for 4L, A V-Belts, 1.75" OD, for 3/4" Shaft Diameter



\$6.23 Each
In stock
6245K14

Each

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Pulley Type	Standard
Bore Type	Finished
For Belt Type	V
For V-Belt Type	Standard
For Belt Trade Size	4L, A
OD	1.75"
For Shaft Diameter	3/4"
ID	3/4"
Pitch Diameter	1.5"
Width	21/32"
Overall Width	1 13/64"
Groove Depth	0.5"
Shaft Mount Type	Set Screw
For Shaft Type	Keyed
Keyway	
Width	3/16"
Depth	3/32"

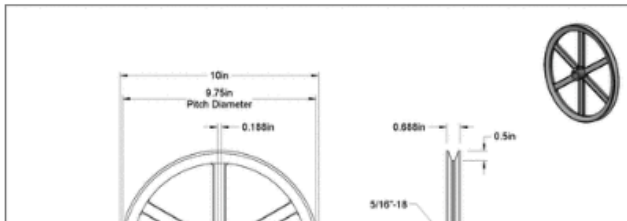
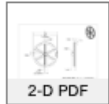
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Problem: Finding the pulley ratios and belt to run the spindle from the motor

240RPM Spindle Pulley

Pulley

for 4L, A V-Belts, 10" OD, for 3/4" Shaft Diameter



\$40.14 Each
In stock
6245K953

☐ Each

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Pulley Type	Standard
Bore Type	Finished
For Belt Type	V
For V-Belt Type	Standard
For Belt Trade Size	4L, A
OD	10"
For Shaft Diameter	3/4"
ID	3/4"
Pitch Diameter	9.75"
Width	45/64"
Overall Width	1 11/32"
Groove Depth	0.5"
Shaft Mount Type	Set Screw
For Shaft Type	Keyed
Keyway	
Width	3/16"
Depth	3/32"

These pulleys are very convenient because they are already keyed.

PICKING THE TYPE OF BELT

Types of belt drives to consider

Belt Type	Joint	Size Range	Center Distance
Flat	Yes	$t = 0.75 - 5 \text{ mm}$	No upper limit
Round	Yes	$d = 3.175 - 19.05$	No upper limit
V	None	$b = 8 - 19 \text{ mm}$	Limited
Timing	None	$p > 2 \text{ mm}$	Limited

Flat Belt:

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Problem: Finding the pulley ratios and belt to run the spindle from the motor

Flat belt would be pretty simple but there would be a tendency to slip under load, I would have to apply really strong tension if I were to maintain consistency when it comes to the spindle speed. Efficiency is okay.

Round Belt:

Not much surface contact due to its round shape. meaning there would be low friction and therefore the efficiency would be hindered. Having less friction would also mean it is more likely to slip, which again isn't what we want.

V Belt:

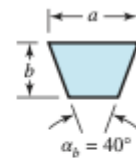
The V shape is ideal to as it could wedge between the pulley and provide good friction → this would mean higher efficiency. They might be slightly more expensive than a flat belt but if the cost of the best wouldn't create much of a dent in the total cost. This would also be pretty suitable to change in my step down pulley.

Timing Belt:

Timing belt would probably have the highest efficiency as it has teeth and would be very precise in its application. This is the most expensive option out of the 4. It would also be a little complicated and includes the added trouble of designing the pulley for precise alignment and fit. This might be an overkill.

Table 17–9 Standard V-Belt Sections

Belt Section	Width a , mm	Thickness b , mm	Minimum Sheave Diameter, mm	kW Range, One or More Belts
A	12	8.5	75	0.2–7.5
B	16	11	135	0.7–18.5
C	22	13	230	11–75
D	30	19	325	37–186
E	38	25	540	75 and up



I have decided to pick the balanced but still effective option of **an A section V belt** as its width would suffice.

Things to watch out for;

- 8:1 is the largest attempt at single reduction drive, 6:1 is a reasonable maximum.
- Wrap angle should not be less than 90 degrees
- Max center distances of pulley should be about 15-20 times the pitch diameter of smaller pulley

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Problem: Finding the pulley ratios and belt to run the spindle from the motor

The center distance between my motor shaft and spindle shaft is **275 mm**.

I can now calculate my belt lengths:

$$L = 2C + \pi(D_2 + D_1)/2 + (D_2 - D_1)^2/4C$$

For bigger pulley ratio

$$L = 2(275) + \frac{\pi(254 + 44.45)}{2} + \frac{(254 - 44.45)^2}{4(275)} = 1058.72 \text{ mm}$$

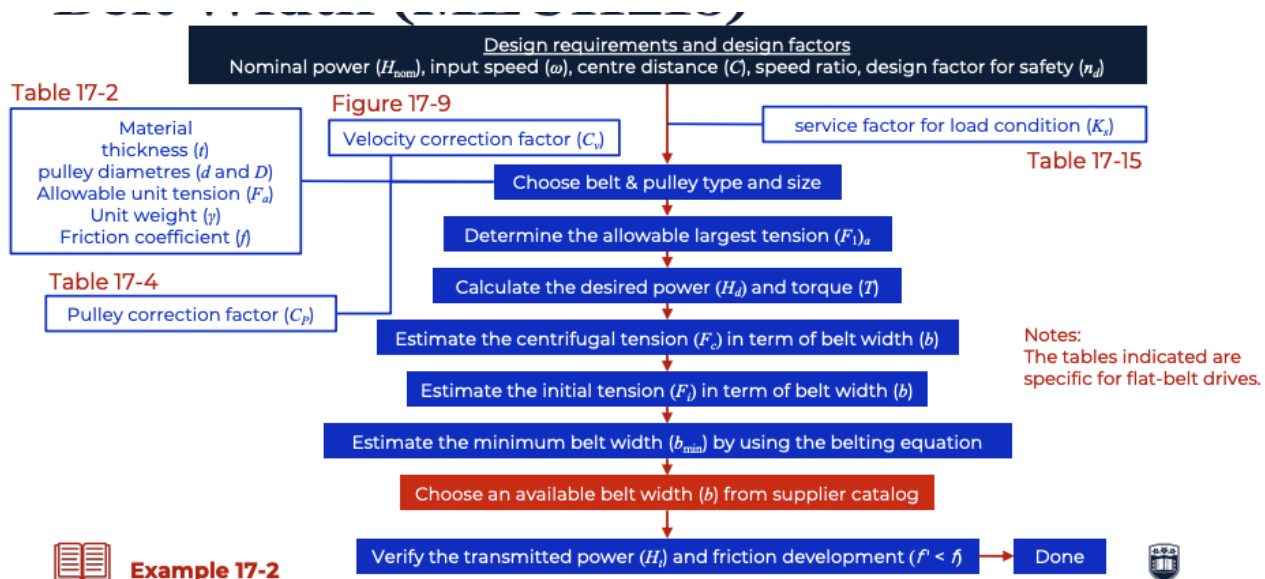
For smaller pulley ratio

$$L = 2(275) + \frac{\pi(50.8 + 44.45)}{2} + \frac{(50.8 - 44.45)^2}{4(275)} = 699.66 \text{ mm}$$

As I plan on using one belt for both set of pulleys; and there is a significant difference between the belt lengths for the different set of pulleys, I would need to implement a **tension adjustment system** to ensure that the tension is maintained - if not either one set of pulleys could end up being too tight or too loose which would cause problems. I will implement this on my CAD (i'm thinking of something like a sliding motor mount).

For now I will find a **belt length** that's somewhat in **between 700 and 1059 mm (27.6" and 41.7")**.

Other belt calculations



Angle of wrap correction factor, K_1

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Problem: Finding the pulley ratios and belt to run the spindle from the motor

$(D-d)/C = 0.76 \rightarrow 0.8$ (Using Table 17-13)

$\frac{D-d}{C}$	ϕ , deg	K_1	
		VV*	V-Flat
0.00	180	1.00	0.75
0.10	174.3	0.99	0.76
0.20	166.5	0.97	0.78
0.30	162.7	0.96	0.79
0.40	156.9	0.94	0.80
0.50	151.0	0.93	0.81
0.60	145.1	0.91	0.83
0.70	139.0	0.89	0.84
0.80	132.8	0.87	0.85
0.90	126.5	0.85	0.85

K_b and K_c parameter for my A section V-belt (using Table 17-16)

Table 17-16 Some V-Belt Parameters*

Belt Section	K_b	K_c
A	220	0.561

$$\gamma = 10 \text{ kN/m}^3 \text{ (Table 17-2)}$$

Sheave Pitch Diameter = 60mm (Table 17-12)

Allowable unit tension, $F_a = 10000 \text{ N/m}$ (Table 17-2)

Coefficient of friction, $f' = 0.5$ (Shigley's Handbook)

$C_v = 1$ (polyamide)

$C_p = 0.73$ or 0.86 (Small pulley diameter 40 - 200)

$K_s = 1.1$ (Uniform - Normal Torque Characteristics)

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Problem: Finding the pulley ratios and belt to run the spindle from the motor

The flat-belt tensions shown in Figure 17–12 ignored the tension induced by bending the belt about the pulleys. This is more pronounced with V belts, as shown in Figure 17–14.

The centrifugal tension F_c is given by

$$F_c = K_c \left(\frac{V}{1000} \right)^2 \quad (17-21)$$

where K_c is from Table 17–16.

The power that is transmitted per belt is based on $\Delta F = F_1 - F_2$, where

$$\Delta F = \frac{63\,025 H_d / N_b}{n(d/2)} \quad (17-22)$$

then from Equation (17–8) the larger tension F_1 is given by

$$F_1 = F_c + \frac{\Delta F \exp(f' \phi)}{\exp(f' \phi) - 1} \quad (17-23)$$

From the definition of ΔF , the smaller tension F_2 is

$$F_2 = F_1 - \Delta F \quad (17-24)$$

From Equation (j) in Section 17–2

$$F_i = \frac{F_1 + F_2}{2} - F_c \quad (17-25)$$

The factor of safety is

$$n_{fs} = \frac{H_d N_b}{H_{nom} K_s} \quad (17-26)$$

$$H_{nom} = 750 \text{ W (motor)}$$

$$n_{fs} = 1.05$$

$$H_d = H_{nom} K_s n_d = 866.25 \text{ W}$$

$$T = \frac{H_d}{2\pi n} = \frac{866.25}{2\pi * \frac{3450}{60}} = 2.398 \text{ Nm}$$

Estimate centrifugal tension F_c in terms of width b :

$$\text{Unit weight of belt, } \omega = \gamma b t = 10000 * b * 10^{-3} = 28b \text{ N/m}$$

$$\text{Belt speed, } V = \pi d n = \pi * 0.04 * \frac{3450}{60} = 7.2257 \text{ m/s}$$

$$\text{Centrifugal tension, } F_c = \frac{\omega V^2}{g}$$

$$(F_1)_a - F_2 = \frac{2T}{d}$$

$$b_{min} = 8 \text{ mm}$$

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Problem: Finding the pulley ratios and belt to run the spindle from the motor

$$\frac{(F_1)_a - F_c}{F_2 - F_c} = \exp(f\phi)$$

for 240 RPM spindle speed:

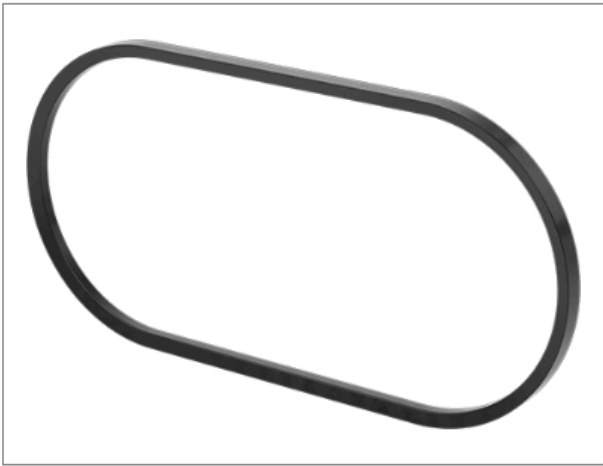
$$\phi = \theta_d = \pi - 2\sin^{-1}\left(\frac{D-d}{2C}\right) = \pi - 2\sin^{-1}\left(\frac{254-44.45}{2*1000}\right) = 2.93 \text{ rad}$$

w = 12.7 mm (1/2"), t = 8 mm

I have chosen the following EPDM Rubber belt with a width of 38.1mm from McMaster Carr

V-Belt

Trade Number A35, 37" Outer Circumference, EPDM



\$14.05 Each

In stock

6186K134

☐ Each

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Belt Type	V
V-Belt Type	Standard
Belt Trade Size	A
Belt Trade No.	A35
Width	1/2"
Thickness	5/16"
Outer Circumference	37"
Number of Bands	1
Material	EPDM Rubber
Reinforcement Material	Polyester Plastic
Color	Black

 2-D PDF


My initial plan was to use a belt that's somewhere in the middle between the required belt lengths of 700 and 1059 mm (27.6" and 41.7"). This would be accompanied by a motor slider which would change the center distance as needed to fit the belt appropriately for the different pulley ratios.

However, I have thought about designing the motor slider and concluded that this would be complicating the design unnecessarily considering we only have 2 pulley ratios to begin with. I think the more **simpler solution** to this would just be to use **two separate belts** for the different pulley ratios. I will find two new belts that fit better for each of the pulley ratios.

28"

Problem: Finding the pulley ratios and belt to run the spindle from the motor

V-Belt

Trade Number A28, 30" Outer Circumference, EPDM



\$12.57 Each
In stock
6186K126

Each

ADD TO ORDER



3-D STEP

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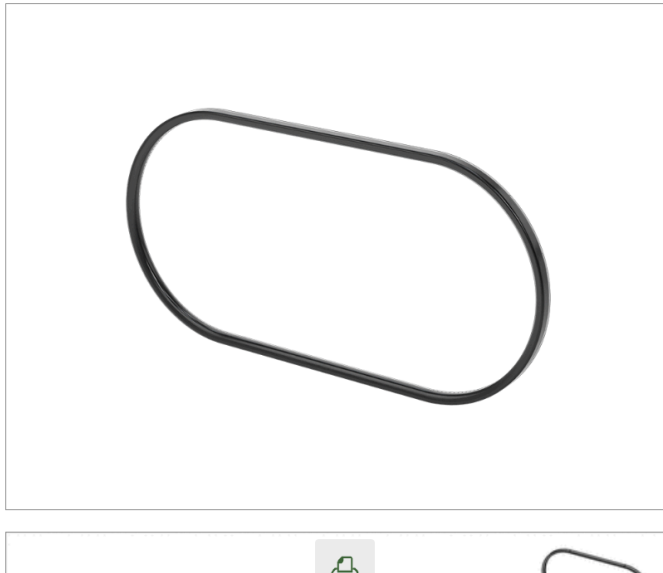
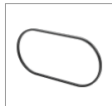
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Belt Type	V
V-Belt Type	Standard
Belt Trade Size	A
Belt Trade No.	A28
Width	1/2"
Thickness	5/16"
Outer Circumference	30"
Number of Bands	1
Material	EPDM Rubber
Reinforcement Material	Polyester Plastic
Color	Black

42"

V-Belt

Trade Number A42, 44" Outer Circumference, EPDM



\$16.14 Each
In stock
6186K142

Each

ADD TO ORDER



3-D STEP

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Belt Type	V
V-Belt Type	Standard
Belt Trade Size	A
Belt Trade No.	A42
Width	1/2"
Thickness	5/16"
Outer Circumference	44"
Number of Bands	1
Material	EPDM Rubber
Reinforcement Material	Polyester Plastic
Color	Black

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Problem: 2 gear ratios need to be found for the changeable gears used between the motor and leadscrew

Desired feed rates as per previous calculations: (mm/rev)

Turning - **0.35** (240 RPM), **0.05** (2000 RPM)

Drilling - **0.22** (240 RPM), **0.03** (2000 RPM)

Threading - I will be attempting to thread **1 mm**, **1.25 mm** and **1.5 mm** pitches

Speed ratios (Gear Ratio)

$$\text{Speed ratio} = \frac{\omega_{\text{out}}}{\omega_{\text{in}}}$$

I have picked the leadscrew pitch to be 1 mm, I can now calculate my desired speed ratios

Turning:

- Feed of 0.35 @ 240 RPM : $\frac{1}{0.35} = 2.86$
- Feed of 0.05 @ 2000 RPM : $\frac{1}{0.05} = 20$

Threading:

- 1 mm : $\frac{1}{1} = 1$
- 1.25 mm : $\frac{1}{1.25} = 0.8$
- 1.5 mm : $\frac{1}{1.5} = 0.66$

$$m = \frac{d}{N} = \frac{\text{Pitch diameter in } mm}{\text{Number of teeth}}$$

$$p = \frac{\pi d}{N} = \pi m$$

I had realised an important change to be made to my design, I will return to gears after making these changes. [ATTEMPT 2 - Revising Layout]

Problem: Shaft layout to initiate pulley/gearing ratios and shaft design

Sections:

- 1) Shaft Assembly Design
- 2) Bearing selection + tolerancing
- 3) Shaft component and coupling selection
- 4) Shaft profile conceptual design
- 5) Sizing

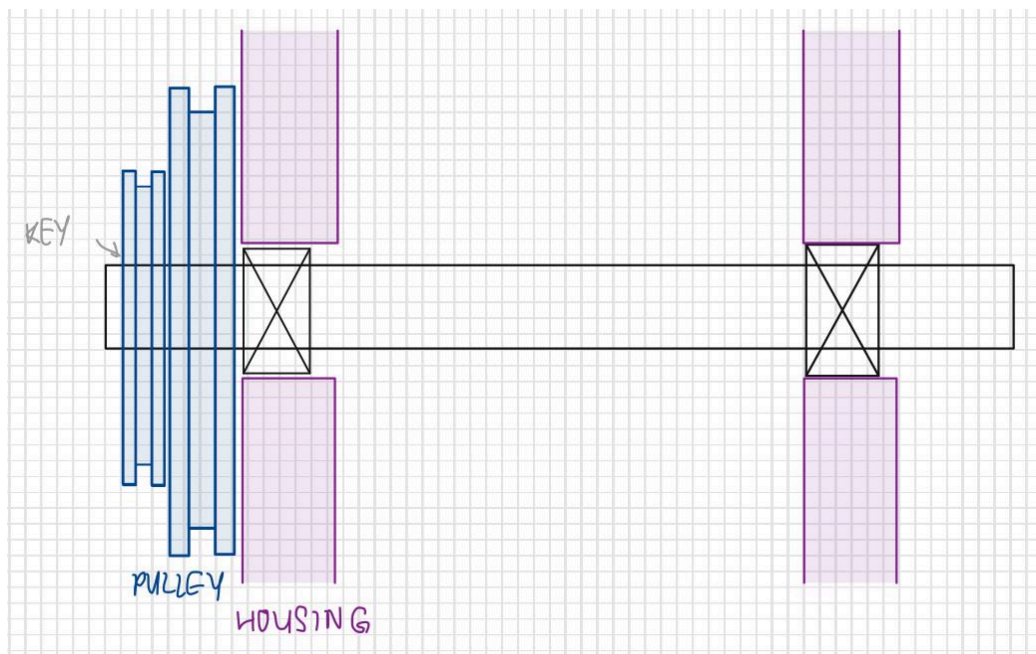
3 Shafts to be designed:

- Spindle Shaft
- Motor Shaft
- Lead Screw Shaft

1) Shaft Assembly Design

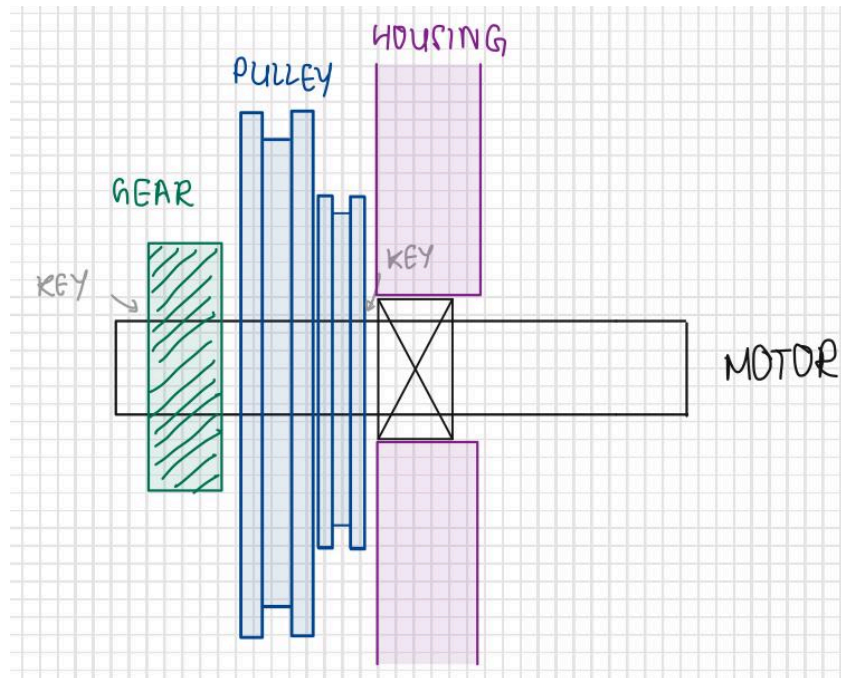
Axial layout:

Spindle Shaft

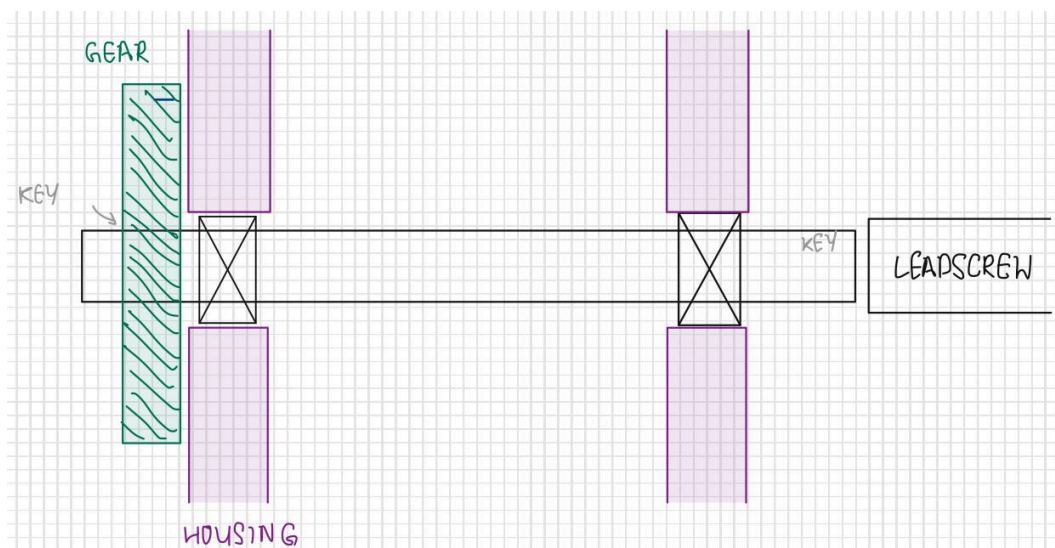


Problem: Shaft layout to initiate pulley/gearing ratios and shaft design

Motor Shaft



Lead screw Shaft



Note - Axial space between components is desirable for lubricant flow, consider assembly and disassembly of parts

UPDATE - This layout has been changed in the later design sheets due to problems encountered and mentioned later.

Only have one bearing to carry the axial load, to allow greater tolerances on shaft dimensions.

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Problem: Shaft layout to initiate pulley/gearing ratios and shaft design

For the spindle shaft and leadscrew shaft, I will add a shoulder on the right side so the **right hand side bearing will bear the axial load**. For the motor this won't be necessary as the motor is mounted to the housing.

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Problem: Change in layout to make it more feasible

ATTEMPT 2 - Revising Layout

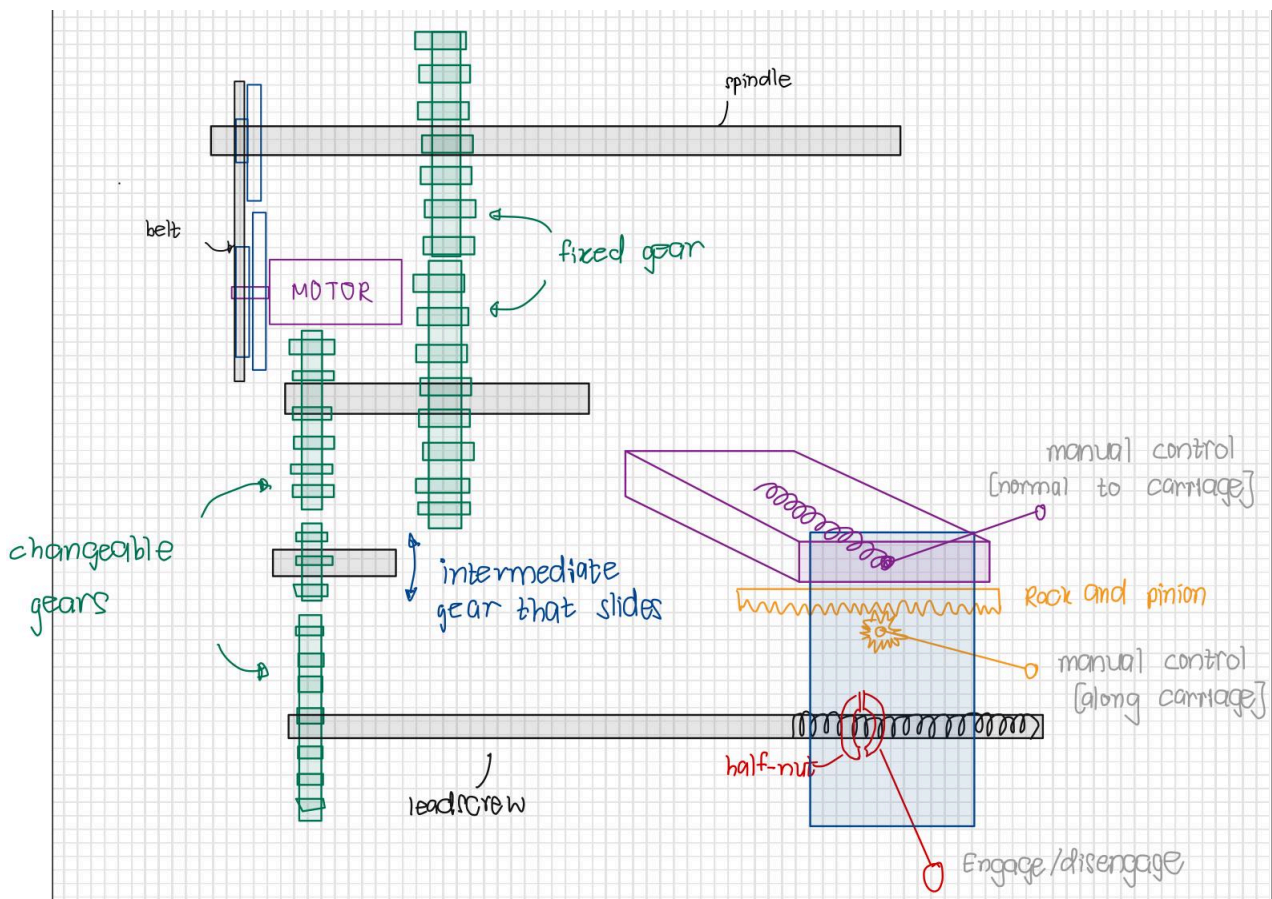
I have realised that connecting the motor to lead screw using changeable gears wouldn't be ideal because this doesn't account for slippage, which would cause a problem when the timing is off between the spindle and leadscrew (especially when turning).

My new design will connect the changeable gears **between the leadscrew and spindle** instead of leadscrew and motor.

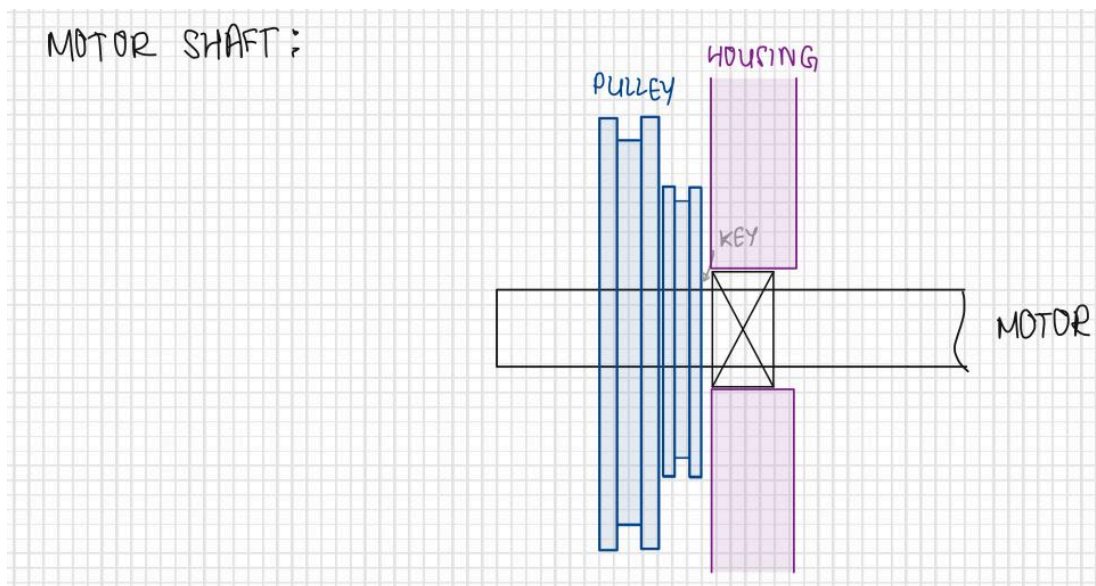
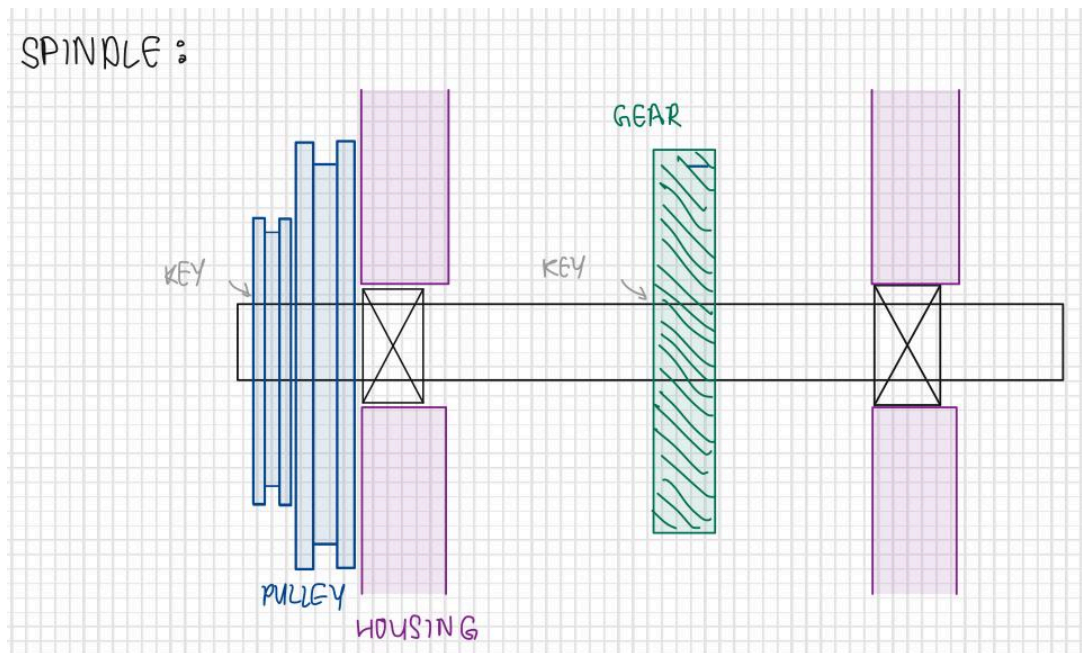
I will also **add another intermediate shaft** so my gear ratios are not significantly big.

In addition to this I will also need an intermediate gear that meshes between the changeable gears (gears between intermediate shaft and leadscrew) so the change in center distance wouldn't affect the meshing between different sized gears and gear ratios. I have attached a diagram for better understanding.

Redesigned Layout:

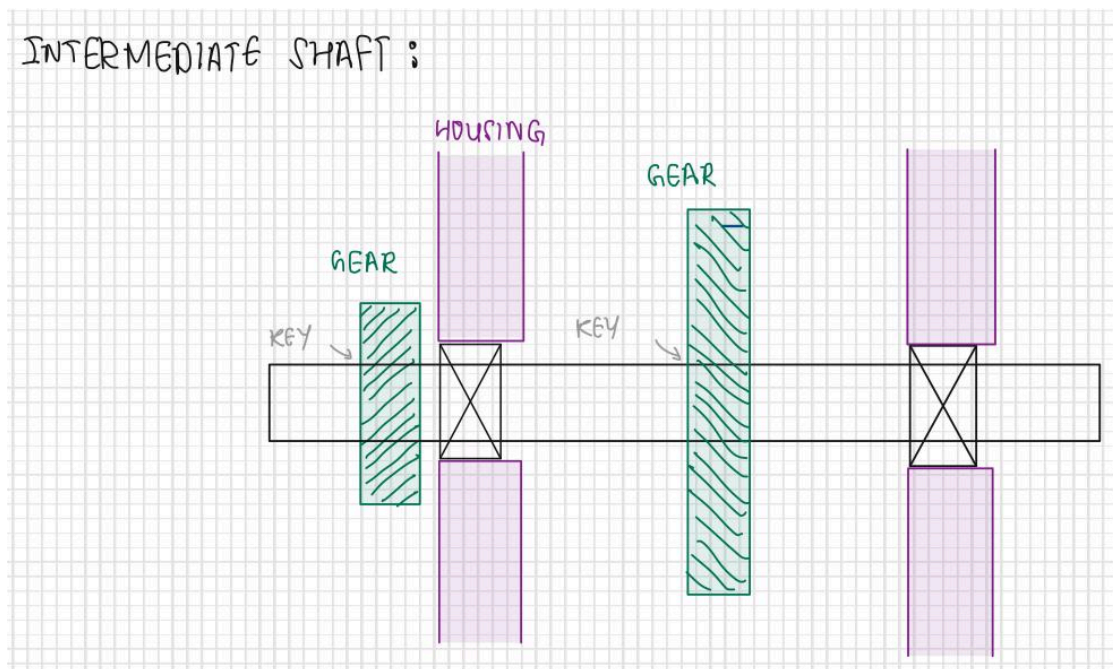


Problem: Change in layout to make it more feasible

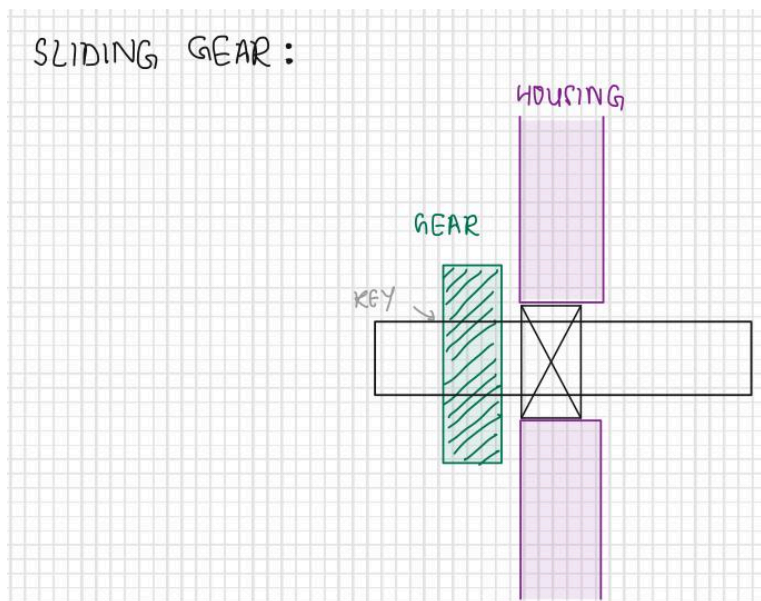


Problem: Change in layout to make it more feasible

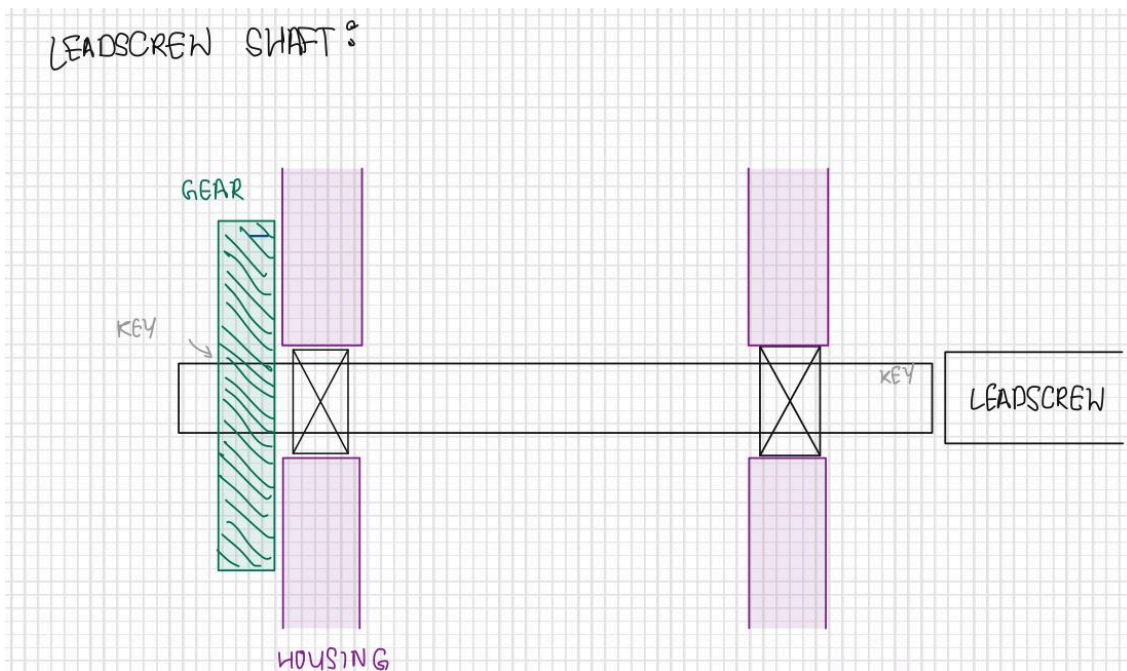
INTERMEDIATE SHAFT :



SLIDING GEAR :



Problem: Change in layout to make it more feasible



Now I can return back to gearing ratios

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Gear Ratios	Designer: Yasan Gunathilaka	Ref: A8
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Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

Notes from previous gear ratio design sheet:

Desired feed rates as per previous calculations: (mm/rev)

Turning - **0.35** (240 RPM), **0.05** (2000 RPM)

Drilling - **0.22** (240 RPM), **0.03** (2000 RPM)

Threading - I will be attempting to thread **1 mm**, **1.25 mm** and **1.5 mm** pitches

Speed ratios (Gear Ratio)

$$\text{Speed ratio} = \frac{\omega_{\text{out}}}{\omega_{\text{in}}}$$

I have picked the leadscrew pitch to be 1 mm, I can now calculate my desired speed ratios

Turning:

- Feed of 0.35 @ 240 RPM : $\frac{1}{0.35} = 2.86$
- Feed of 0.05 @ 2000 RPM : $\frac{1}{0.05} = 20$

Threading:

- 1 mm : $\frac{1}{1} = 1$
- 1.25 mm : $\frac{1}{1.25} = 0.8$
- 1.5 mm : $\frac{1}{1.5} = 0.66$

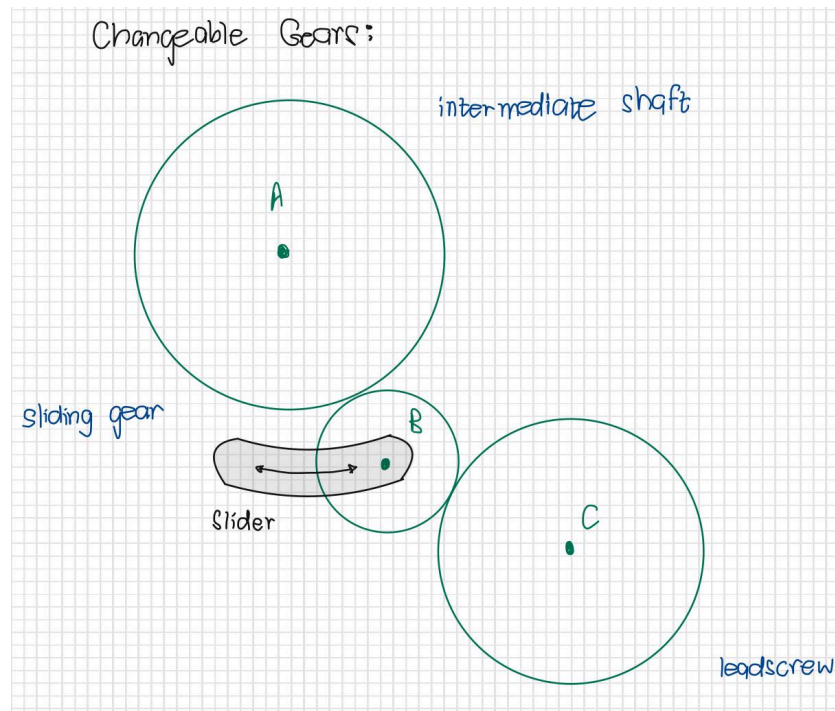
$$m = \frac{d}{N} = \frac{\text{Pitch diameter in mm}}{\text{Number of teeth}}$$

$$p = \frac{\pi d}{N} = \pi m$$

(As aforementioned we will not be using the leadscrew for drilling, the carriage would instead be run manually)

With the new layout, my changeable gears would look like this:

Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft** and **leadscrew**



The inner gear of my intermediate shaft (not A) and the gear of my spindle would be fixed. So I can approach this by finding a suitable ratio between those 2 gears, and then finding N_A , N_B and N_C . I need to ensure that these teeth are within the min and max number of gear teeth to prevent interference:

For standard spur gears with a 20° pressure angle, the general rule is to have at least 17 to 21 teeth on a gear, or around 21 teeth for unmodified gears. I will stick to **21**.

Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

For turning; 0.05 (2000 RPM); speed ratio = 20, $N_A + N_B = 120$

Let $\frac{N_S}{N_I} = 1$, $20 = \left(-\frac{N_S}{N_I}\right)\left(-\frac{N_A}{N_B}\right)\left(-\frac{N_B}{N_C}\right)$

$20 = \left(\frac{N_S}{N_I}\right)\left(\frac{N_A}{N_C}\right)$

$20 = \frac{N_A}{N_C} \Rightarrow N_A = 20 N_C$

If $N_C = 21$, $N_A = \underline{420}$ X

$\frac{N_S}{N_I} = 2$, $20 = (2)\left(\frac{N_A}{N_C}\right)$

$10 = \frac{N_A}{N_C} \Rightarrow N_A = 10 N_C$

If $N_C = 21$, $N_A = \underline{210}$ X

$\frac{N_S}{N_I} = 4$, $20 = (4)\left(\frac{N_A}{N_C}\right)$

$5 = \frac{N_A}{N_C} \Rightarrow N_A = 5 N_C$

If $N_C = 21$, $N_A = 105$

$N_B = \underline{15}$ X

$\frac{N_S}{N_I} = 5$, $20 = (5)\left(\frac{N_A}{N_C}\right)$

$4 = \frac{N_A}{N_C} \Rightarrow N_A = 4 N_C$

If $N_C = 21$, $N_A = 84$

$N_B = 36$ ✓ YAY!

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Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

Similarly I will use the excel sheet - [F5 - Pulley, Belt & Gearing] to calculate the number of teeth for the other feed rates

Turning	0.05 (2000 RPM)		Turning	0.35 (240 RPM)			
Na + Nb	120		Na + Nb	120			
Ns/Ni	5		Ns/Ni	5			
Speed Ratio	20		Speed Ratio	2.86			
Nc	21		Nc	36			
Na	84		Na	21			
Nb	36		Nb	99			
Threading	1 mm		Threading	1.25 mm		Threading	1.5 mm
Na + Nb	120		Na + Nb	120		Na + Nb	120
Ns/Ni	5		Ns/Ni	5		Ns/Ni	5
Speed Ratio	1		Speed Ratio	0.8		Speed Ratio	0.67
Nc	105		Nc	150		Nc	158
Na	21		Na	24		Na	21
Nb	99		Nb	96		Nb	99

I will now pick a module for all my gears so I can get the diameter of the gears for their respective number of teeth:

$$m = \frac{d}{N}$$

I will use my biggest difference in number of teeth between gears to get a worst case scenario: Nc = 158, Na = 21

Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

Gear module: $m = \frac{d}{N} \Rightarrow d = mN$

let $m=1$; $N_C = 158$, $d_C = 158 \text{ mm}$
 $N_A = 21$, $d_A = 21 \text{ mm}$ ← Too small

let $m=1.5$ $N_C = 158$, $d_C = 237 \text{ mm}$ ✓ Good in the middle
 $N_A = 21$, $d_A = 31.5 \text{ mm}$ ✓

let $m=2$; $N_C = 158$, $d_C = 316 \text{ mm}$ A little too big
 $N_A = 21$, $d_A = 42 \text{ mm}$ to my liking

$m=1.5$ works well and smaller the better °.

Force Analysis

Spindle power from motor,

$$P_c = P_m E = 750 \text{ W} \times 0.9 = 675 \text{ W}$$

$$H = T\omega = \left(\frac{W_t d}{2}\right)\omega = 675 \text{ W}$$

Force applied to the gear on the spindle;

- At 2000 RPM:

$$W_t = \frac{60000H}{\pi d n} = \frac{60000 \times 0.675}{\pi \times d \times 2000} = \frac{7.446}{d} \text{ kN}$$

Using a pressure angle of 20° :

$$W_r = W_t \tan(\phi) = \frac{7.446}{d_s} \times \tan(20^\circ) = \frac{2.346}{d_s} \text{ kN}$$

- At 240 RPM:

Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

$$W_t = \frac{60000H}{\pi d_s n} = \frac{60000 \times 0.675}{\pi \times d_s \times 240} = \frac{53.715}{d_s} \text{ kN}$$

Using a pressure angle of 20° :

$$W_r = W_t \tan(\phi) = \frac{53.715}{d_s} \times \tan(20^\circ) = \frac{19.551}{d_s} \text{ kN}$$

The difference torques and forces acting on each gear can then be determined

- 1) Intermediate Gear (non-changeable) - Connect to the spindle

$$W_t = -F_{bt} = -\frac{7.446}{d_s} \text{ kN}$$

$$W_r = -F_{br} = -\frac{19.551}{d_s} \text{ kN}$$

$$T_{b2} = \frac{4d_s W_t}{2d_s} = 14.892 \text{ kNm}$$

- 2) Intermediate Gear (changeable) - Gear A:

$$T_A = -T_{b2} = \frac{4d_s W_t}{2d_s} = 14.892 \text{ kNm}$$

$$F_{At} = \frac{2T_A}{d_A} = \frac{29.784}{d_A} \text{ kN}$$

$$F_{Ar} = F_{At} \times \tan(20^\circ) = \frac{10.840}{d_A} \text{ kN}$$

- 3) Sliding Gear - Gear B

$$T_b = 0 \text{ (Idler gear therefore no torque applied to shaft)}$$

$$F_{Bt} = -F_{At} = -\frac{29.784}{d_A} \text{ kN}$$

$$F_{Br} = -F_{Ar} = -\frac{10.840}{d_A} \text{ kN}$$

- 4) Lead Screw Gear - Gear C:

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Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

$$T_C = \frac{F_{Bt} d_c}{2} = - \frac{14.892 d_c}{d_A} \text{ kNm}$$

$$F_{Ct} = - F_{Bt} = \frac{29.784}{d_A} \text{ kN}$$

$$F_{Cr} = - F_{Br} = \frac{10.840}{d_A} \text{ kN}$$

That concludes my gear force analysis, I now have all the gear specifications for my 10 different changeable gears - I will make a table for the user so they know which gear set up to use for their different applications:

Gear	Used For	Number of Teeth	Largest Forces & Torques Endured
Gear A	Turning (0.05 - 2000 RPM)	84	(when $d_A = 31.5 \text{ mm}$) $F_t = 0.946 \text{ kN}$ $F_r = 0.344 \text{ kN}$ $T = 14.892 \text{ kNm}$
	Turning (0.35 - 240 RPM)	21	
	Threading - 1 mm	21	
	Threading - 1.25 mm	24	
	Threading - 1.5 mm	21	
Gear B	Turning (0.05 - 2000 RPM)	21	(when $d_A = 31.5 \text{ mm}$) $F_t = 0.946 \text{ kN}$ $F_r = 0.344 \text{ kN}$ $T = 0 \text{ kNm}$
	Turning (0.35 - 240 RPM)	36	
	Threading - 1 mm	105	
	Threading - 1.25 mm	150	
	Threading - 1.5 mm	158	
Gear C	Turning (0.05 - 2000 RPM)	36	(when $d_A = 31.5 \text{ mm}$, $d_C = 148.5 \text{ mm}$) $F_t = 0.946 \text{ kN}$ $F_r = 0.344 \text{ kN}$ $T = 15.323 \text{ kNm}$
	Turning (0.35 - 240 RPM)	99	
	Threading - 1 mm	99	
	Threading - 1.25 mm	96	
	Threading - 1.5 mm	99	

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Problem: 2 gear ratios need to be found for the changeable gears used between the **intermediate shaft and leadscrew**

I will now list the 9 different gears that would be provided and their specifications:

Number of Teeth	Pitch Diameter (m = 1.5 mm)
21	31.5
24	36
36	54
84	126
96	144
99	148.5
105	157.5
150	225
158	237

I can pick my spindle gear and its connecting gear with a suitable pitch diameter and number of teeth. It will need to have the previously chosen speed ratio of 1:5 and $m = 1.5$

Spindle Gear:

- Number of teeth = 105
- Pitch Diameter = 157.5 mm

Intermediate Gear (non-changeable) - Connect to the spindle:

- Number of teeth = 21
- Pitch Diameter = 31.5 mm

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Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

Shaft Layouts:

I will be working on my spindle first. I will be using **Shigly's Mechanical Engineering Design** as a guide to shaft design.

Common torque transfer elements include keys, splines, setscrews, pins... I **will be using keys** for ease and familiarity; and it would be easy assembly and disassembly. I will need to take into consideration the shaft profile when it comes to assembly and disassembly, generally keeping the largest diameter in the center.

Fluctuating stresses due to bending and torsion of a round cross section shaft:

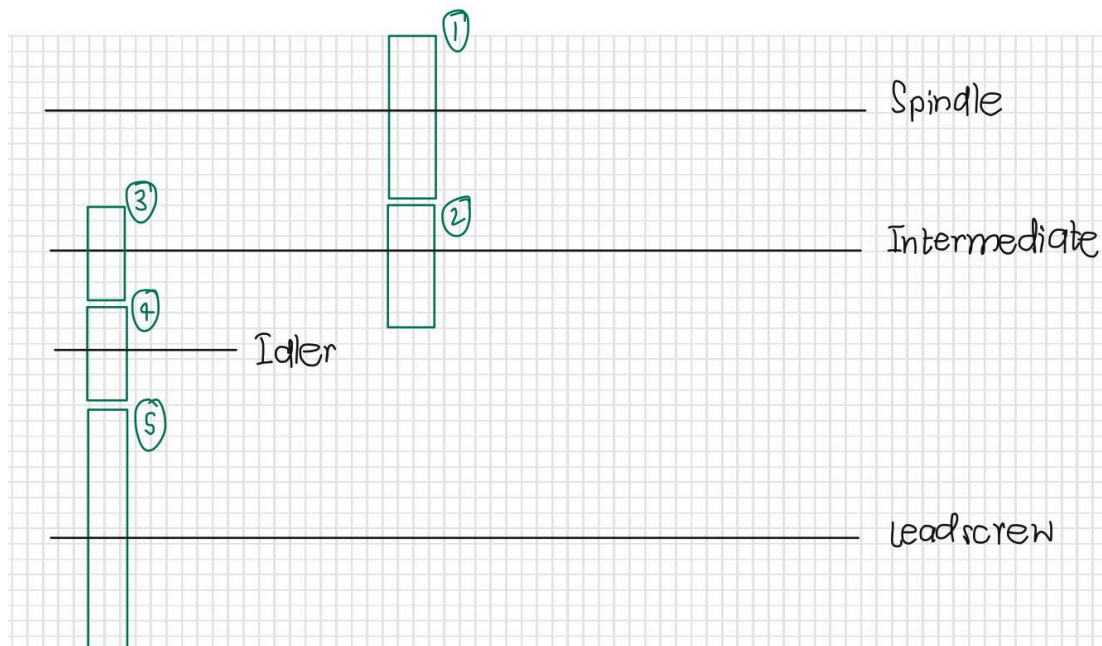
$$\sigma_a = K_f \frac{32M_a}{\pi d^3} \quad \sigma_m = K_f \frac{32M_m}{\pi d^3} \quad (7-2)$$

$$\tau_a = K_{fs} \frac{16T_a}{\pi d^3} \quad \tau_m = K_{fs} \frac{16T_m}{\pi d^3} \quad (7-3)$$

$$\sigma'_a = (\sigma_a^2 + 3\tau_a^2)^{1/2} = \left[\left(\frac{32K_f M_a}{\pi d^3} \right)^2 + 3 \left(\frac{16K_{fs} T_a}{\pi d^3} \right)^2 \right]^{1/2} \quad (7-4)$$

$$\sigma'_m = (\sigma_m^2 + 3\tau_m^2)^{1/2} = \left[\left(\frac{32K_f M_m}{\pi d^3} \right)^2 + 3 \left(\frac{16K_{fs} T_m}{\pi d^3} \right)^2 \right]^{1/2} \quad (7-5)$$

I will make a quick labelling of all my gears for ease of force analysis on the shafts

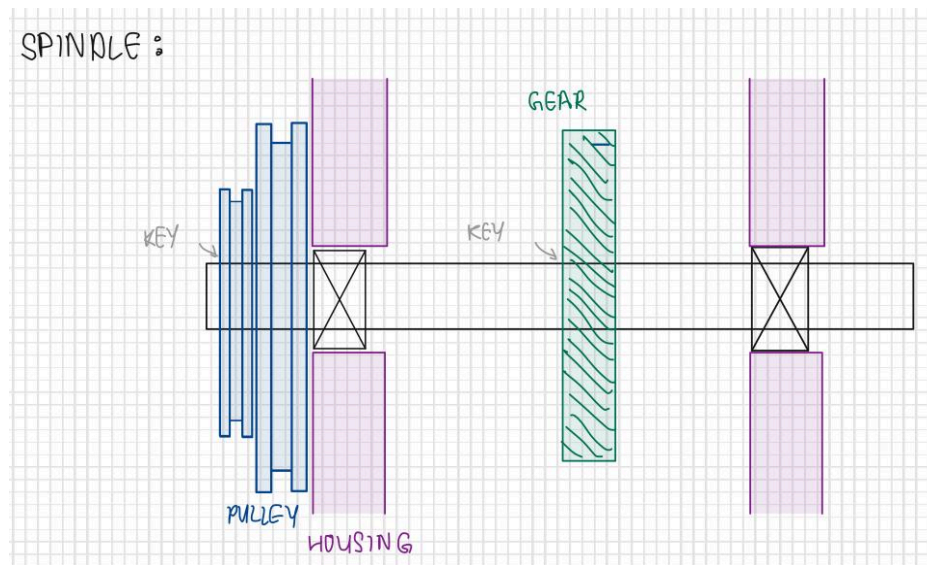


Analysis at 240 RPM - Condition with highest forces on shaft

Forces are found using the equations derived from gears and pulleys [from design sheet A10]

Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

1) Spindle Shaft



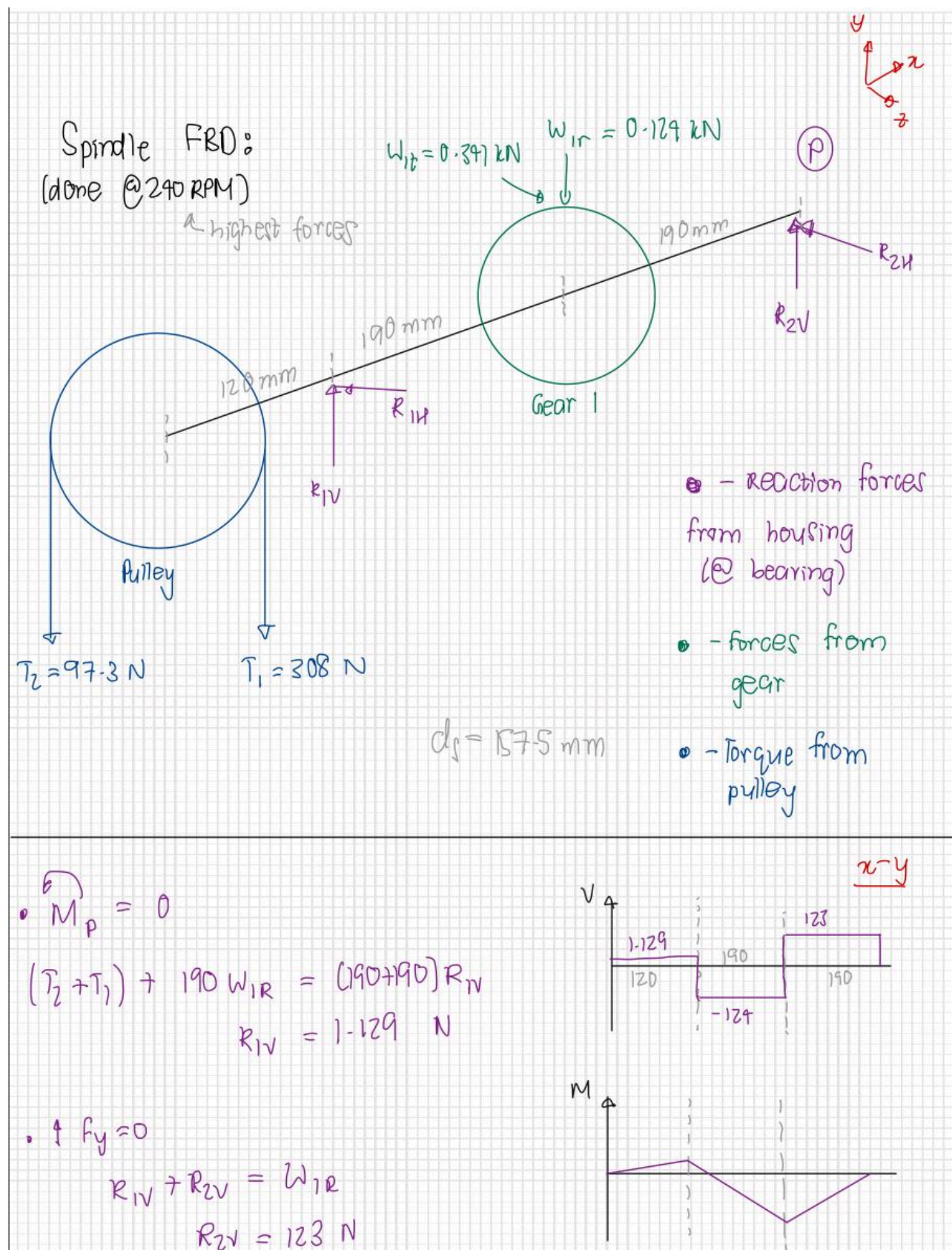
Forces at the gear 1:

$$W_{lt} = \frac{7.446}{d_s} = \frac{53.715}{157.5} = 0.341 \text{ kN}$$

$$W_{lr} = \frac{19.551}{d_s} = \frac{19.551}{157.5} = 0.124 \text{ kN}$$

I can now perform free body diagram analysis to get reaction forces:

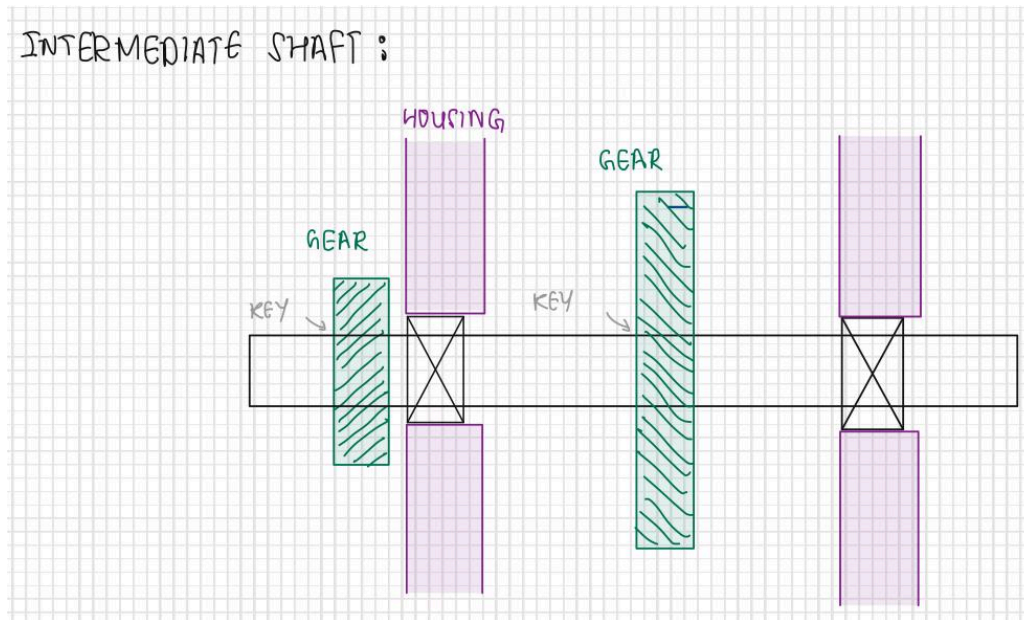
Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces



Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

Similarly I will do the free body analysis for the other shafts.

2) Intermediate Shaft



- Gear 2

$$W_{2t} = - \frac{7.446}{d_s} = \frac{7.446}{157.5} = 0.0473 \text{ kN}$$

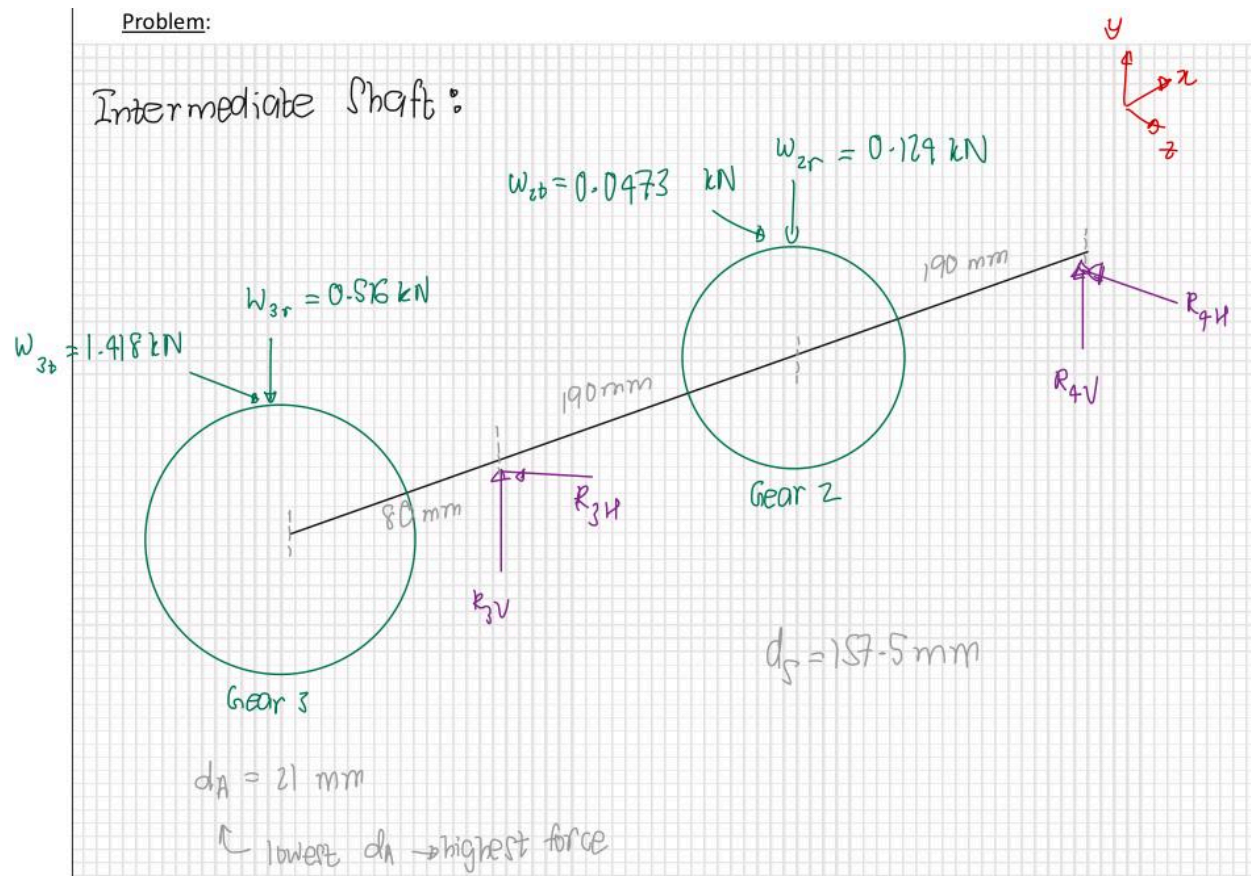
$$W_{2r} = \frac{19.551}{d_s} = \frac{19.551}{157.5} = 0.124 \text{ kN}$$

- Gear 3

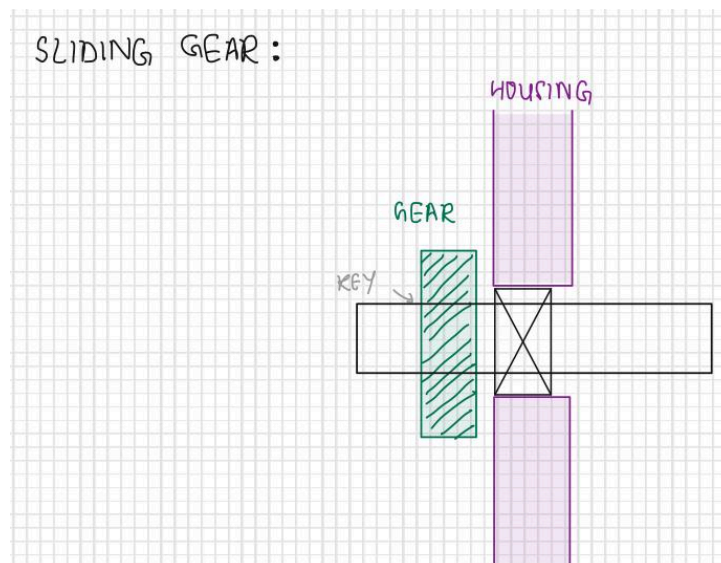
$$W_{3t} = \frac{29.784}{d_A} = \frac{29.784}{21} = 1.418 \text{ kN}$$

$$W_{3r} = W_{3t} \times \tan(20^\circ) = \frac{10.840}{d_A} = \frac{10.840}{21} = 0.516 \text{ kN}$$

Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces



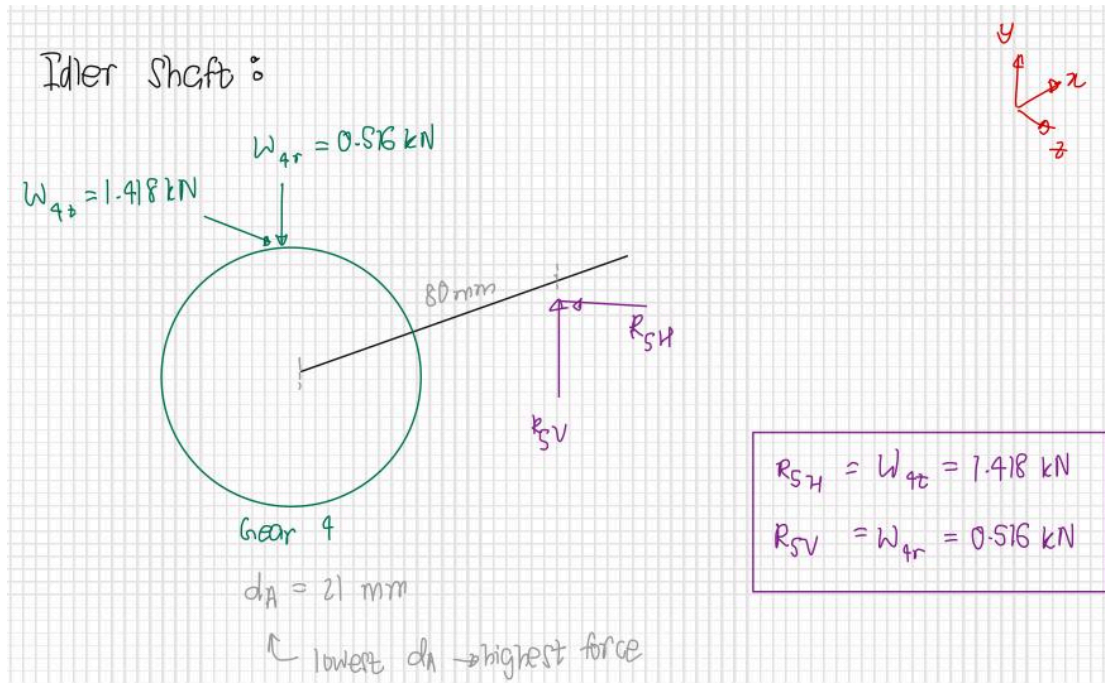
3) Idler Shaft



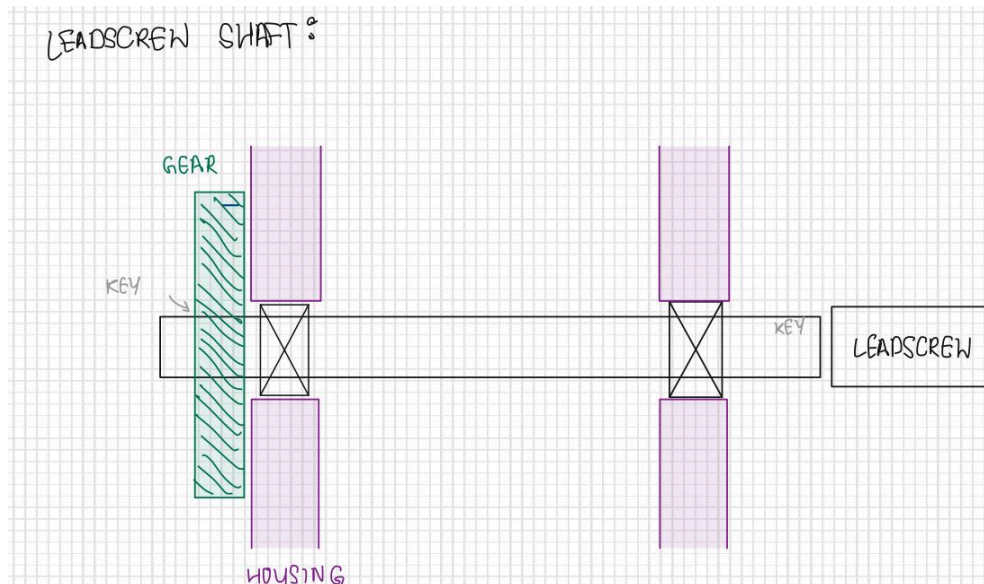
$$W_{4t} = -W_{3t} = -\frac{29.784}{d_A} = 1.418 \text{ kN}$$

Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

$$W_{4r} = -W_{3r} = -\frac{10.840}{d_A} = 0.516 \text{ kN}$$



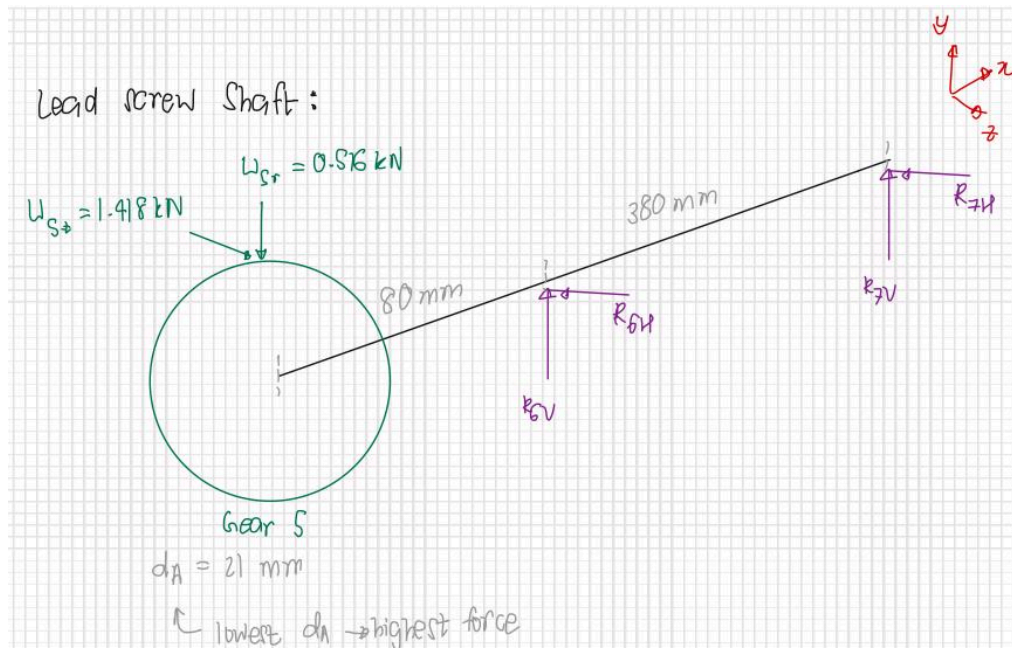
4) Lead Screw Shaft



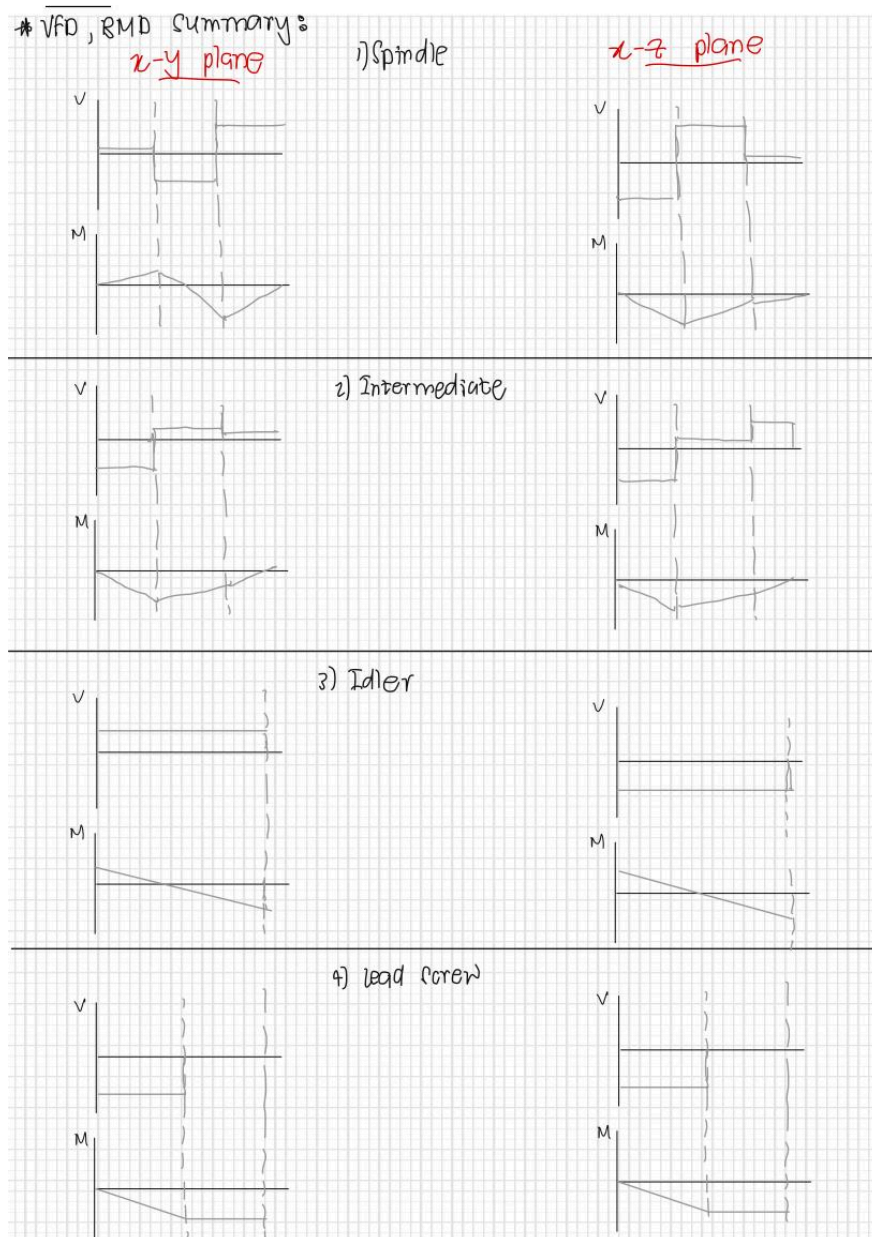
$$W_{5t} = -W_{4t} = \frac{29.784}{d_A} = 1.418 \text{ kN}$$

$$W_{5r} = -W_{4r} = \frac{10.840}{d_A} = 0.516 \text{ kN}$$

Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces



Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces



No FEA Analysis was needed as my shaft designs were kept very simple and could therefore be analysed just with force analysis.

Using my VFDs and BMDs, I have found the points with max force/stress and their values so I can now determine my minimum shaft diameters:

I will be referring to **Example 7-2, from Shigley's Machinery Handbook [pg. 386]** as a guidance for my minimum shaft diameter calculations.

1) Spindle shaft:

$$T_m = 47,276 \text{ Nmm (From design sheet A10 - Gear ratios)}$$

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Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

$$M_a = \sqrt{36.73^2 + 0.585^2} = 36,735 \text{ Nmm}$$

Assume generous fillet radius for gear.

As actual dimensions haven't been found yet, I will use Table 7-1 to estimate:

$$K_f = 1.7, K_{fs} = 1.5$$

For a quick conservative first, it is assumed,

$$K_f = K_b, K_{fs} = K_{ts}$$

Choosing an inexpensive steel, 1020 CD steel with $S_{ut} = 470 \text{ MPa}$

$$k_a = a S_{ut}^b = (3.04)(470^{-0.217}) = 0.8$$

$$k_b = 0.9 \text{ (to be checked later)}$$

$$k_c = k_d = k_e = 1$$

$$S_e = k_a k_b k_c k_d k_e S_e', S_e' = 0.5 S_{ut}$$

$$S_e = (0.8)(0.9)(1)(1)(1)(470) = 169.2$$

$$n = 1.5$$

Using the DE-Goodman criterion of equation 7-8;

$$d = \left(\frac{16n}{\pi} \left(\frac{2(K_f M_a)}{S_e} + \frac{(3(K_{fs} T_m)^2)^{1/2}}{S_{ut}} \right) \right)^{1/3}$$

$$d_{min} = 19.7 \text{ mm}$$

Similarly, I have used the **Excel Sheet [F6 - Shaft Sizing]** to calculate the minimum diameters for the other shafts:

Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

Kf	1.7	a	3.04
Kfs	1.5	b	-0.217
Sut	470	kb	0.9
ka	0.7998864656	kc,kd,ke	1
		n	1.5
Spindle Shaft			
Tm	47276		
Ma	36735		
Se	169.2		
dmin (mm)	19.7		
Intermediate Shaft			
Tm	47276		
Ma	143061		
Se	169.2		
dmin (mm)	28.8		
Idler Shaft			
Tm	0		
Ma	58055		
Se	169.2		
dmin (mm)	20.7		
Lead screw Shaft			
Tm	47726		
Ma	143061		
Se	169.2		
dmin (mm)	28.8		

Shaft	Minimum Diameter (mm)	Chosen Diameter (mm)
Spindle Shaft	19.7	20 (increased to 50)
Intermediate Shaft	28.8	30
Idler Shaft	20.7	25
Lead Screw Shaft	28.8	30 (increased to 40)

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Shaft Design	Designer: Yasan Gunathilaka	Ref: A11
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Problem: Shaft design and analysis need to be done to ensure the shaft can endure the forces

The chosen diameter for the spindle was 20 mm, but upon bearing selections, my 2 bearings diameters are:

Bearing 1: Inner - 35 mm, Outer - 47 mm

Bearing 2: Inner - 25 mm, Outer - 47 mm

Considering the assembly of the shaft, I would want my shaft diameter (at the center) to be the largest diameter, therefore I will have to **increase my spindle diameter to 50 mm**.

All shafts will be assembled from right to left and are designed such that they can be assembled and disassembled in such a way.

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Bearing Selection	Designer: Yasan Gunathilaka	Ref: A12
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Problem: The 4 shafts would need bearings to be mounted to the housing

Within that process an approach is to:

1. Estimate the parameters for the relevant **Equivalent Dynamic Bearing Load** by taking typical values for the estimated bearing size.
2. Calculate the **required Dynamic Load Rating, C**.
3. Find a bearing with that rating and use its **parameters for Equivalent Dynamic Bearing Load** to calculate the expected life.
4. **Iterate as necessary** to ensure a suitable bearing is selected.
5. Check the other criteria are met:
esp. minimum load and maximum static load.

NOTE:

Requirements for bearing selection:

- 1) Bearing type
- 2) Bearing life
- 3) Bearing/shaft fit

Where possible I have placed my load carrying components (gear) in between bearings. The **bearing on the left hand side will bear my axial load** (locating bearings). The **other bearing is simply press-fit** onto the shaft with no shoulder.

I will be using roller bearings where possible as they support and guide with minimal friction. And they are more suited for shafts. I will be referring to **MECH215 Rolling Bearing slides** for my bearing analysis.

I will also be using the **SKF rolling bearings catalogue** for my bearing selections.

Calculating Bearing Life - Basic Rating Life in ISO

Equations:

$$L_{10} = \left(\frac{C}{P}\right)^p, L_{10h} = \left(\frac{10^6}{60n}\right)L_{10}$$

C - basic dynamic load rating (kN)

P - equivalent dynamic radial load (kN)

n - rotational speed (2000 RPM - worst case)

p = 3 for ball bearing and 10/3 for roller bearings

I have estimated that the hobbyist would use the lathe for about 6 months on average (increased to 8 months just in case), at about 5 hours each day for a life span of about 10 years. They wouldn't need it running constantly, I assume they would only be using it for small projects they'd be working on.

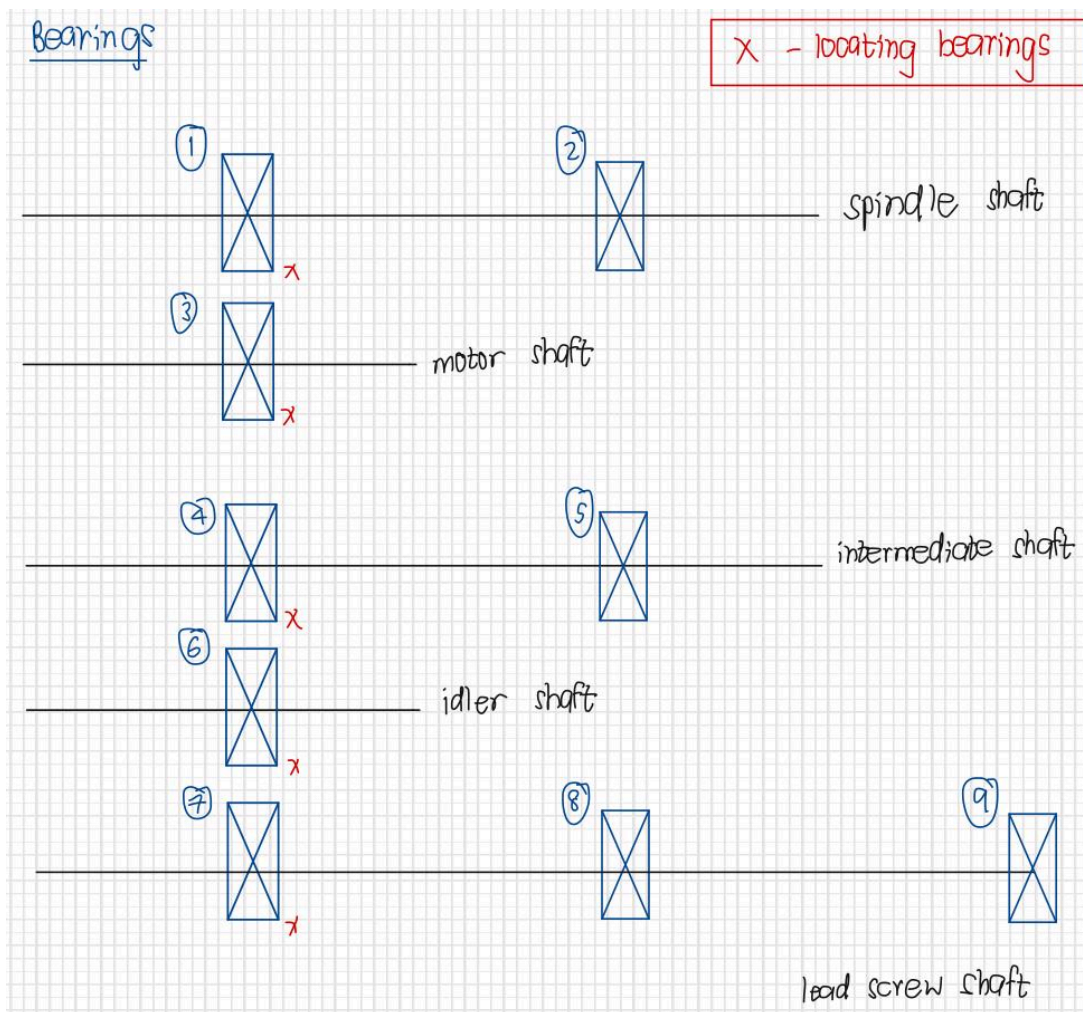
Problem: The 4 shafts would need bearings to be mounted to the housing

$$L_{10h} = 10 \text{ yrs} \times 8 \text{ months} \times 30 \text{ days/month} \times 4 \text{ hrs/day} = 9600 \text{ total operating hours}$$

Calculating Bearing Size

I will do **size selection based on rating life** instead of static load as I have rotating shafts with normal speeds, good lubrication and somewhat smooth loads

I have 9 bearings, similar to my gear layouts - I will make a layout and label the bearings for ease of understanding.



From above equation,

$$C = P \left(\frac{60n}{10^6} L_{10h} \right)^{\frac{1}{p}}$$

I will calculate C for the worst conditions (2000 RPM) and then do trial and reiteration with different SKF bearings to ensure that the C_{new} is less than the C_{calc} .

Problem: The 4 shafts would need bearings to be mounted to the housing

$$F_a/F_r \leq e \rightarrow P = F_r$$

$$F_a/F_r > e \rightarrow P = X F_r + Y F_a$$

I will use Excel Sheet: **F7 - Bearing Selection** for my calculations:

L10h	9600			
n	2000			
SPINDLE SHAFT			MOTOR SHAFT	
Bearing 1	Deep Groove Ball		Bearing 3	Roller bearing
p	3.00		p	3.33
Fr	0.301	Calculated	P (Fr)	3.390
Fa	1.22	Spindle thrust at drilling [F4]	C	28.095
f0	14			
C0	24			
f0Fa/C0	0.71			
C bearing	43.30			
e	0.26			
X	0.56			
Y	1.70			
Fa/Fr	4.05	Fa/Fr > e		
P	2.24			
C	23.51	C < C bearing		
Bearing 2	Roller bearing			
p	3.33			
P (Fr)	0.795			
C	6.589			

$$F_a/F_r \leq e \rightarrow P = F_r$$

$$F_a/F_r > e \rightarrow P = X F_r + Y F_a$$

Problem: The 4 shafts would need bearings to be mounted to the housing

Attempt 1

①

d	D	B	C	C ₀							
15	24	5	1,9	1,1	0,048	-	17 000	0,0074	► 61802-2RS1	-	
	24	5	1,9	1,1	0,048	60 000	30 000	0,0074	► 61802-2Z	-	
	24	5	1,9	1,1	0,048	60 000	38 000	0,0065	► 61802	-	
	28	7	4,36	2,24	0,095	-	16 000	0,016	► 61902-2RS1	-	
	28	7	4,36	2,24	0,095	56 000	28 000	0,016	► 61902-2RZ	-	
	28	7	4,36	2,24	0,095	56 000	28 000	0,016	► 61902-2Z	-	
	28	7	4,36	2,24	0,095	56 000	34 000	0,016	► 61902	-	
	32	8	5,85	2,85	0,12	50 000	32 000	0,027	► 16002	-	
	32	8	5,85	2,85	0,12	50 000	26 000	0,025	► 16002-2Z	16002-Z	
	32	9	5,85	2,85	0,12	50 000	32 000	0,03	► 6002	-	
	32	9	5,85	2,85	0,12	-	14 000	0,03	► 6002-2RSH	6002-RSH	
	32	9	5,85	2,85	0,12	50 000	26 000	0,03	► 6002-2RSL	6002-RSL	

15	17,8	-	-	22,2	0,3	17	17,8	22	0,3	0,015	14
	17,8	-	-	22,2	0,3	17	17,8	22	0,3	0,015	14
	17,8	-	21,3	-	0,3	17	-	22	0,3	0,015	14
	18,8	-	-	25,3	0,3	17	18,3	26	0,3	0,02	14
	18,8	-	-	25,3	0,3	17	18,3	26	0,3	0,02	14
	18,8	-	-	25,3	0,3	17	18,3	26	0,3	0,02	14
	18,8	-	-	25,3	0,3	17	-	26	0,3	0,02	14
	20,5	-	-	28,2	0,3	17	-	30	0,3	0,02	14
	20,5	-	-	28,2	0,3	17	20,1	30	0,3	0,02	14
	20,5	-	-	28,2	0,3	17	-	30	0,3	0,025	14
	-	18,3	-	28,2	0,3	17	18,5	30	0,3	0,025	14
	-	18,3	-	28,2	0,3	17	18,5	30	0,3	0,025	14

Calculation factors for deep groove ball bearings

Single row and double row bearings
Normal clearance

$f_0 F_d / C_0$	e	X	Y
0,172	0,19	0,56	2,3
0,345	0,22	0,56	1,99
0,689	0,26	0,56	1,71
1,03	0,28	0,56	1,55
1,38	0,3	0,56	1,45
2,07	0,34	0,56	1,31
3,45	0,38	0,56	1,15
5,17	0,42	0,56	1,04
6,89	0,44	0,56	1

$$\frac{f_0 F_d}{C_0} = 7,63$$

→ interpolate

Problem: The 4 shafts would need bearings to be mounted to the housing

Attempt 2

Calculation factors for deep groove ball bearings			
Single row and double row bearings Normal clearance			
$f_0 F_d / C_0$	e	X	Y
0,172	0,19	0,56	2,3
0,345	0,22	0,56	1,99
0,689	0,26	0,56	1,71
1,03	0,28	0,56	1,55
1,38	0,3	0,56	1,45
2,07	0,34	0,56	1,31
3,45	0,38	0,56	1,15
5,17	0,42	0,56	1,04
6,89	0,44	0,56	1

$$\frac{f_0 F_d}{C_0} = 3.42$$

$$d = 20$$

$$f_0 = 1$$

$$C_0 = 5$$

$$C_{bear} = 9.95$$

$$e = 0.39 \quad X = 0.56 \quad Y = 1.15$$

$$C_{calc} = 76.52 > C_{bear}$$

X

Attempt 3

Calculation factors for deep groove ball bearings			
Single row and double row bearings Normal clearance			
$f_0 F_d / C_0$	e	X	Y
0,172	0,19	0,56	2,3
0,345	0,22	0,56	1,99
0,689	0,26	0,56	1,71
1,03	0,28	0,56	1,55
1,38	0,3	0,56	1,45
2,07	0,34	0,56	1,31
3,45	0,38	0,56	1,15
5,17	0,42	0,56	1,04
6,89	0,44	0,56	1

$$\frac{f_0 F_d}{C_0} = 0.71$$

$$d = 40$$

$$f_0 = 14$$

$$C_0 = 24$$

$$C_{bear} = 43.3$$

$$e = 0.26 \quad X = 0.56 \quad Y = 1.7$$

$$C_{calc} = 23.51 < 43.3$$

✓

Problem: The 4 shafts would need bearings to be mounted to the housing

Interpolation	
$y = y_1 + (x - x_1) \frac{(y_2 - y_1)}{(x_2 - x_1)}$	
x	0.71
e	0.26
x1	0.69
y1	0.26
x2	1.03
y2	0.28
x	0.71
X	0.56
x1	0.69
y1	0.56
x2	1.03
y2	0.56
x	0.71
Y	1.70
x1	0.69
y1	1.71
x2	1.03
y2	1.55

INTERMEDIATE SHAFT		IDLER SHAFT		LEAD SCREW SHAFT	
Bearing 4	Roller bearing	Bearing 6	Roller bearing	Bearing 7	Roller bearing
p	3.33	p	3.33	p	3.33
P (Fr)	2.320	P (Fr)	1.996	P (Fr)	2.010
C	19.228	C	16.542	C	16.658
Bearing 5	Roller bearing			Bearing 8	Roller bearing
p	3.33			p	3.33
P (Fr)	3.470			P (Fr)	0.486
C	28.758			C	4.028
				Bearing 9	Roller bearing
				p	3.33
				P (Fr)	0.486
				C	4.028

Problem: The 4 shafts would need bearings to be mounted to the housing

Principal dimensions			Basic load ratings		Fatigue	Speed ratings		Mass	Designations	Alternative
d	D	B	C	C ₀	load limit	Reference	Limiting		Bearing with	standard cage ¹⁾
mm			kN		kN	speed	speed	kg	--	
15	35	11	12,5	10,2	1,22	22 000	26 000	0,047	► NU 202 ECP	PHA
	35	11	12,5	10,2	1,22	22 000	26 000	0,048	► NJ 202 ECP	PHA
	40	12	20	14,3	1,73	20 000	22 000	0,066	► N 203 ECP	PH
17	40	12	20	14,3	1,73	20 000	22 000	0,068	► NU 203 ECP	PHA
	40	12	20	14,3	1,73	20 000	22 000	0,069	► NJ 203 ECP	PHA
	40	12	20	14,3	1,73	20 000	22 000	0,072	► NUP 203 ECP	PHA
20	40	16	27,5	21,6	2,65	20 000	22 000	0,087	► NU 2203 ECP	--
	40	16	27,5	21,6	2,65	20 000	22 000	0,093	► NJ 2203 ECP	--
	40	16	27,5	21,6	2,65	20 000	22 000	0,097	► NUP 2203 ECP	--
25	47	14	28,5	20,4	2,55	17 000	20 000	0,12	► N 303 ECP	--
	47	14	28,5	20,4	2,55	17 000	20 000	0,12	► NJ 303 ECP	--
	47	14	28,5	20,4	2,55	17 000	20 000	0,12	► NU 303 ECP	--
20	47	14	28,5	22	2,75	17 000	19 000	0,11	► N 204 ECP	--
	47	14	28,5	22	2,75	17 000	19 000	0,11	► NJ 204 ECP	ML, PHA
	47	14	28,5	22	2,75	17 000	19 000	0,11	► NU 204 ECP	ML, PHA
25	47	18	34,5	27,5	3,45	17 000	19 000	0,14	► NUP 204 ECP	ML, PHA
	47	18	34,5	27,5	3,45	17 000	19 000	0,14	► NJ 2204 ECP	--
	47	18	34,5	27,5	3,45	17 000	19 000	0,14	► NU 2204 ECP	--
30	52	15	35,5	26	3,25	15 000	18 000	0,14	► NU 304 ECP	--
	52	15	35,5	26	3,25	15 000	18 000	0,15	► N 304 ECP	--
	52	15	35,5	26	3,25	15 000	18 000	0,15	► NJ 304 ECP	--
35	52	15	35,5	26	3,25	15 000	18 000	0,16	► NUP 304 ECP	--
	52	21	47,5	38	4,8	15 000	18 000	0,21	► NU 2304 ECP	--
	52	21	47,5	38	4,8	15 000	18 000	0,22	► NJ 2304 ECP	--
40	52	21	47,5	38	4,8	15 000	18 000	0,22	► NUP 2304 ECP	--
	52	12	14,2	13,2	1,4	18 000	18 000	0,082	► NU 1005	--
	52	15	32,5	27	3,35	15 000	16 000	0,13	► N 205 ECP	--
45	52	15	32,5	27	3,35	15 000	16 000	0,13	► NU 205 ECP	J, ML, PH, PHA
	52	15	32,5	27	3,35	15 000	16 000	0,14	► NJ 205 ECP	J, ML, PH, PHA
	52	15	32,5	27	3,35	15 000	16 000	0,14	► NUP 205 ECP	J, ML, PH, PHA
50	52	18	39	34	4,25	15 000	16 000	0,16	► NU 2205 ECP	ML, PH
	52	18	39	34	4,25	15 000	16 000	0,17	► NJ 2205 ECP	ML, PH
	52	18	39	34	4,25	15 000	16 000	0,17	► NUP 2205 ECP	ML, PH
55	62	17	46,5	36,5	4,55	12 000	15 000	0,23	► N 305 ECP	--

I was CADing and I **found an error** I have missed: As per my specification, my spindle bore diameter has to be 18 mm. This means I wouldn't be able to have my bearing 2 inner diameter be 15 mm (smaller than spindle bore diameter).

I will pick my new Bearing 2 to be 25 mm

25	47	12	14,2	13,2	1,4	18 000	18 000	0,082	► NU 1005	--
	52	15	32,5	27	3,35	15 000	16 000	0,13	► N 205 ECP	--
	52	15	32,5	27	3,35	15 000	16 000	0,13	► NU 205 ECP	J, ML, PH, PHA

Additionally I will need to **readjust my Spindle Shaft Diameter**. I will return design sheet A11 to make the change.

Problem: The 4 shafts would need bearings to be mounted to the housing

I have also noticed that my bearing one with 40 mm inner diameter and 80 mm outer diameter is **abnormally big** even though the calculations seem to add up. I tried with a slightly small bearing: 35 mm inner diameter and 47 mm outer diameter - but the calculations don't add up there.

I think I may have **made a mistake** just for my Bearing 1 calculation but I haven't got the time to reiterate and see why I need such a big bearing. For now I will stick to the biggest reasonable size of 35 mm inner diameter and 47 mm outer diameter even though the calculation doesn't match up:

Attempt 4

Calculation factors for deep groove ball bearings

Single row and double row bearings
Normal clearance

$f_0 F_a / C_0$	e	X	Y
0,172	0,19	0,56	2,3
0,345	0,22	0,56	1,99
0,689	0,26	0,56	1,71
1,03	0,28	0,56	1,55
1,38	0,3	0,56	1,45
2,07	0,34	0,56	1,31
3,45	0,38	0,56	1,15
5,17	0,42	0,56	1,04
6,89	0,44	0,56	1

$$\frac{f_0 F_a}{C_0} = 5,7$$

$$d = 35 \text{ mm}$$

$$f_0 = 14$$

$$C_0 = 3,75$$

$$C_{\text{bear}} = 4,36$$

→ interpolate

$$e = 0,42 \quad X = 0,56 \quad Y = 1,33$$

$$C_{\text{calc}} = 18,72 < 4,36 \quad \times$$

35	47	7	4,36	3,35	0,14	-	8 500	0,022	▶ 61807-2RS1	-
	47	7	4,36	3,35	0,14	30 000	15 000	0,03	▶ 61807-2RZ	-
	47	7	4,36	3,35	0,14	30 000	18 000	0,029	▶ 61807	-

35	38,2	-	-	44,4	0,3	37	38	45	0,3	0,015	14
	38,2	-	-	44,4	0,3	37	38	45	0,3	0,015	14
	38,2	-	42,8	-	0,3	37	-	45	0,3	0,015	14

EDIT - When CADing I found out that I could simplify my design further by not have bearing 8 and just leaving a hole in the housing with some clearance. Bearing 8 is not necessary as we have bearing 7 and 9 to hold the shaft in place. **Bearing 8 has been removed.**

The 8 bearings (7 roller bearings and 1 deep groove ball bearing) have been chosen :)

FACULTY OF ENGINEERING & INFORMATION SCIENCES School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Lubrication	Designer: Yasan Gunathilaka	Ref: A13
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Problem: The lathe would need to be lubricated to keep it running smoothly

I have never previously done anything related to lubrication so I will need to do some research to learn about it.

Lubrication widely used to control friction and wear during service.

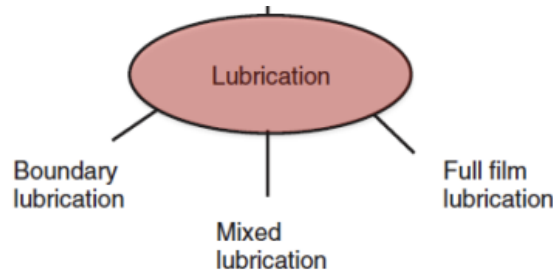


Table 1.1 Typical friction coefficients for different types of contact conditions

Case	Friction coefficient	Type of motion
Unlubricated journal bearing	0.1–0.2	Sliding
Oil lubricated journal bearings	0.01–0.09	Sliding
Oil lubricated roller bearings	0.001–0.005	Rolling
Unlubricated roller bearings ^a	0.0005–0.001	Rolling

^a Only valid for low loads

Reason for lubrication - reduce friction, wear, reduce noise and vibrations, cool components, prevent corrosion, separate moving parts, carry away contaminants and debris, act as a sealant - all of which are important for my hobbyist lathe.

Types of lubricants to explore:

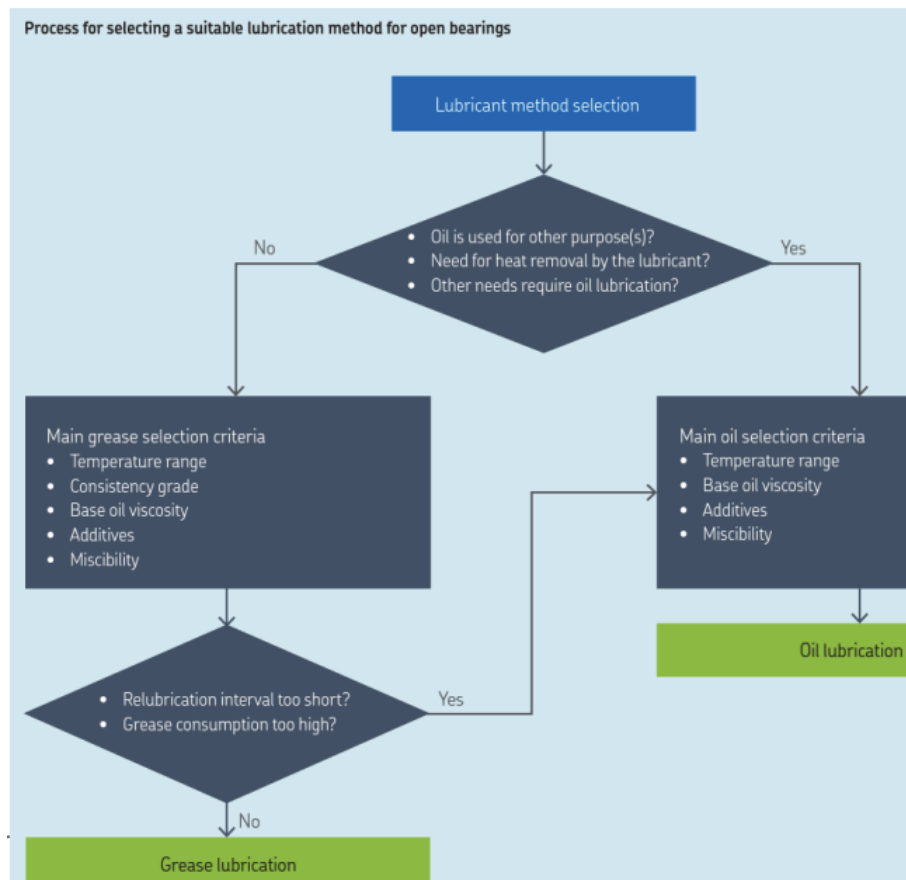
Liquid: Animal oils, Vegetable oils, Petroleum oils, Synthetic

Semi-solid (Grease): formed by emulsifying oil and fat with thickening agents; soda based, lithium based, calcium based

Solid: Graphite, MoS₂

FACULTY OF ENGINEERING & INFORMATION SCIENCES UNIVERSITY OF WOLLONGONG School of Mechanical Materials Mechatronic and Biomedical Engineering - MECH311 Project: Detailed Design System: Hobbyist Lathe	TASK: Lubrication	Designer: Yasan Gunathilaka	Ref: A13
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Problem: The lathe would need to be lubricated to keep it running smoothly



To be considered - Right type, quality, amount, time, place

2 categories of lubricant application: All-loss, reuse

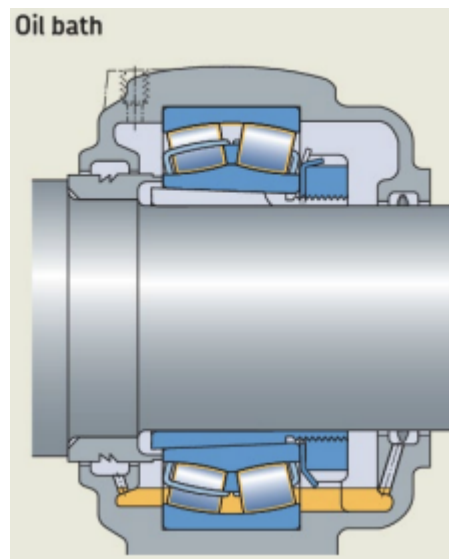
LUBRICATING THE LATHE

Areas to consider:

- Spindle bearings - As the spindle would be undergoing a lot of rotational motion

Light machine oil would be sufficient, I will be using an oil bath method, similar to one shown in the picture

Problem: The lathe would need to be lubricated to keep it running smoothly



- Lead Screw + Carriage - both components would be engaging with each other which could lead to wear and tear

I intend to use light grease, as this is more suitable for components with linear motion and heavier load. As the carriage and leadscrew is not very accessible once it is assembled, light grease would only need to be applied when the lathe is undergoing maintenance, about once every 2 months should be sufficient based on the usage by the hobbyist.

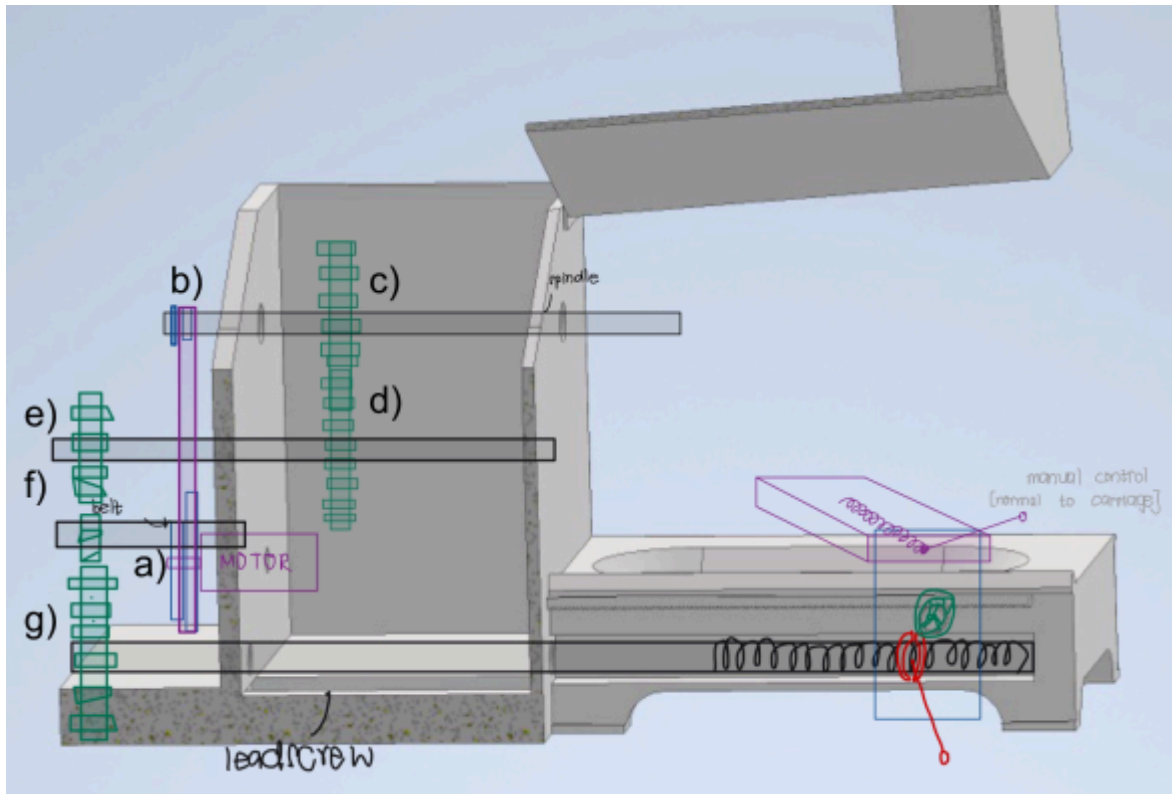
- Gearbox

As we are using changeable gears, I wouldn't be able to add a constant reuse method. However it's very easy to access the changeable gears as they would constantly be replaced when changing gear ratios. A simple gear oil would be ideal, and it can be applied whenever the lubrication seems to be missing when the gears are changed.

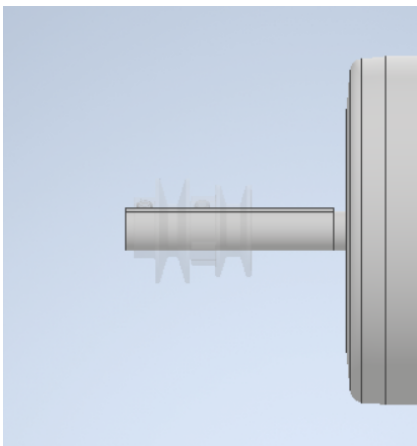
- Pulleys

Pulley wouldn't necessarily need lubrication, as our pulley system is somewhat open, it is important to make sure that the belt isn't undergoing wear and tear and that no debris gets in between.

Problem: Different tolerances would be needed for each shaft based on the fitting



a) Motor shaft-pulley tolerance:



In order to maintain zero clearance between the pulley holes and the motor shaft, 'Transition fit (similar fit) - H7/k6' is selected. This is to prevent the pulleys from moving/sliding. It also aids with vibration reduction.

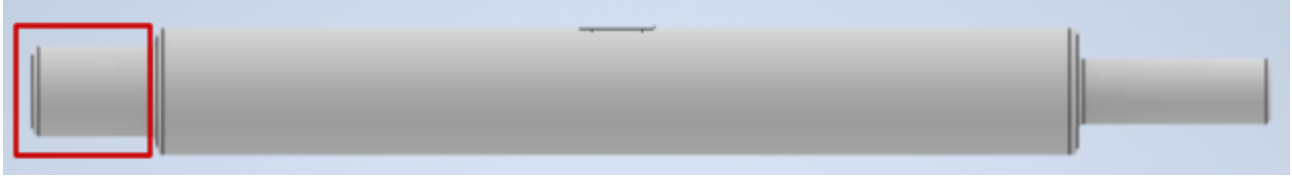
$D = 19.05\text{mm}$

ISO description of fit: **H7k9**

Large hole = $19.05 + 0.021 = 19.071\text{mm}$	Large shaft = $19.05 + 0.015 = 19.065\text{mm}$
Small hole = $19.05 + 0 = 19.05\text{mm}$	Small shaft = $19.05 + 0.002 = 19.052\text{mm}$
Tolerance = 0.021mm	Tolerance = 0.013mm

Problem: Different tolerances would be needed for each shaft based on the fitting

b) Spindle shaft-pulley tolerance:



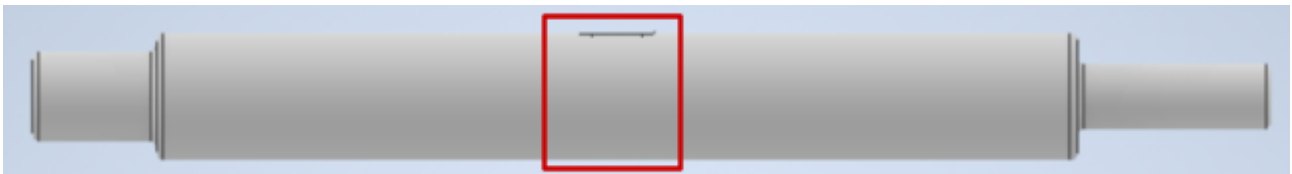
Spindle shaft-pulley tolerance will also have a 'Transition fit(similar fit) - H7k6'. This is to prevent the pulleys from moving/sliding.

$D = 35.00 \text{ mm}$

Iso description of fit: **H7k9**

Large hole = $35.00 + 0.025 = 35.025\text{mm}$	Large shaft = $35.00 + 0.018 = 35.018\text{mm}$
Small hole = $35.00 + 0 = 35.00\text{mm}$	Small shaft = $35.00 + 0.002 = 35.002\text{mm}$
Tolerance = 0.025mm	Tolerance = 0.016mm

c) Spindle shaft-gear tolerance:



Spindle shaft-gear tolerance will also have a 'Transition fit(similar fit) - H7k6'. This is to prevent the spindle gear from moving/sliding along the shaft.

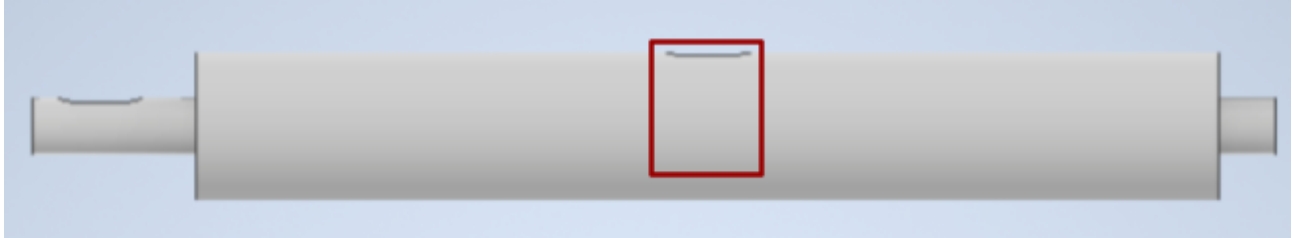
$D = 50.00 \text{ mm}$

ISO description of fit: **H7k9**

Large hole = $50.00 + 0.025 = 50.025\text{mm}$	Large shaft = $50.00 + 0.018 = 50.018\text{mm}$
Small hole = $50.00 + 0 = 50.00\text{mm}$	Small shaft = $50.00 + 0.002 = 50.002\text{mm}$
Tolerance = 0.025mm	Tolerance = 0.016mm

Problem: Different tolerances would be needed for each shaft based on the fitting

d) Intermediate shaft-gear tolerance (intermediate shaft and spindle shaft connection):



Intermediate shaft-gear tolerance will also have a 'Transition fit(similar fit) - H7k6'. This is to prevent the intermediate gear from moving/sliding along the intermediate shaft.

$D = 27.00 \text{ mm}$

ISO description of fit: **H7k9**

Large hole = $27 + 0.025 = 27.025\text{mm}$	Large shaft = $27.00 + 0.018 = 27.018\text{mm}$
Small hole = $27 + 0 = 27.00\text{mm}$	Small shaft = $27.00 + 0.002 = 27.002\text{mm}$
Tolerance = 0.025mm	Tolerance = 0.016mm

e) Intermediate shaft-gear tolerance (intermediate shaft & idler shaft connection):



Intermediate shaft-gear tolerance will also have a 'Transition fit(similar fit) - H7k6'. This is to prevent the intermediate gear from moving/sliding along the intermediate shaft.

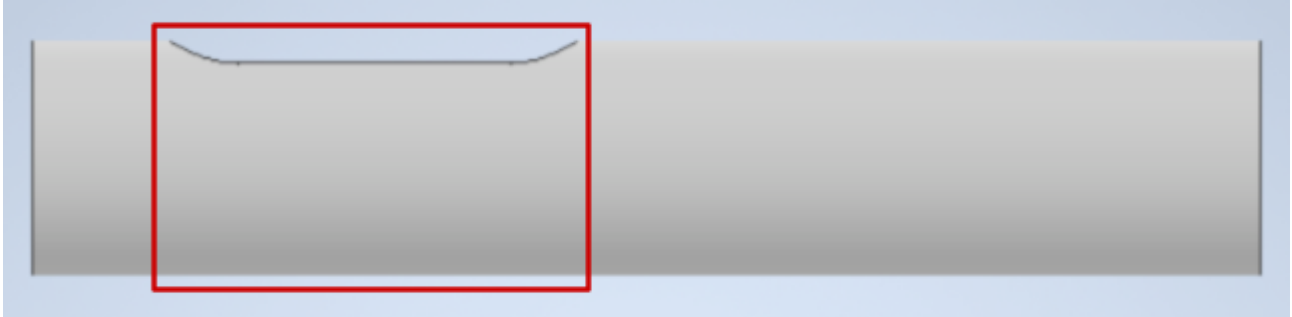
$D = 20.00 \text{ mm}$

ISO description of fit: **H7k9**

Large hole = $20.00 + 0.025 = 20.025\text{mm}$	Large shaft = $20.00 + 0.018 = 20.018\text{mm}$
Small hole = $20.00 + 0 = 20.00\text{mm}$	Small shaft = $20.00 + 0.002 = 20.002\text{mm}$
Tolerance = 0.025mm	Tolerance = 0.016mm

Problem: Different tolerances would be needed for each shaft based on the fitting

f) Idler shaft-gear tolerance (intermediate shaft & idler shaft connection):



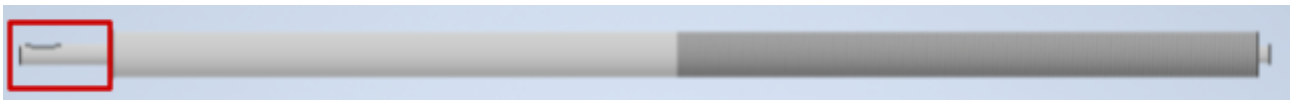
Idler shaft-gear tolerance will also have a 'Transition fit(similar fit) - H7k6'. This is to prevent the idler gear from moving/sliding along the idler shaft.

$D = 17.00 \text{ mm}$

ISO description of fit: **H7k9**

Large hole = $17.00 + 0.021 = 17.021\text{mm}$	Large shaft = $17.00 + 0.015 = 17.015\text{mm}$
Small hole = $17.00 + 0 = 17.00\text{mm}$	Small shaft = $17.00 + 0.002 = 17.002\text{mm}$
Tolerance = 0.021mm	Tolerance = 0.013mm

g) Lead screw-gear tolerance (Leadscrew & idler shaft connection):



Lead screw shaft-gear tolerance will also have a 'Transition fit(similar fit) - H7k6'. This is to prevent the lead screw gear from moving/sliding.

$D = 17.00 \text{ mm}$

ISO description of fit: **H7k9**

Large hole = $17.00 + 0.021 = 17.021\text{mm}$	Large shaft = $17.00 + 0.015 = 17.015\text{mm}$
Small hole = $17.00 + 0 = 17.00\text{mm}$	Small shaft = $17.00 + 0.002 = 17.002\text{mm}$
Tolerance = 0.021mm	Tolerance = 0.013mm